

Class05

Alishba

TITLE HERE

Subtitle

Subsubtitle

stuff in bold

not in bold

indent text

something italics

Q1: Make a ggplot plot

Answer Q1: gherferfer

```
x <- 1:5  
x + 1
```

```
[1] 2 3 4 5 6
```

Let's make a graph

```
library()
```

Fixing Bad code:

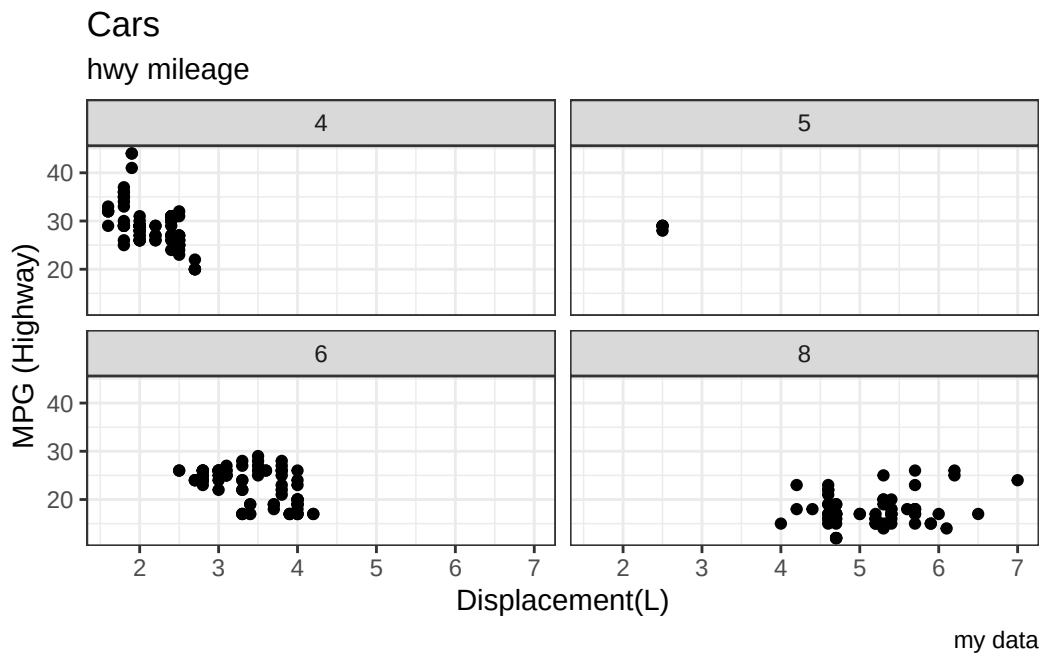
```
4 + 2
```

[1] 6

Plots:

```
# loaded ggplot2 package
library(ggplot2)

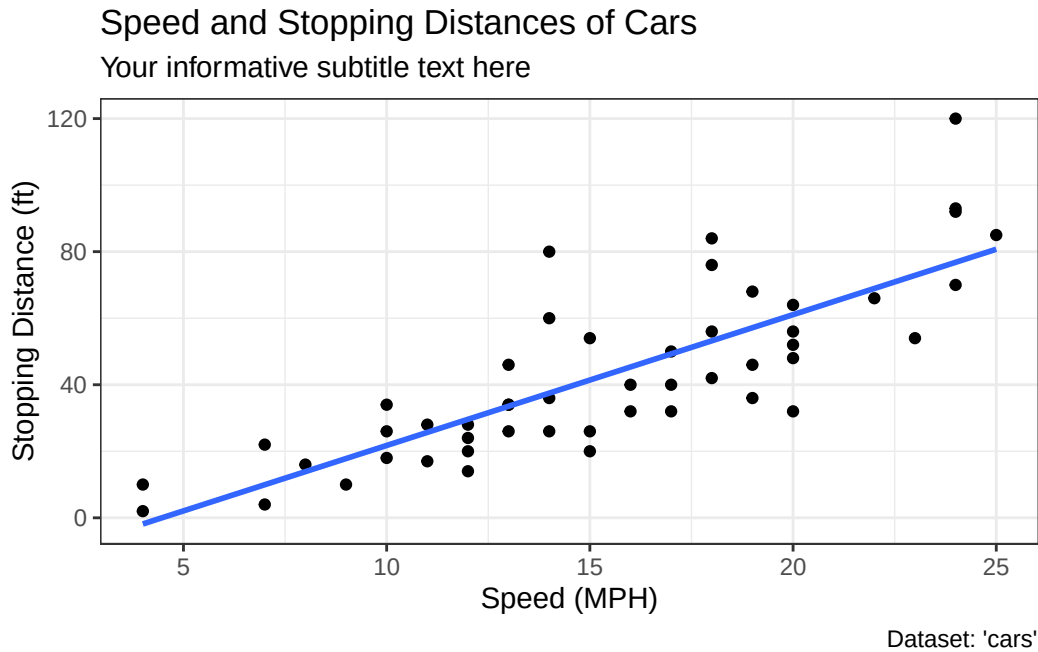
# specified data set, then determined the aesthetics
ggplot(mpg) +
  aes(x=displ, y=hwy) +
  # visualized data using scatterplot; included trendline
  geom_point() +
  theme_bw() +
  labs(title="Cars", subtitle="hwy mileage", caption="my data", x="Displacement(L)", y="MPG (Highway)")
  facet_wrap(~cyl)
```



```
# made another graph using different data set, aesthetics, etc.
library(ggplot2)
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  labs(title="Speed and Stopping Distances of Cars", x="Speed (MPH)", y="Stopping Distance (ft)")
```

```
geom_smooth(method="lm", se=FALSE) +  
theme_bw()
```

`geom\_smooth()` using formula = 'y ~ x'



Q: Which geometric layer should be used to create scatter plots in ggplot2?

A: `geom_point()`

In another data set, I created a plot answered questions about gene expression.

```
# created a new data set on anti-viral drugs  
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"  
genes <- read.delim(url)  
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging

```

5      AATK  0.4711421  0.5598642  unchanging
6 AB015752.4 -3.6808610 -3.5921390  unchanging

```

```

# found number of genes in data set
nrow(genes)

```

```

[1] 5196

```

```

#found column names and number of columns
colnames(genes)

```

```

[1] "Gene"      "Condition1" "Condition2" "State"

```

```

ncol(genes)

```

```

[1] 4

```

```

# determined number of 'up' regulated genes
table(genes$State)

```

```

      down  unchanging      up
      72      4997      127

```

```

#determined fraction of up-regulated genes, rounded it
round( table(genes$State)/nrow(genes) * 100, 2 )

```

```

      down  unchanging      up
      1.39      96.17      2.44

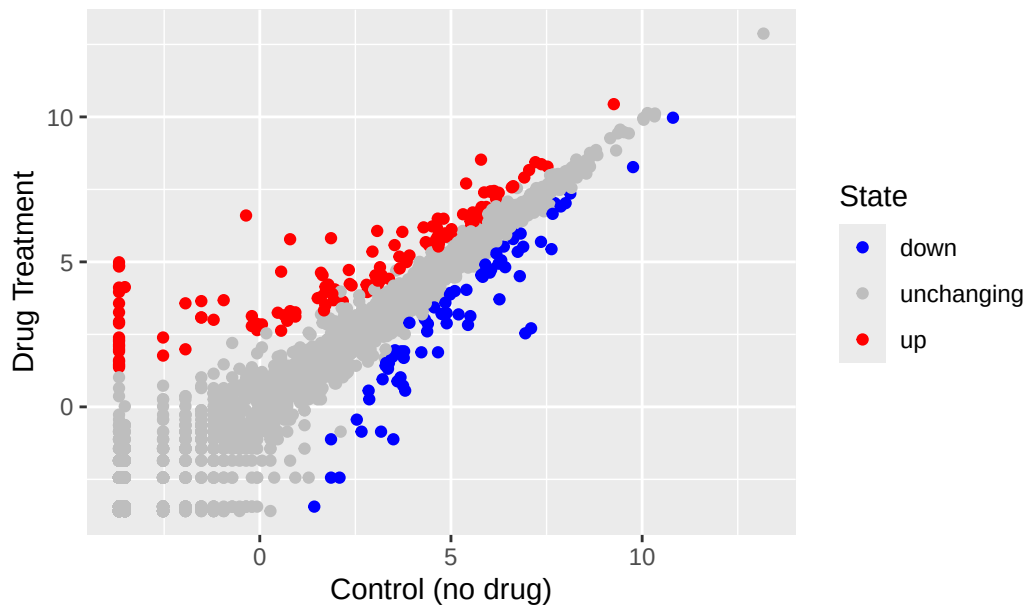
```

```

# created basic scatter plot of data set
p <- ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point()
p + scale_colour_manual( values=c("blue", "gray", "red")) +
  labs(title="Gene Expression Changes Upon Drug Treatment", x="Control (no drug)", y="Drug Treatment")

```

Gene Expression Changes Upon Drug Treatment



Q: Use the `nrow()` function to find out how many genes are in this dataset. What is your answer?

Answer: 5196 genes.

Q: Use the `colnames()` function and the `ncol()` function on the genes data frame to find out what the column names are (we will need these later) and how many columns there are. How many columns did you find?

Answer: 4 columns.

Q: Use the `table()` function on the `State` column of this data.frame to find out how many 'up' regulated genes there are. What is your answer?

Answer: 127 'up' regulated genes.

Using your values above and 2 significant figures. What fraction of total genes is up-regulated in this dataset?

Answer: 2.44% of total genes are up-regulated.

Optional Sections 7-10

Section 7

In this data set, I modeled the economic and demographic data of various countries.

```
# obtained data set
library(gapminder)

# install.packages("dplyr") ## un-comment to install if needed
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

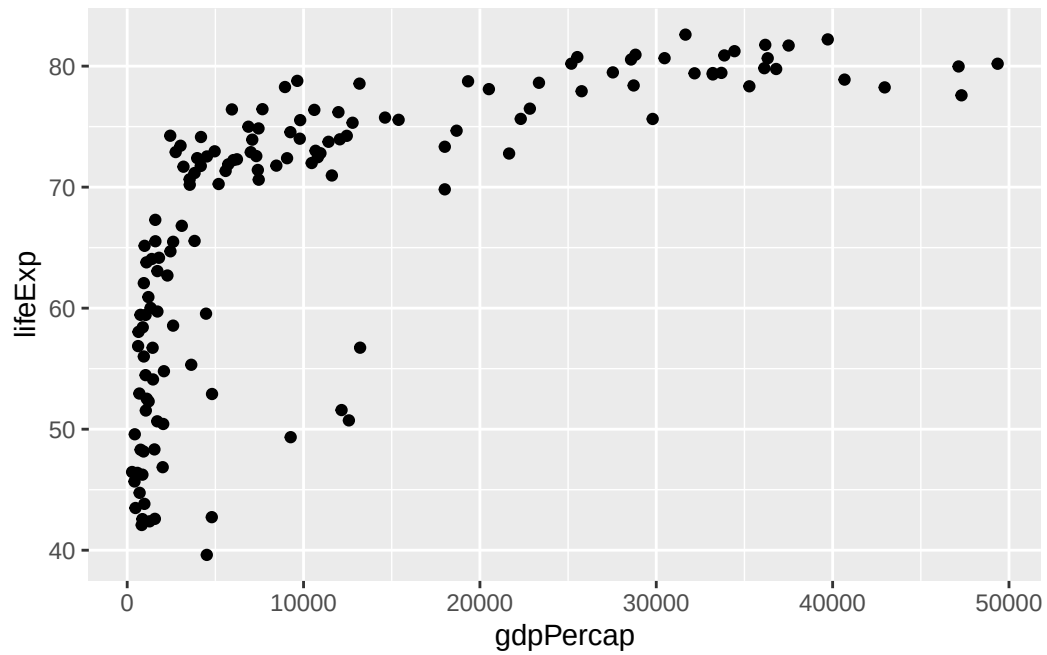
filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

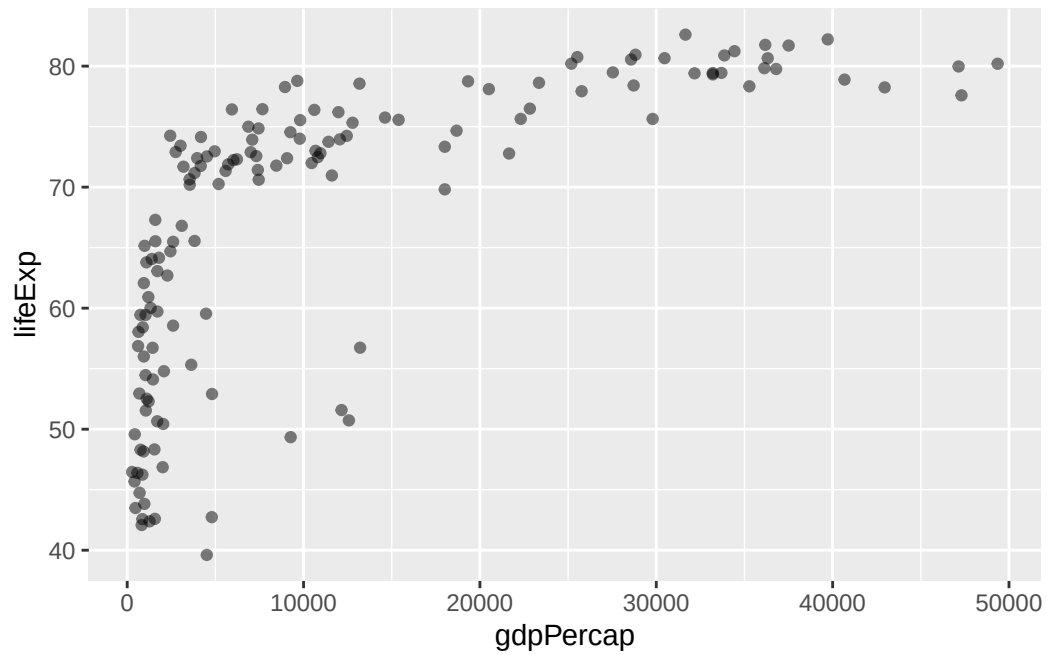
```
gapminder_2007 <- gapminder %>% filter(year==2007)

# created plot of the data set for year 2007
ggplot(gapminder_2007) +
  aes(x=gdpPerCap, y=lifeExp) +
  geom_point()
```

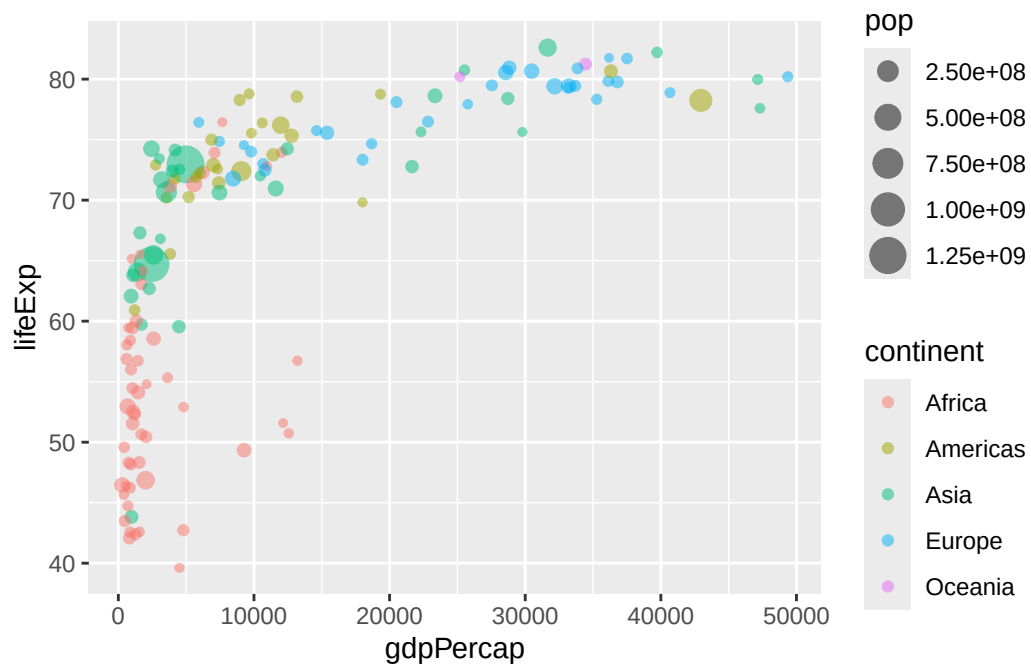


```
# continuation from previous chunk. made points slightly transparent for better visualization

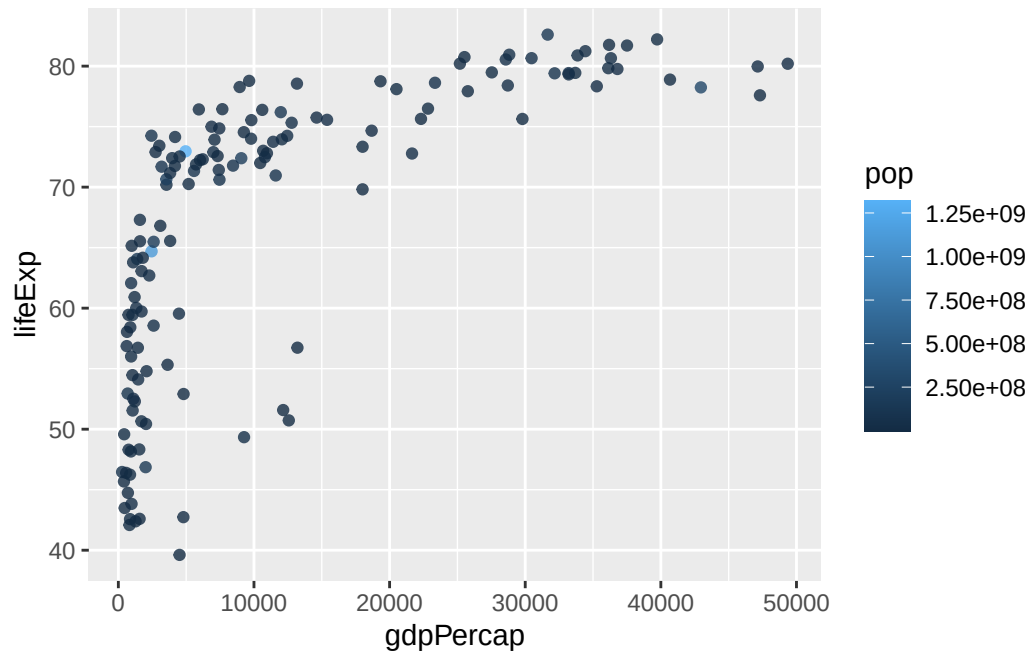
ggplot(gapminder_2007) +
  aes(x=gdpPercap, y=lifeExp) +
  geom_point(alpha=0.5)
```



```
# continuation from previous chunk. added more variables for better visualization.  
  
ggplot(gapminder_2007) +  
  aes(x=gdpPerCap, y=lifeExp, color=continent, size=pop) +  
  geom_point(alpha=0.5)
```

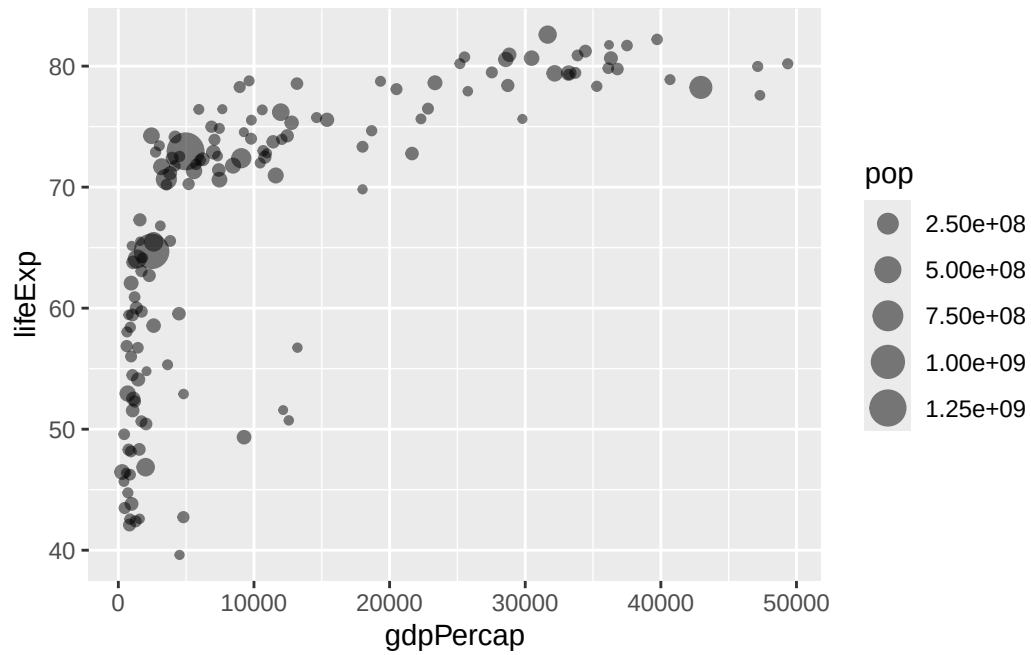


```
# colored points by numeric variable population
ggplot(gapminder_2007) +
  aes(x = gdpPercap, y = lifeExp, color = pop) +
  geom_point(alpha=0.8)
```



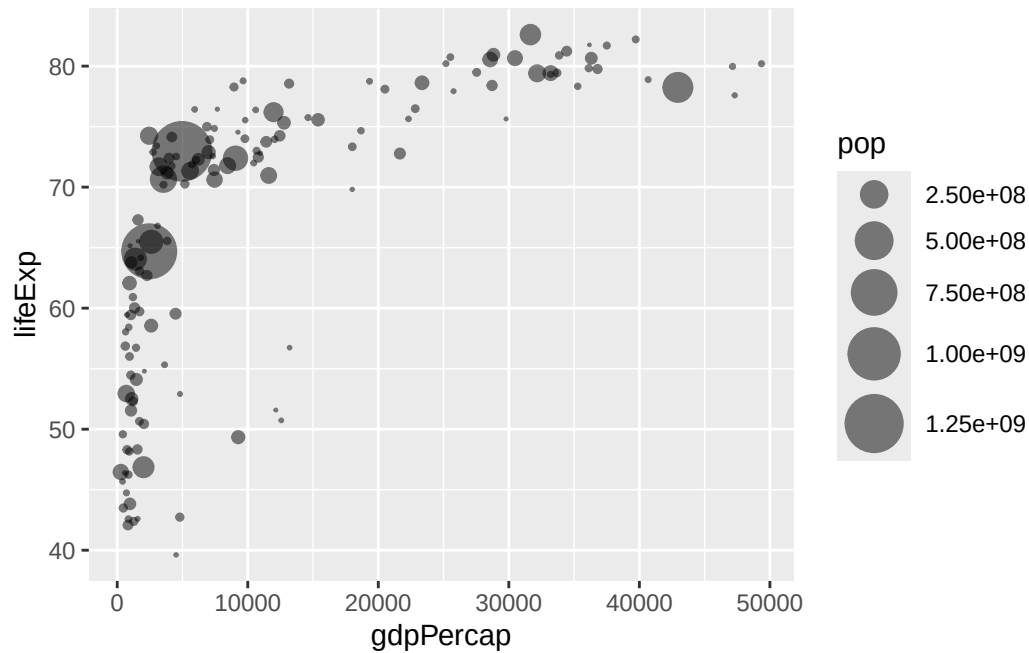
```
# continuation of gapminder_2007 data set. plotted GDP per capita vs. life expectancy. adjust
```

```
ggplot(gapminder_2007) +  
  aes(x = gdpPercap, y = lifeExp, size = pop) +  
  geom_point(alpha=0.5)
```



```
# made point sizes more proportional to population size

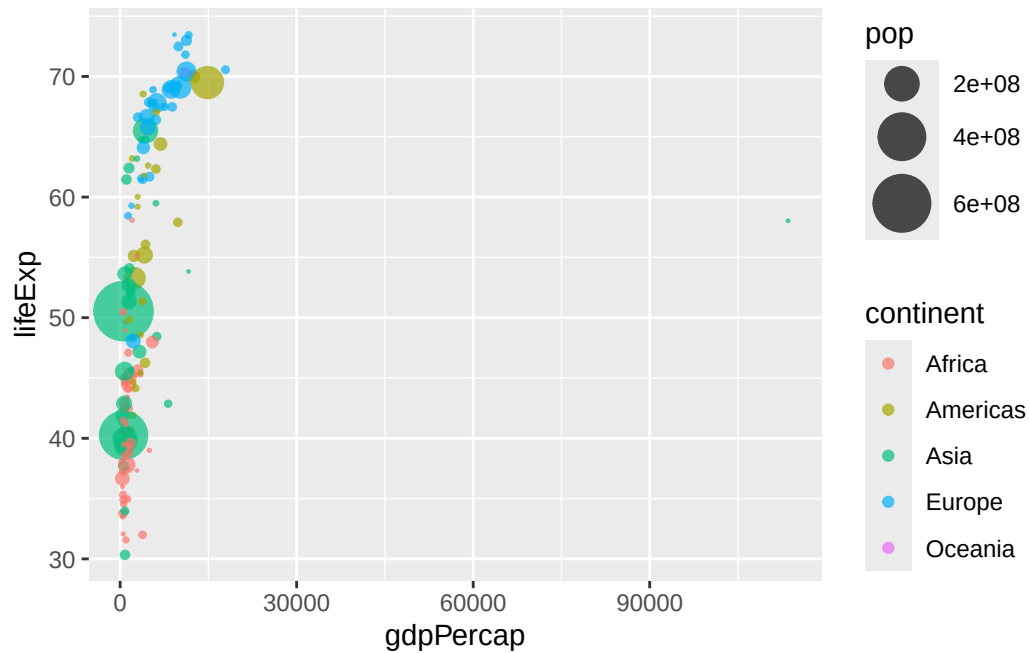
ggplot(gapminder_2007) +
  geom_point(aes(x = gdpPercap, y = lifeExp,
                 size = pop), alpha=0.5) +
  scale_size_area(max_size = 10)
```



The following plots are for the year 1957 instead of 2007.

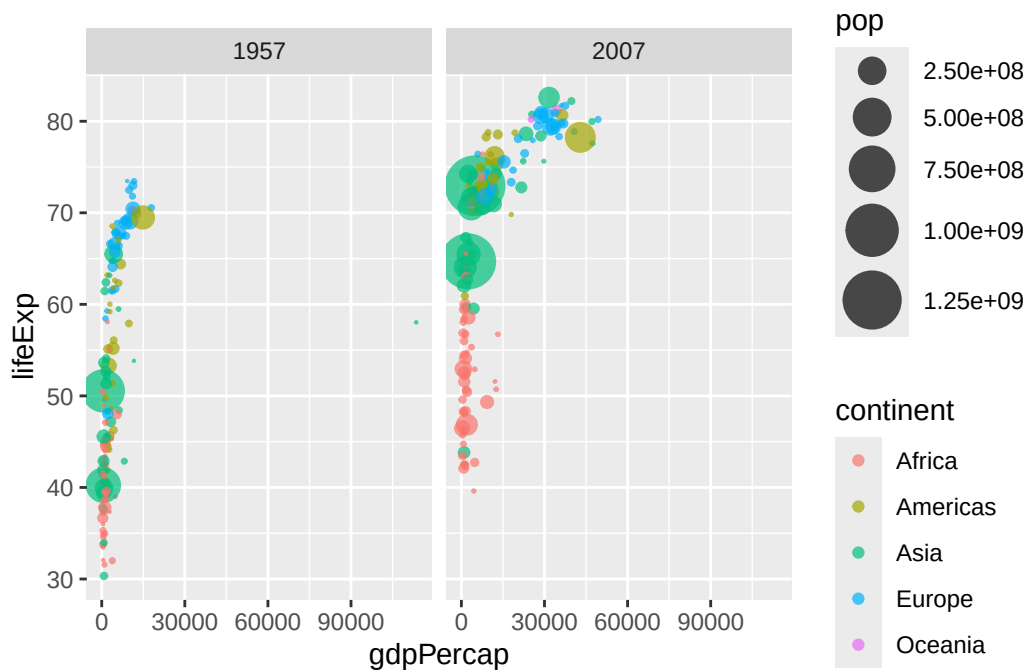
```
# filtered data set to only include the year 1957
gapminder_1957 <- gapminder %>% filter(year==1957)

# created scatter plot of GDP per capita vs. life expectancy. adjusted variables for better v
ggplot(gapminder_1957) +
  aes(x = gdpPercap, y=lifeExp, color=continent, size=pop) +
  geom_point(alpha=0.7) +
  scale_size_area(max_size=10)
```



```
# completed plots for both years 1957 and 2007
gapminder_1957 <- gapminder %>% filter(year==1957 | year==2007)

ggplot(gapminder_1957) +
  geom_point(aes(x = gdpPerCap, y = lifeExp, color=continent,
                 size = pop), alpha=0.7) +
  scale_size_area(max_size = 10) +
  # used facet_wrap() to create two plots
  facet_wrap(~year)
```



Section 8

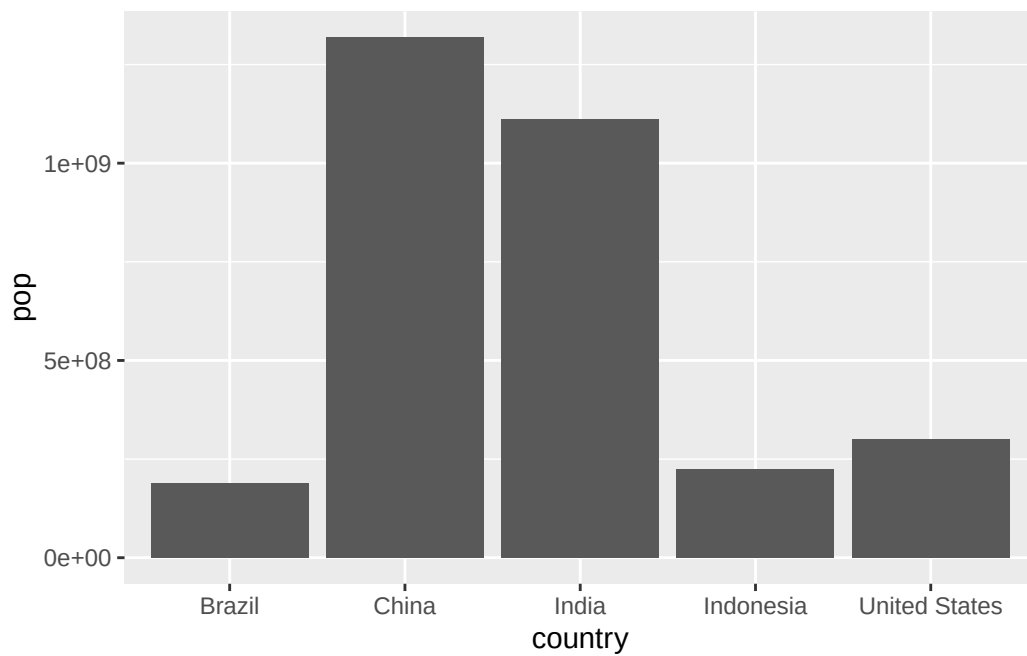
```
# loaded data set showing people in the five biggest countries by population in 2007
gapminder_top5 <- gapminder %>%
  filter(year==2007) %>%
  arrange(desc(pop)) %>%
  top_n(5, pop)

# created bar chart for country vs. population size
gapminder_top5
```

A tibble: 5 x 6

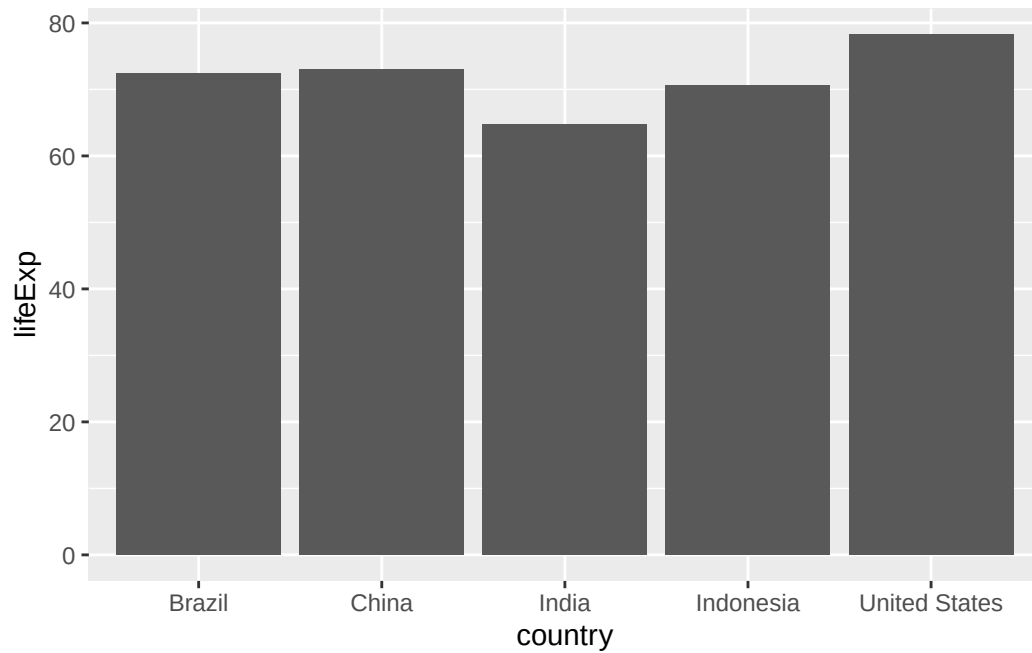
	country	continent	year	lifeExp	pop	gdpPercap
	<fct>	<fct>	<int>	<dbl>	<int>	<dbl>
1	China	Asia	2007	73.0	1318683096	4959.
2	India	Asia	2007	64.7	1110396331	2452.
3	United States	Americas	2007	78.2	301139947	42952.
4	Indonesia	Asia	2007	70.6	223547000	3541.
5	Brazil	Americas	2007	72.4	190010647	9066.

```
ggplot(gapminder_top5) +  
  geom_col(aes(x = country, y = pop))
```

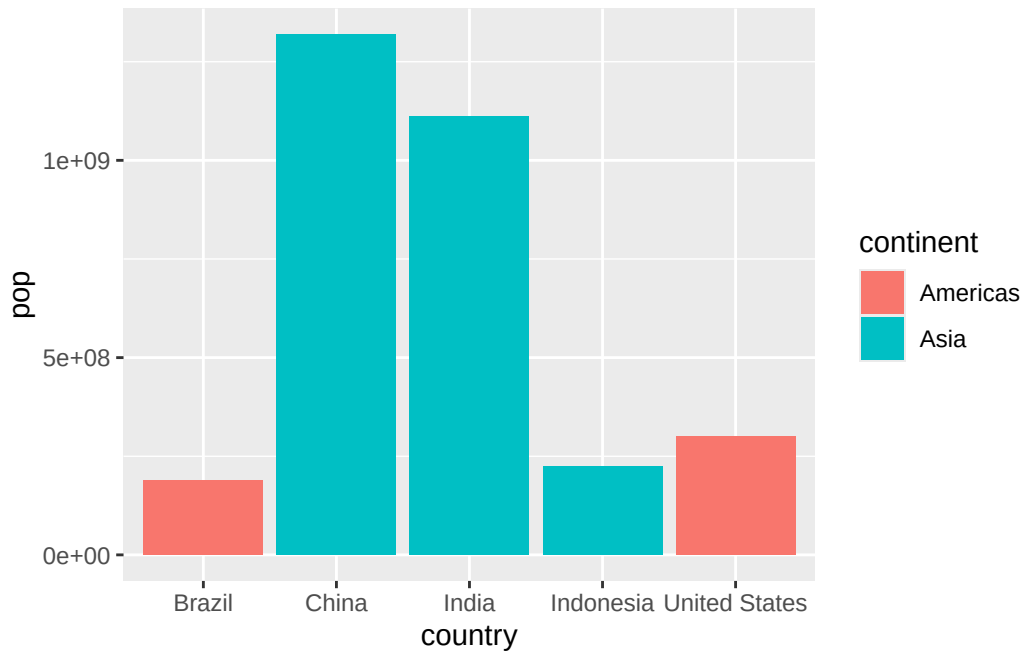


```
# created bar chart showing life expectancy of five most populous countries in 2007
```

```
ggplot(gapminder_top5) +  
  geom_col(aes(x = country, y = lifeExp))
```

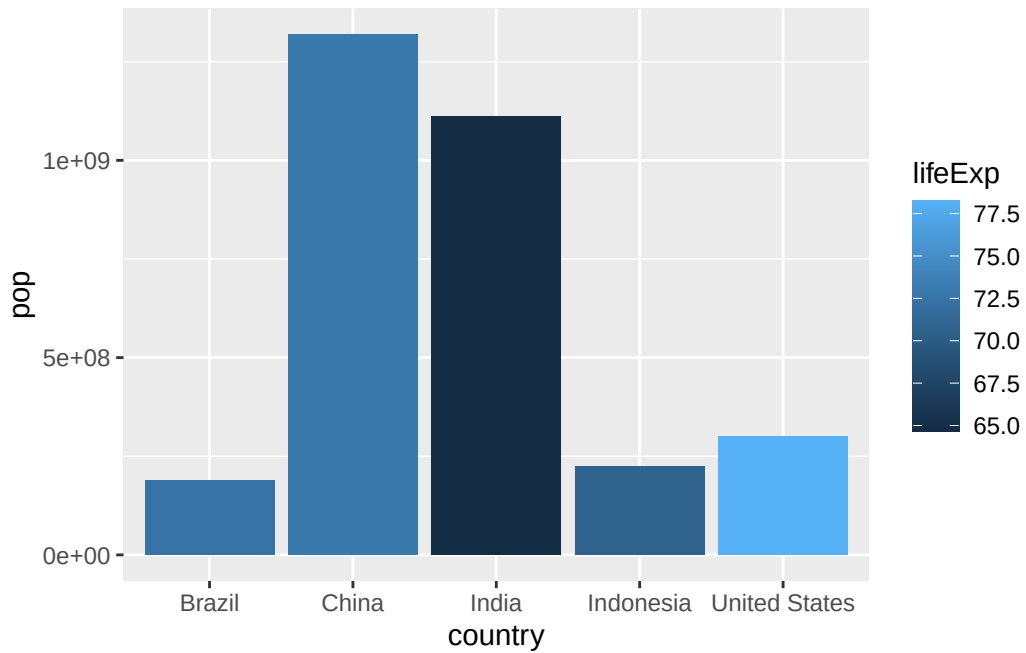


```
# colored the bars of the country vs. population graph  
ggplot(gapminder_top5) +  
  geom_col(aes(x = country, y = pop, fill = continent))
```



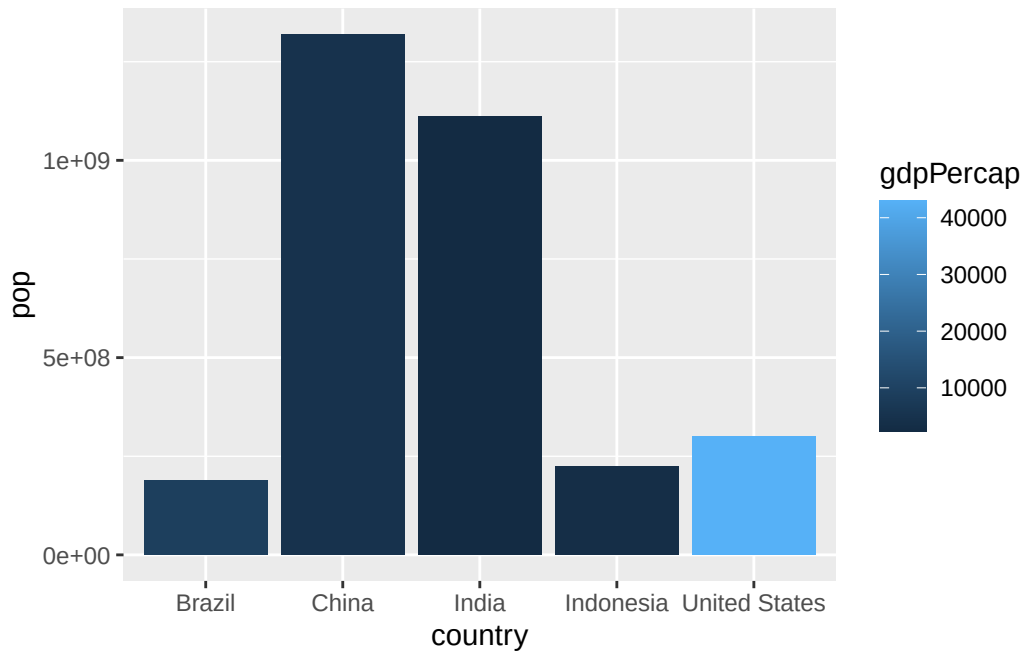
```
# colored the bars using numeric variable of life expectancy
```

```
ggplot(gapminder_top5) +  
  geom_col(aes(x = country, y = pop, fill = lifeExp))
```

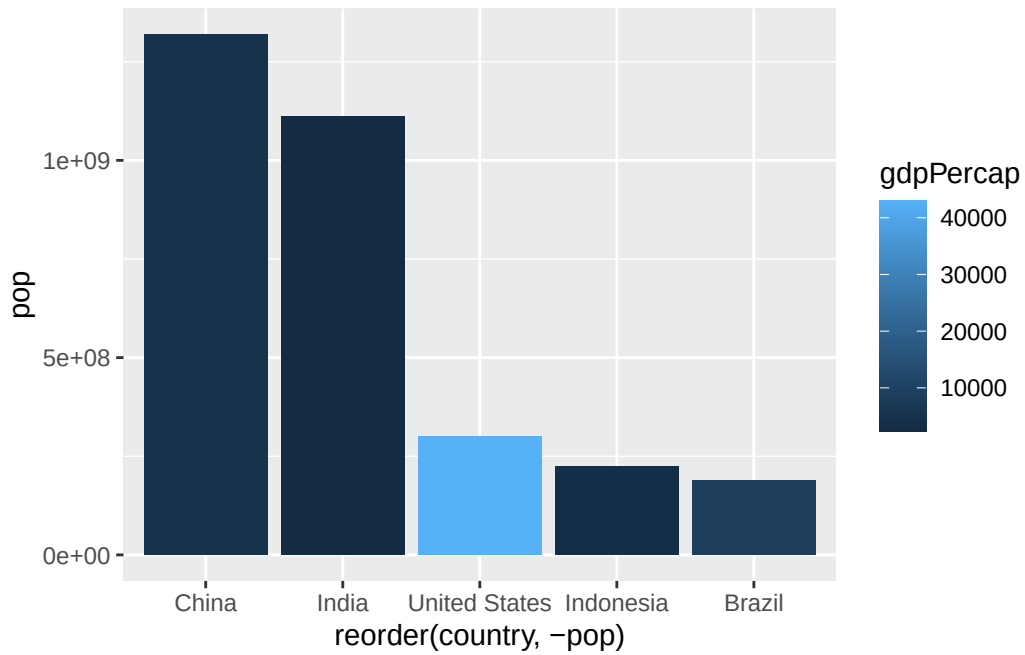


```
# created bar chart of country vs. population. used GDP per capita to color bars

ggplot(gapminder_top5) +
  aes(x=country, y=pop, fill=gdpPercap) +
  geom_col()
```

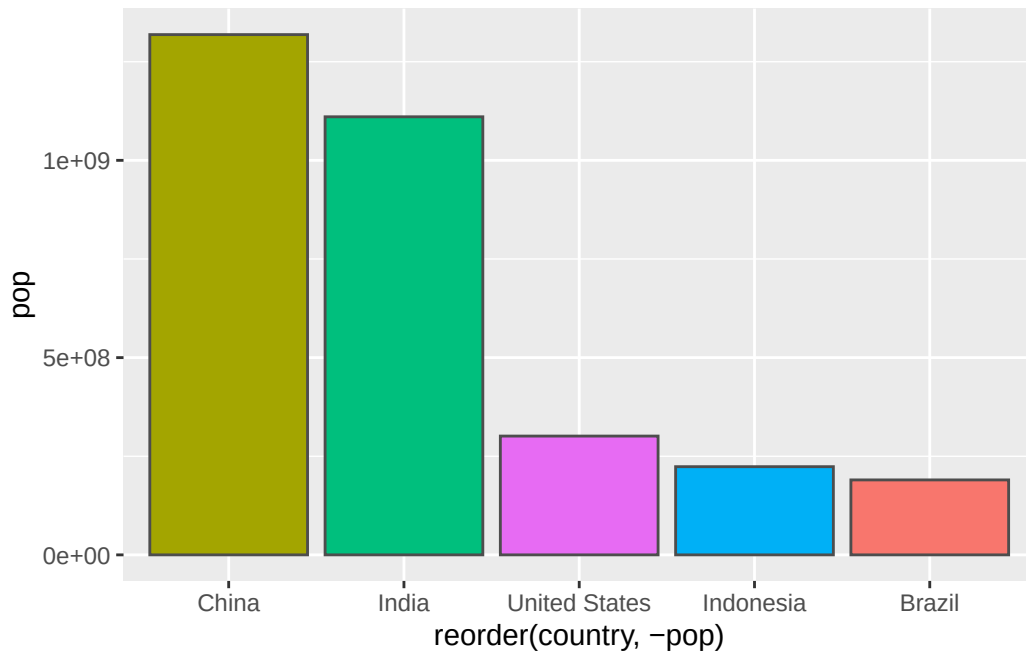


```
# changed order of bars  
  
ggplot(gapminder_top5) +  
  aes(x=reorder(country, -pop), y=pop, fill=gdpPercap) +  
  geom_col()
```



```
# filled by country

ggplot(gapminder_top5) +
  aes(x=reorder(country, -pop), y=pop, fill=country) +
  geom_col(col="gray30") +
  guides(fill="none")
```



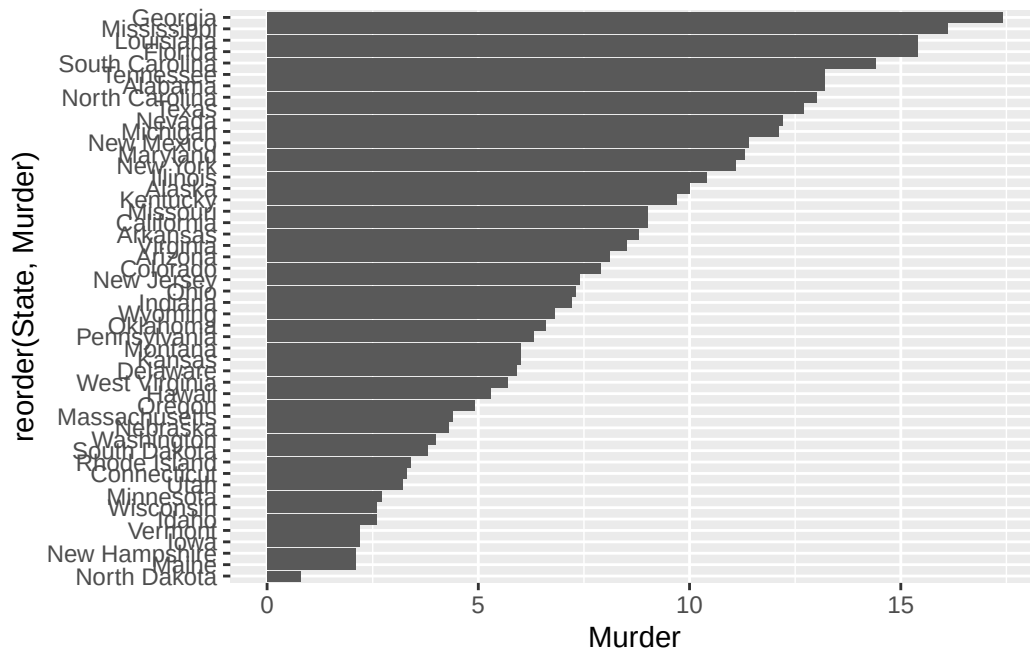
```
# loaded another data set for arrest data in the US
```

```
head(USArrests)
```

	Murder	Assault	UrbanPop	Rape
Alabama	13.2	236	58	21.2
Alaska	10.0	263	48	44.5
Arizona	8.1	294	80	31.0
Arkansas	8.8	190	50	19.5
California	9.0	276	91	40.6
Colorado	7.9	204	78	38.7

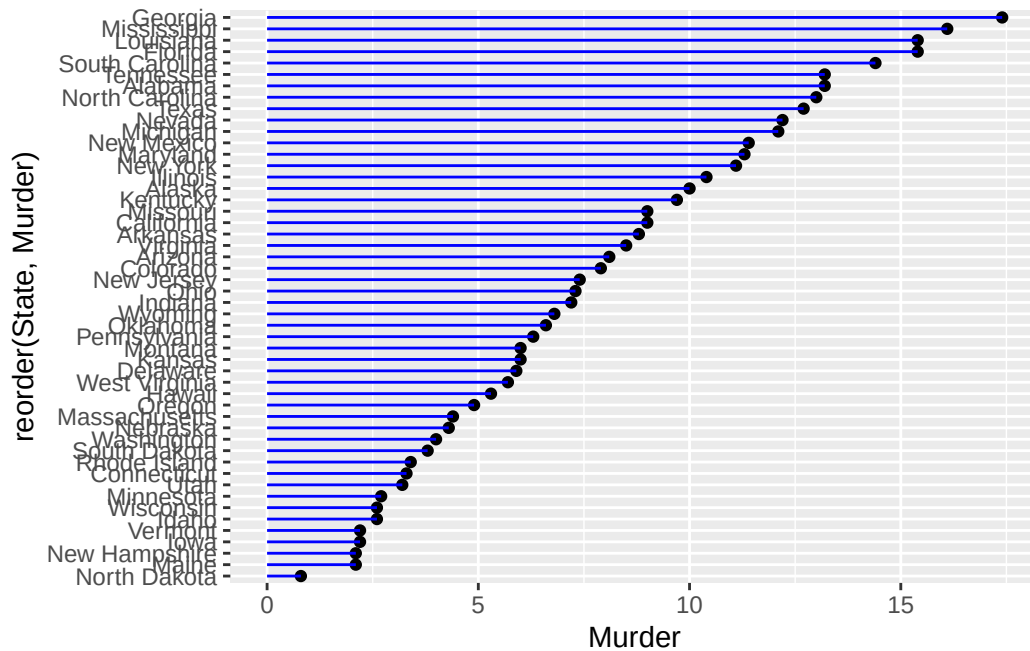
```
# plotted flipped bar chart for data set
```

```
USArrests$State <- rownames(USArrests)
ggplot(USArrests) +
  aes(x=reorder(State,Murder), y=Murder) +
  geom_col() +
  coord_flip()
```



```
# visualized graph better

ggplot(USArrests) +
  aes(x=reorder(State,Murder), y=Murder) +
  geom_point() +
  geom_segment(aes(x=State,
                   xend=State,
                   y=0,
                   yend=Murder), color="blue") +
  coord_flip()
```



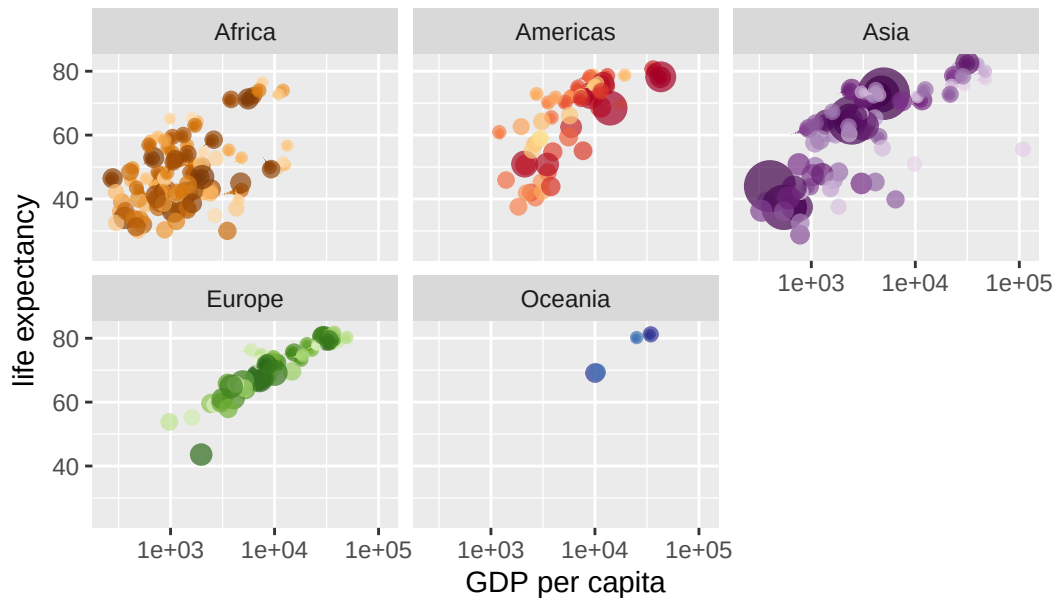
Section 9

```
# loaded package
library(gapminder)
library(gganimate)

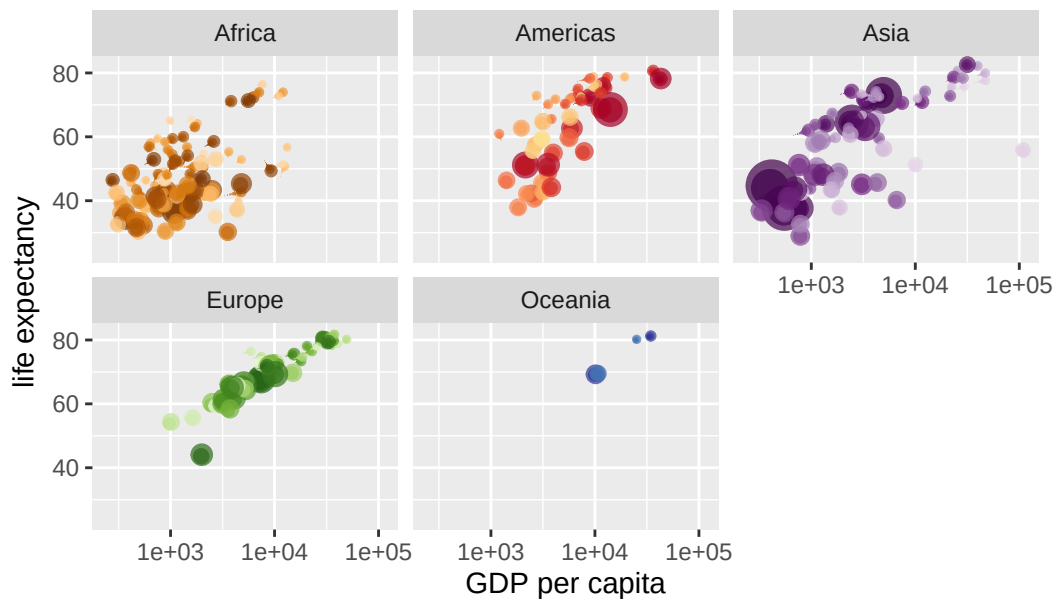
# Setup nice regular ggplot of the gapminder data
ggplot(gapminder, aes(gdpPercap, lifeExp, size = pop, colour = country)) +
  geom_point(alpha = 0.7, show.legend = FALSE) +
  scale_colour_manual(values = country_colors) +
  scale_size(range = c(2, 12)) +
  scale_x_log10() +
  # Facet by continent
  facet_wrap(~continent) +
  # Here comes the gganimate specific bits
  labs(title = 'Year: {frame_time}', x = 'GDP per capita', y = 'life expectancy') +
  transition_time(year) +
  shadow_wake(wake_length = 0.1, alpha = FALSE)
```

Warning in formals(fun): argument is not a function

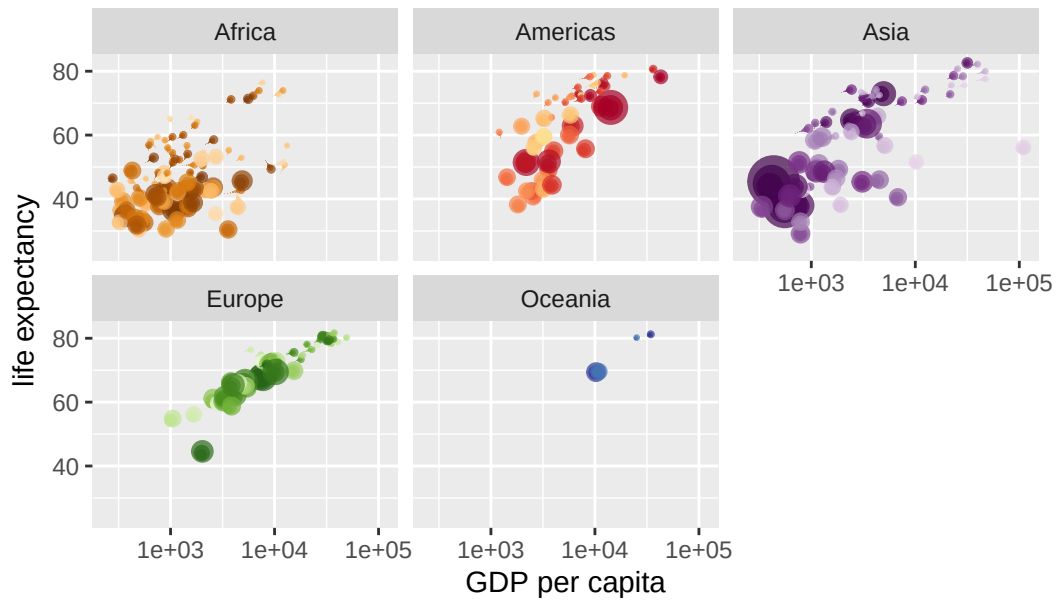
Year: 1952



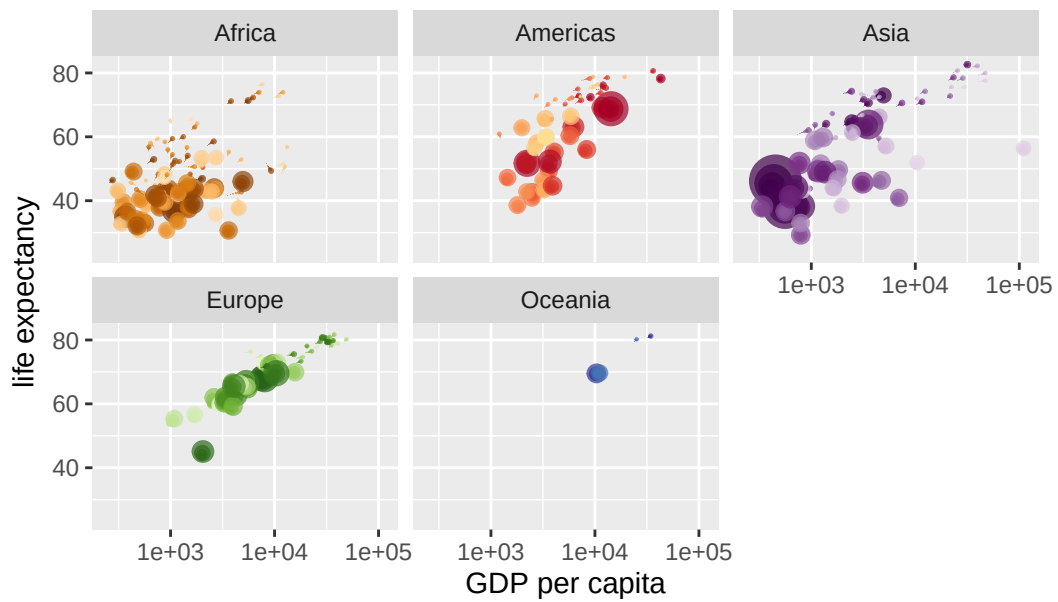
Year: 1953



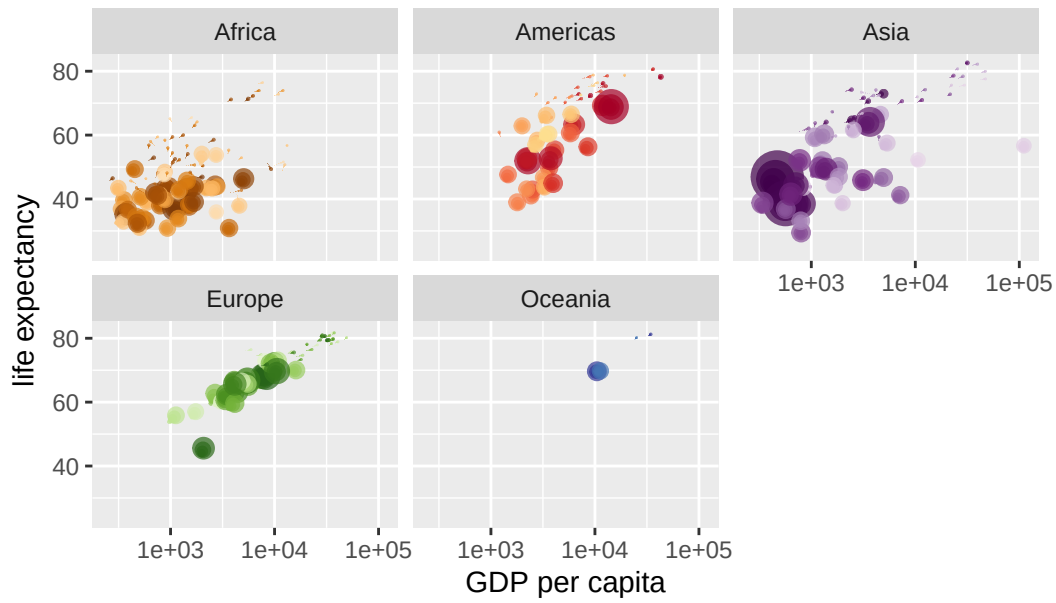
Year: 1953



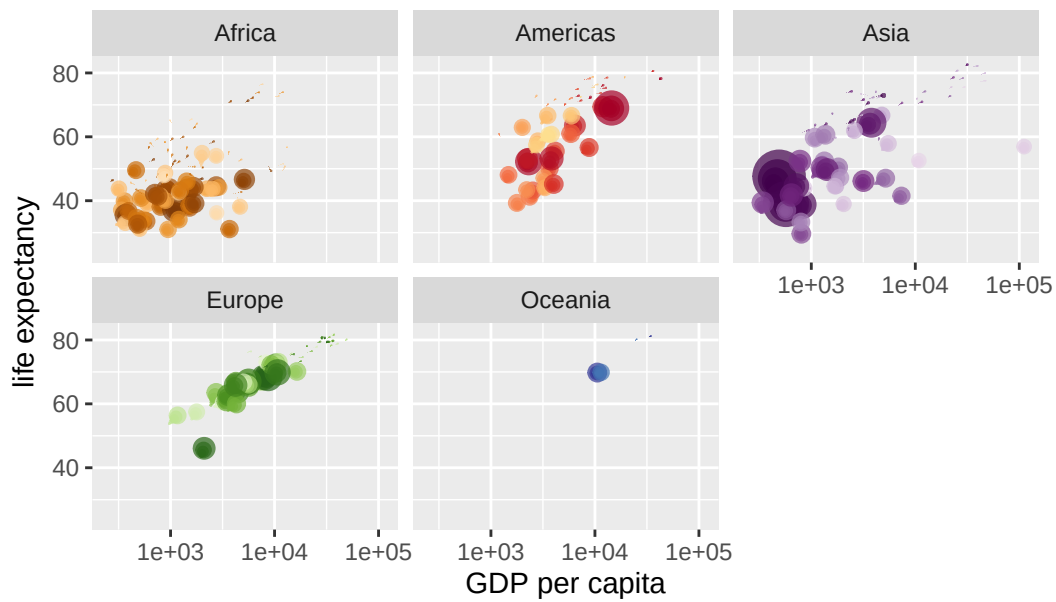
Year: 1954



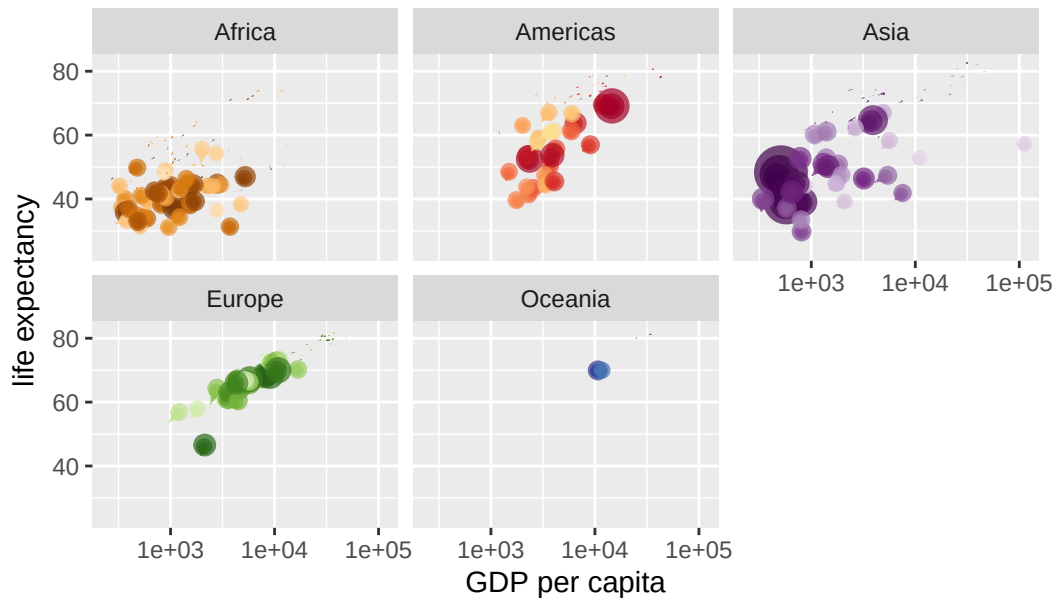
Year: 1954



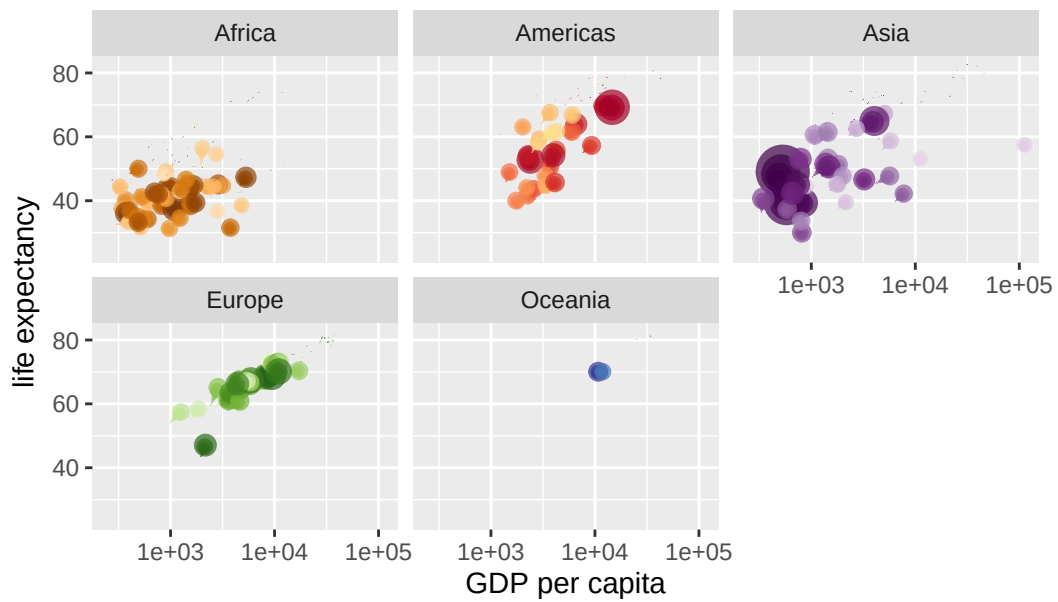
Year: 1955



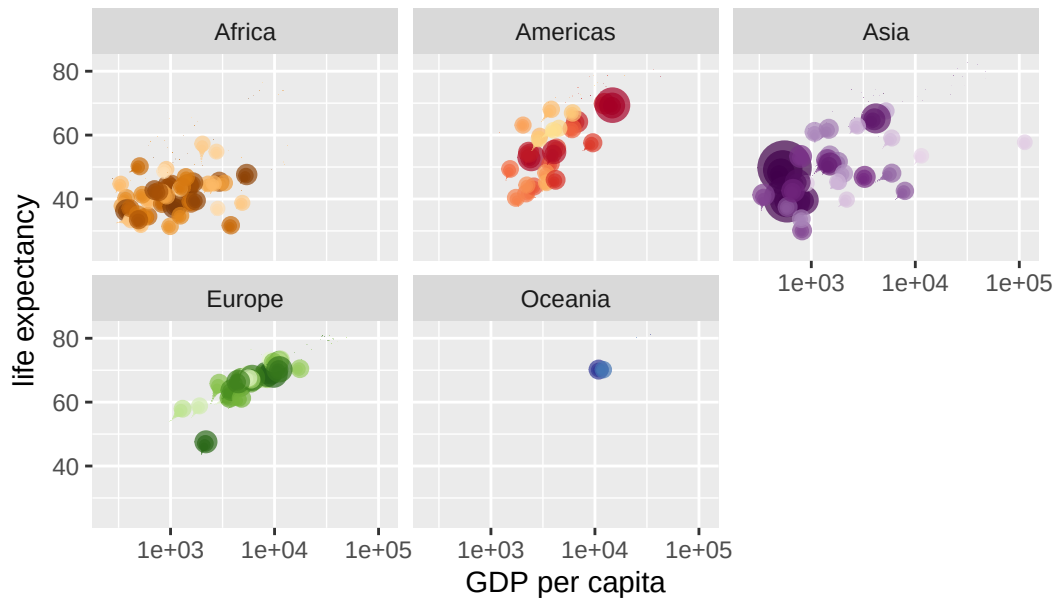
Year: 1955



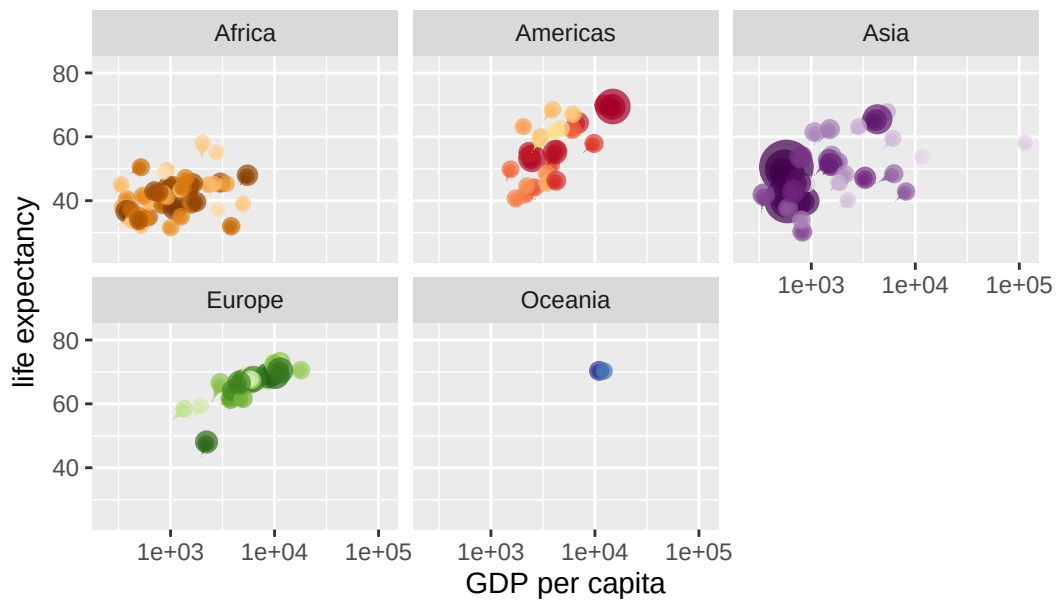
Year: 1956



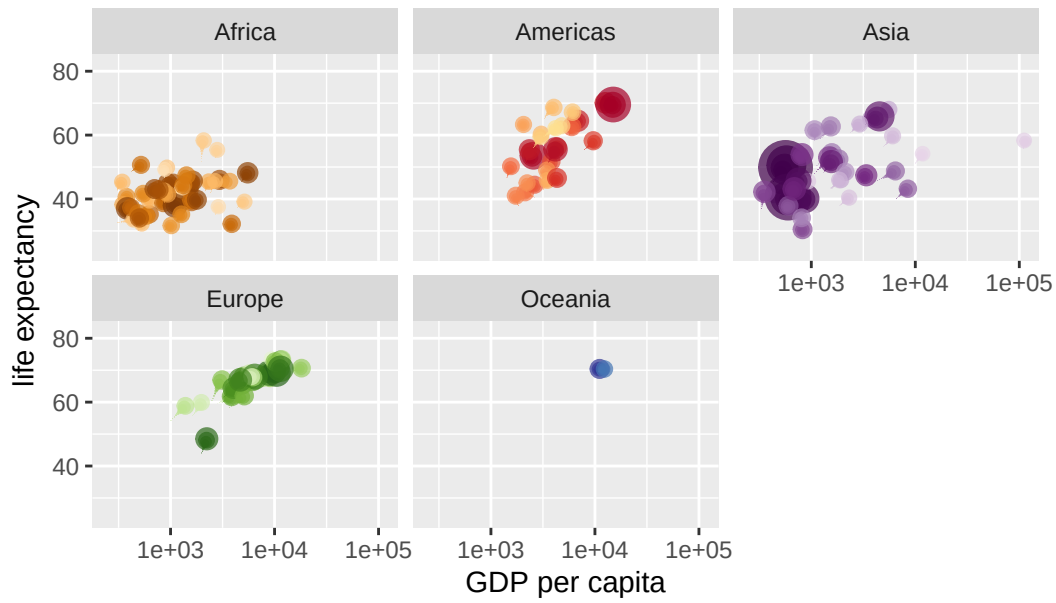
Year: 1956



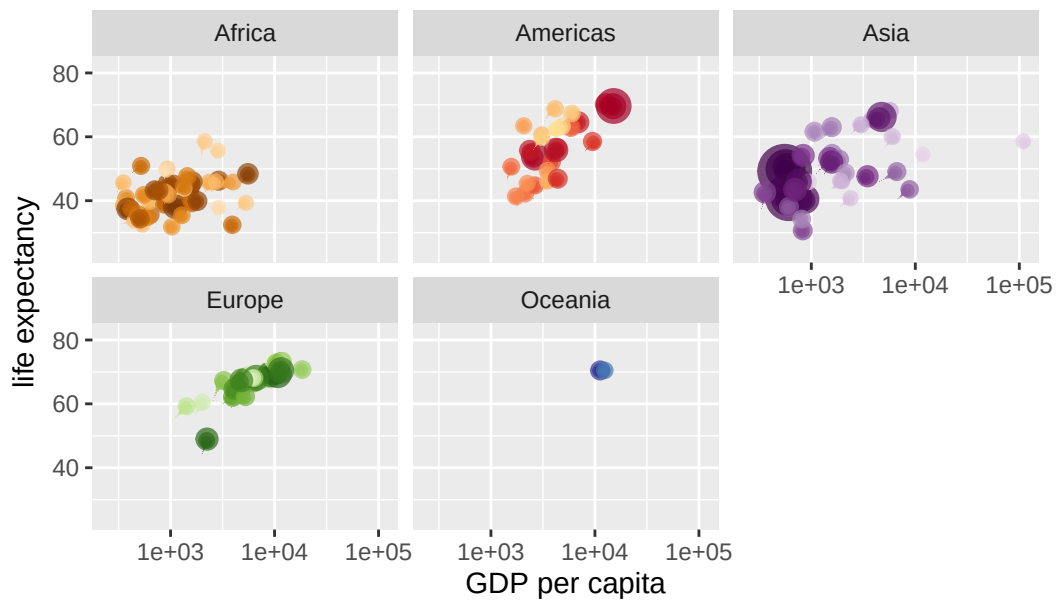
Year: 1957



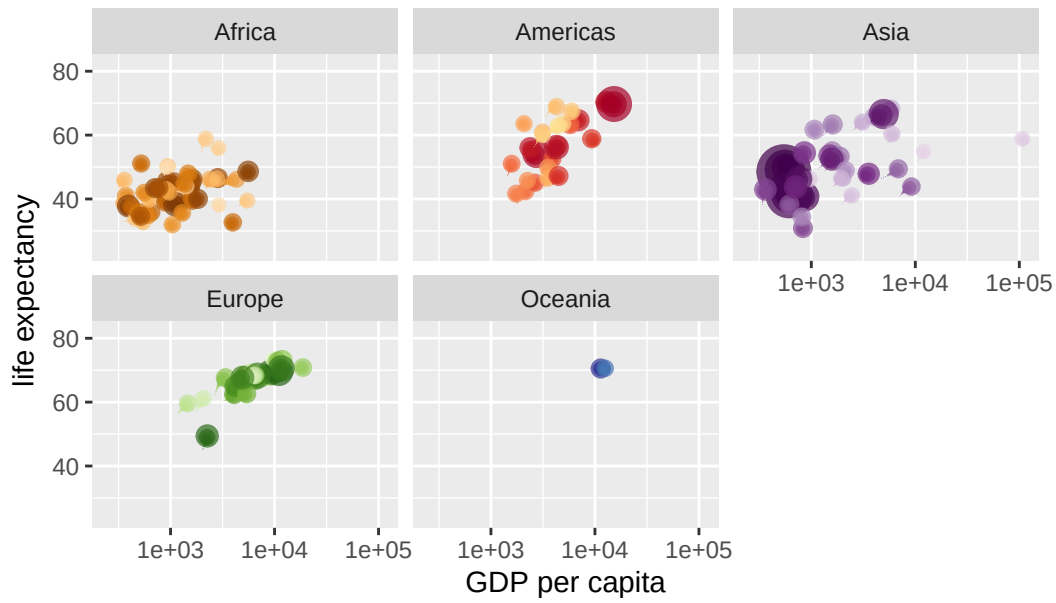
Year: 1958



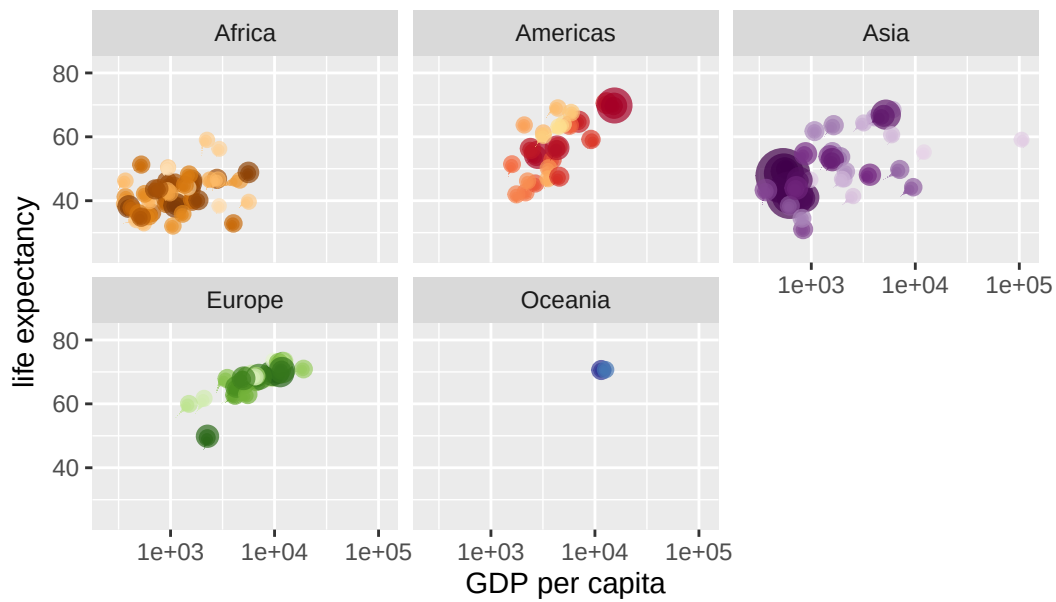
Year: 1958



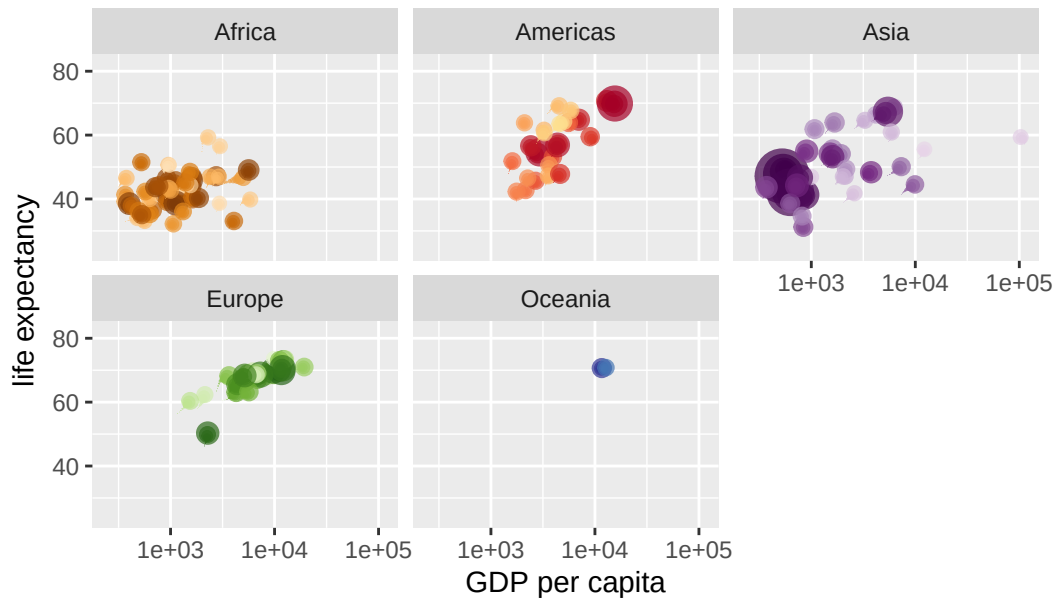
Year: 1959



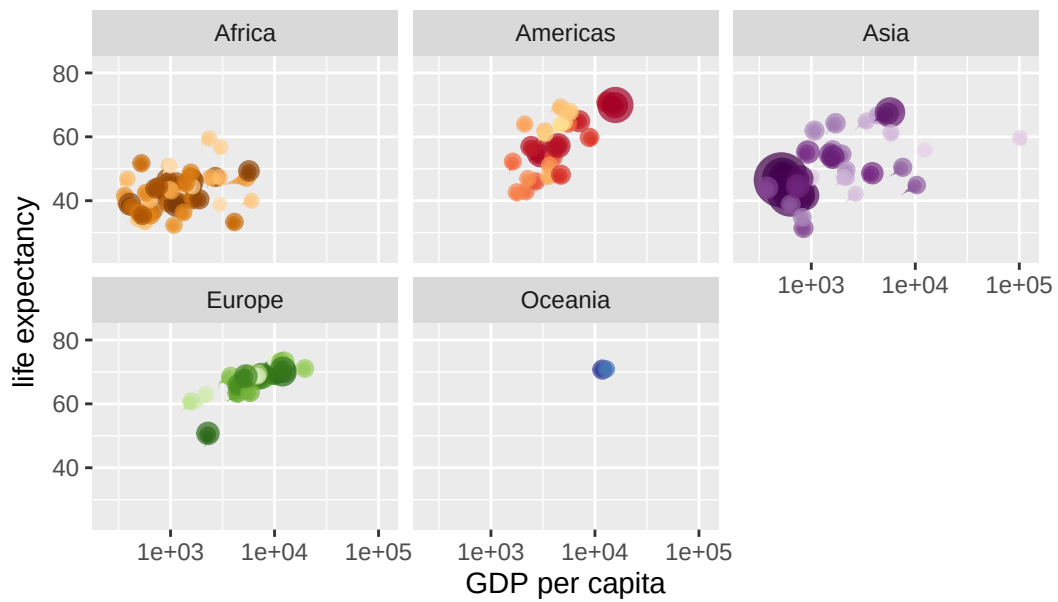
Year: 1959



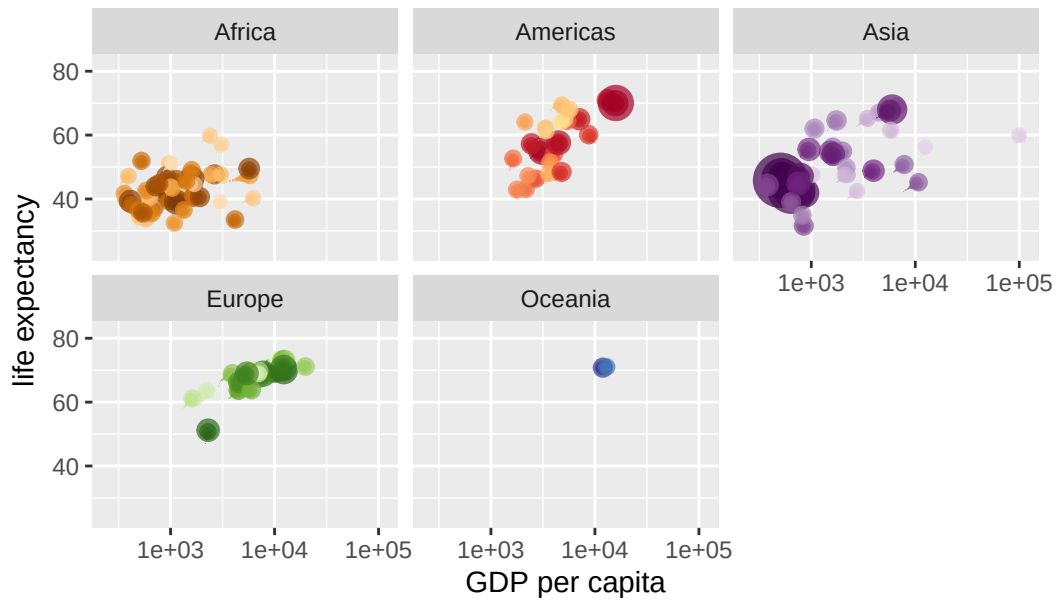
Year: 1960



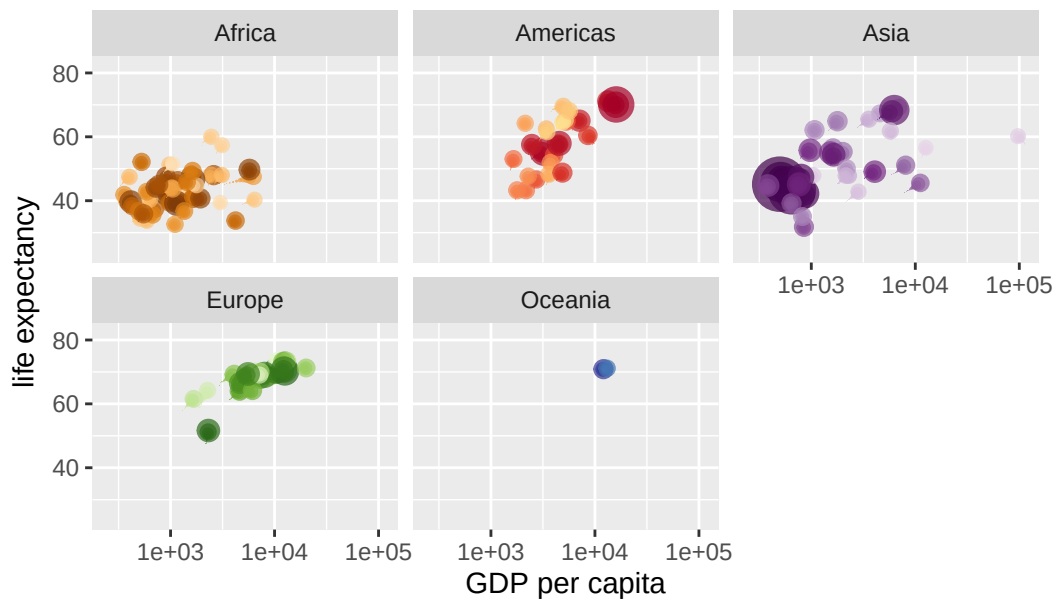
Year: 1960



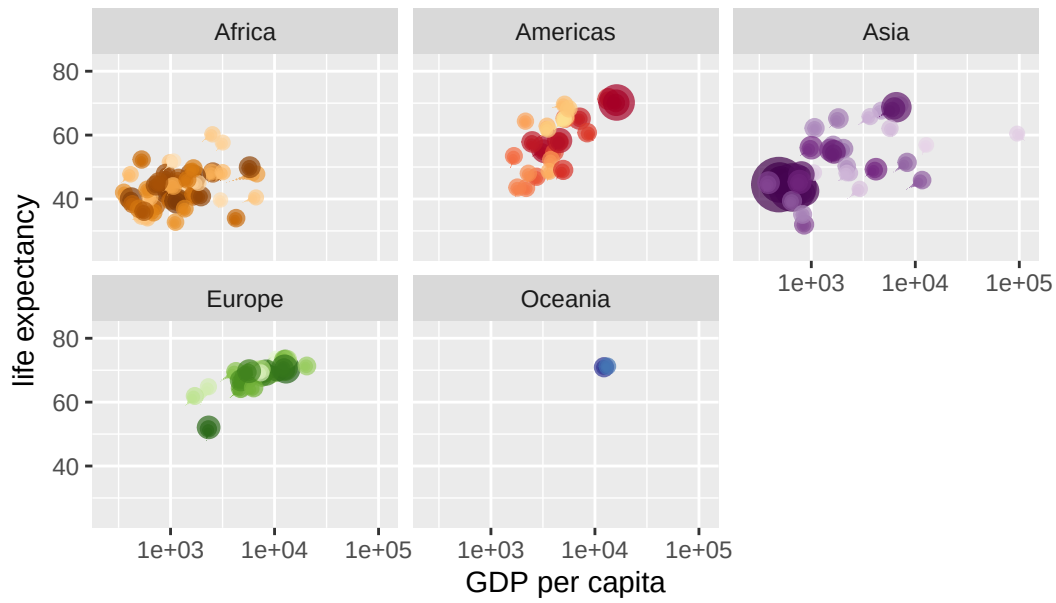
Year: 1961



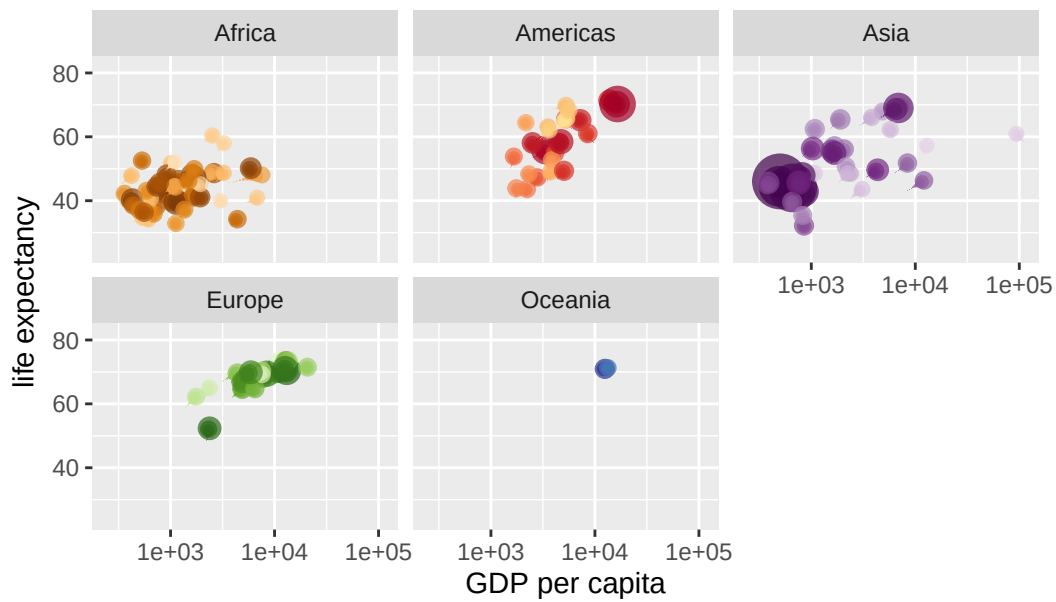
Year: 1961



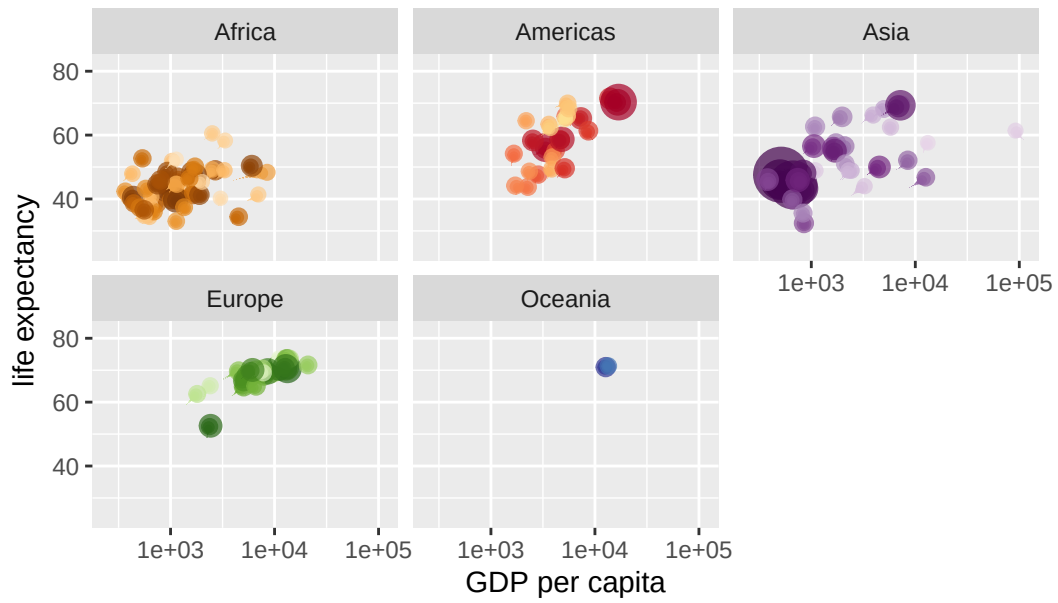
Year: 1962



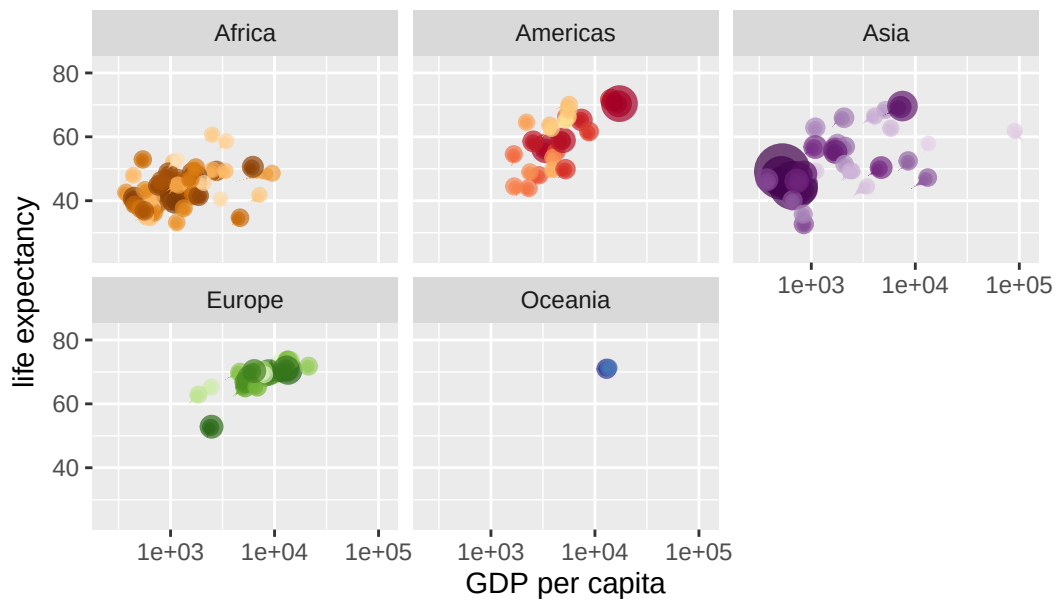
Year: 1963



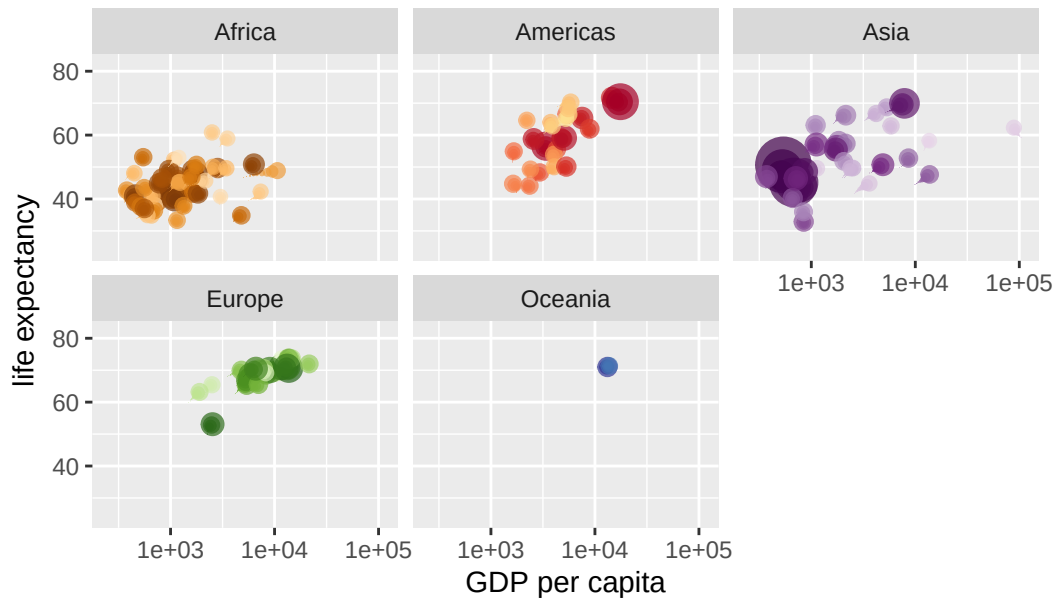
Year: 1963



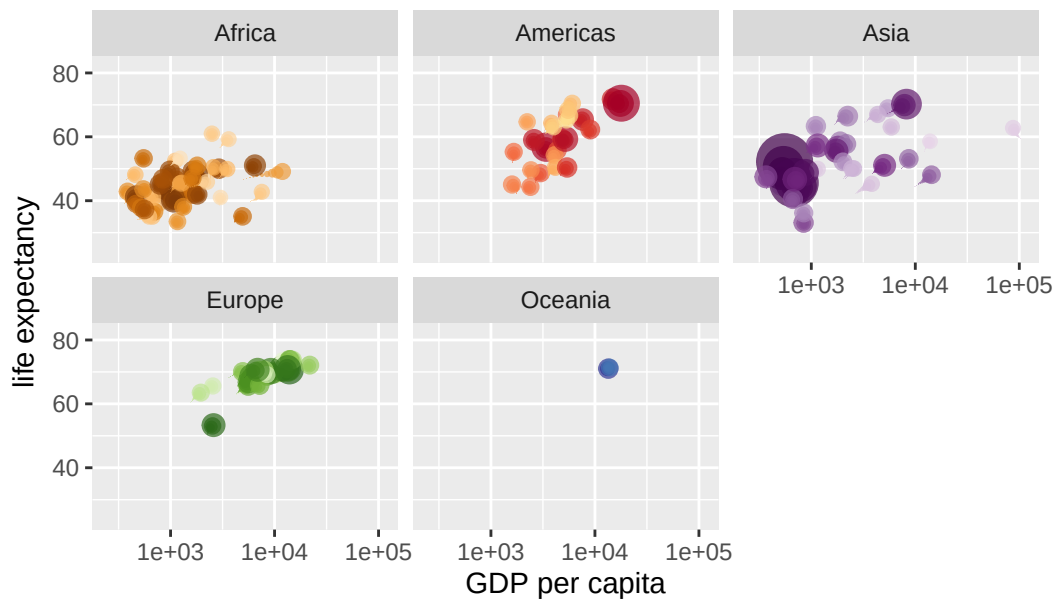
Year: 1964



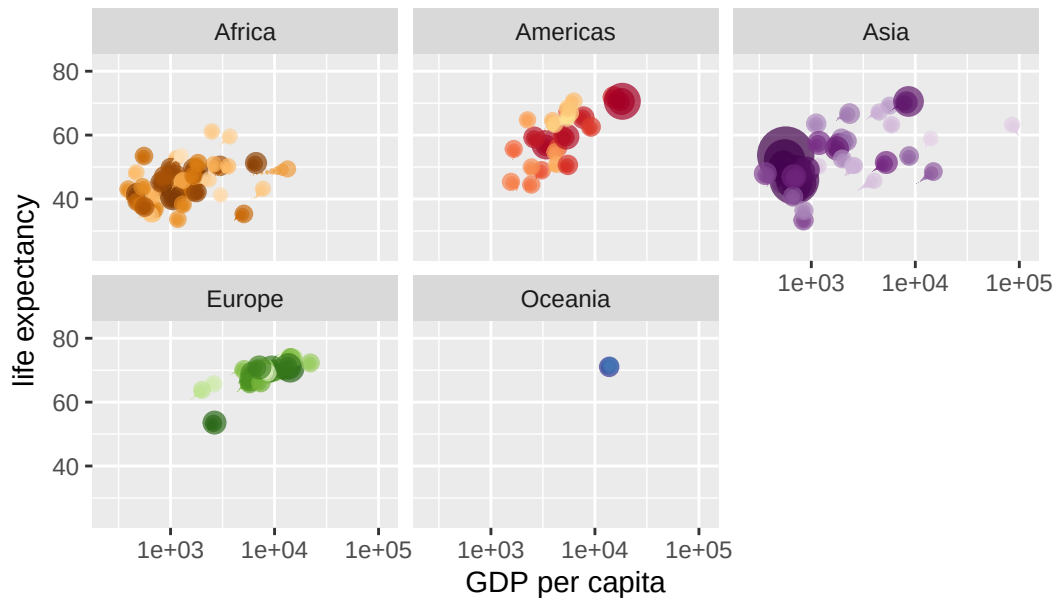
Year: 1964



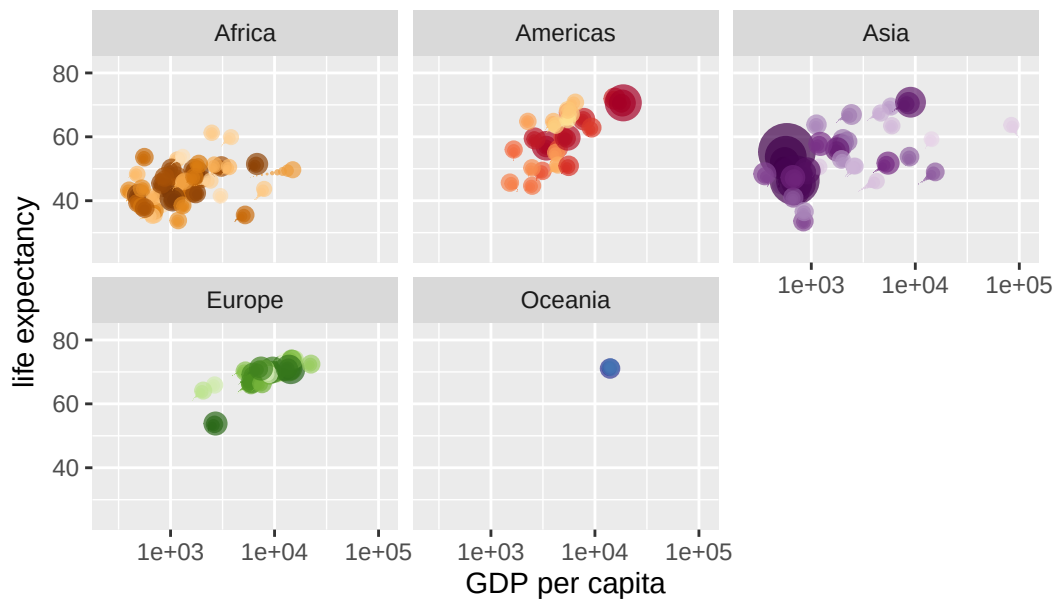
Year: 1965



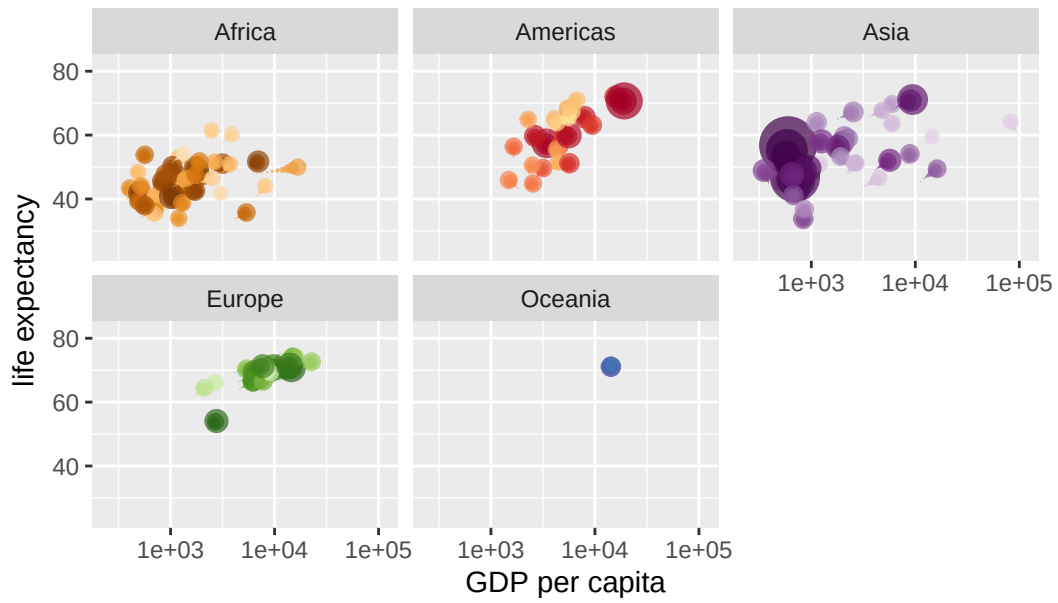
Year: 1965



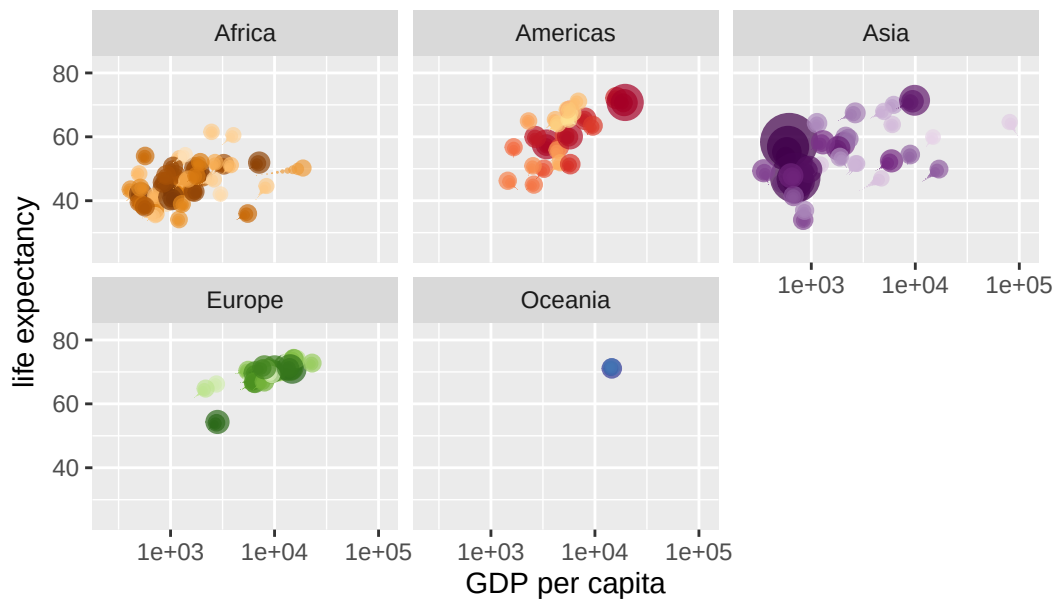
Year: 1966



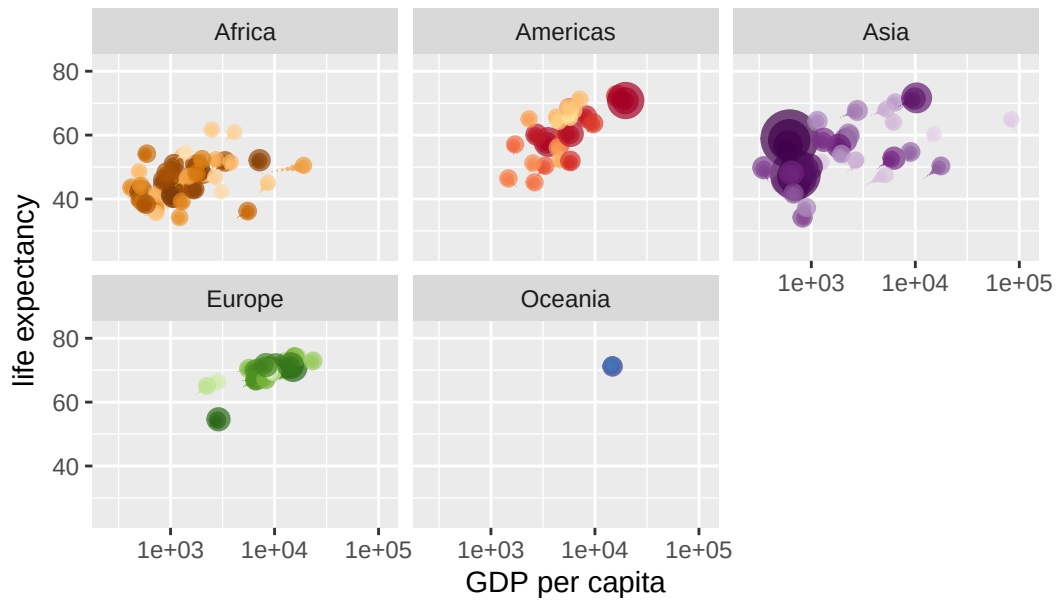
Year: 1966



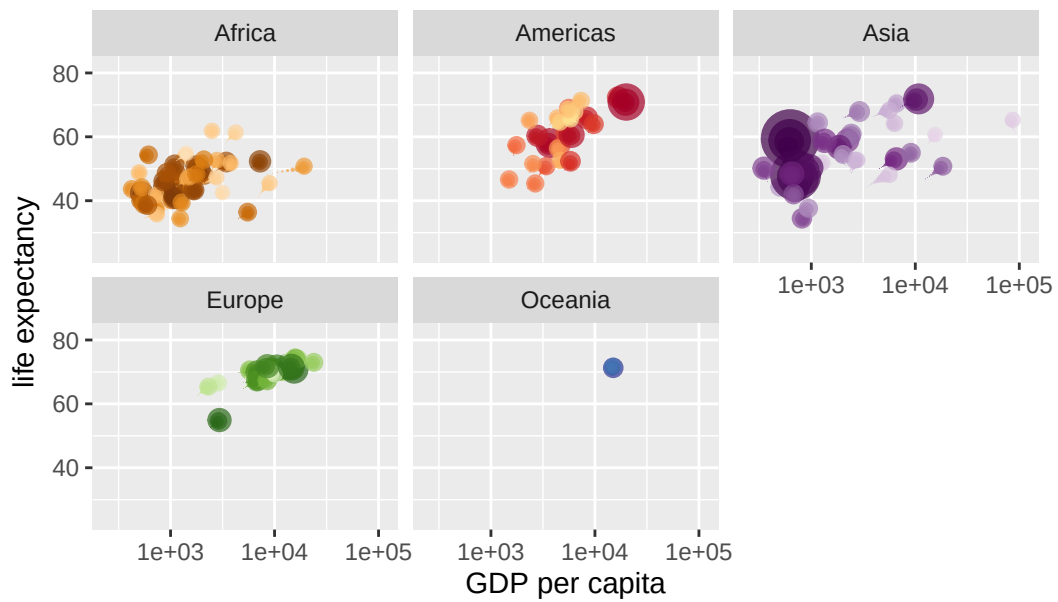
Year: 1967



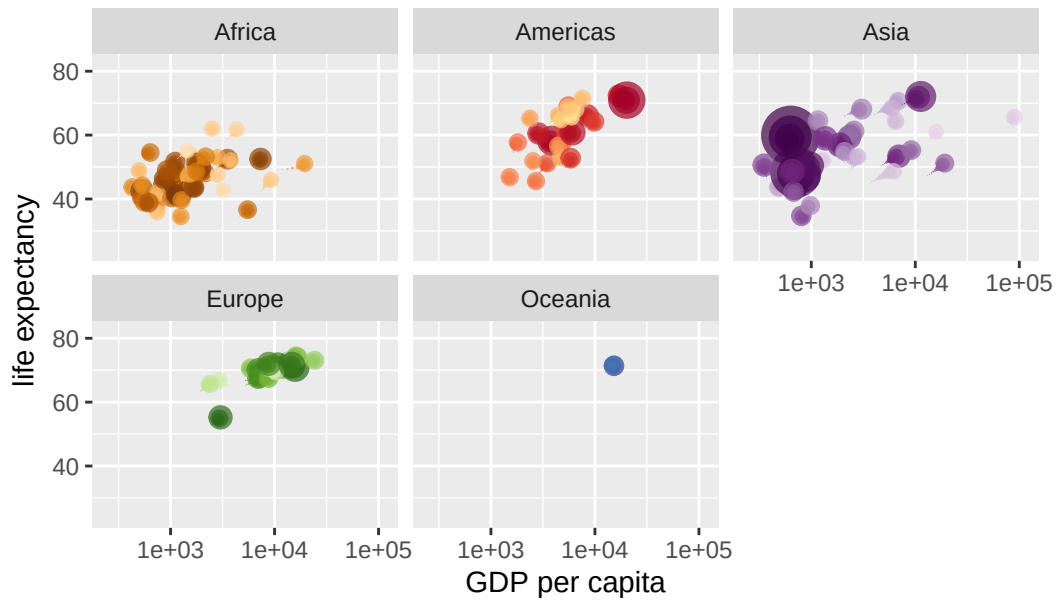
Year: 1968



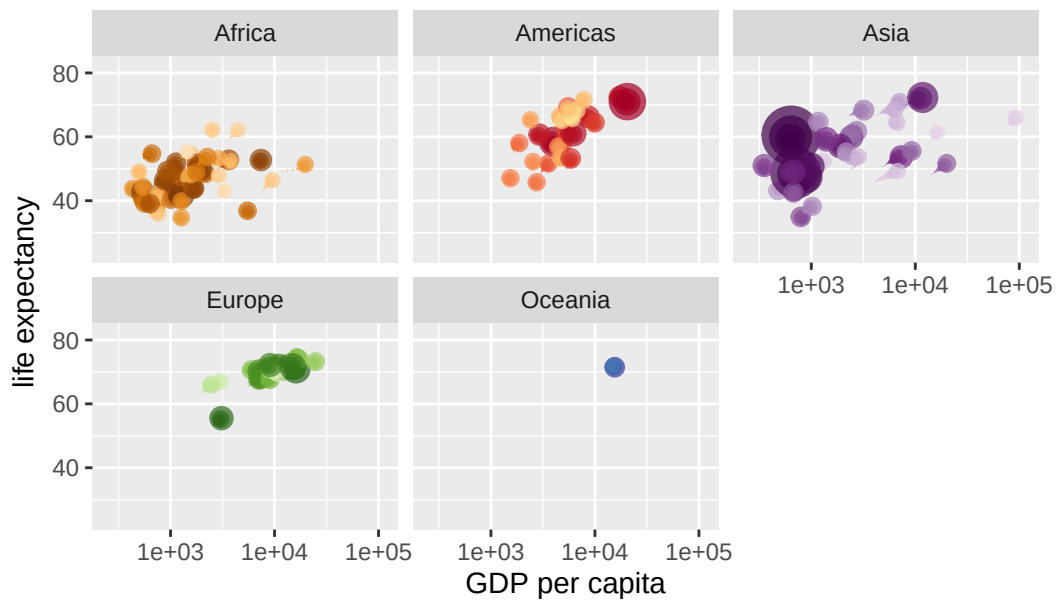
Year: 1968



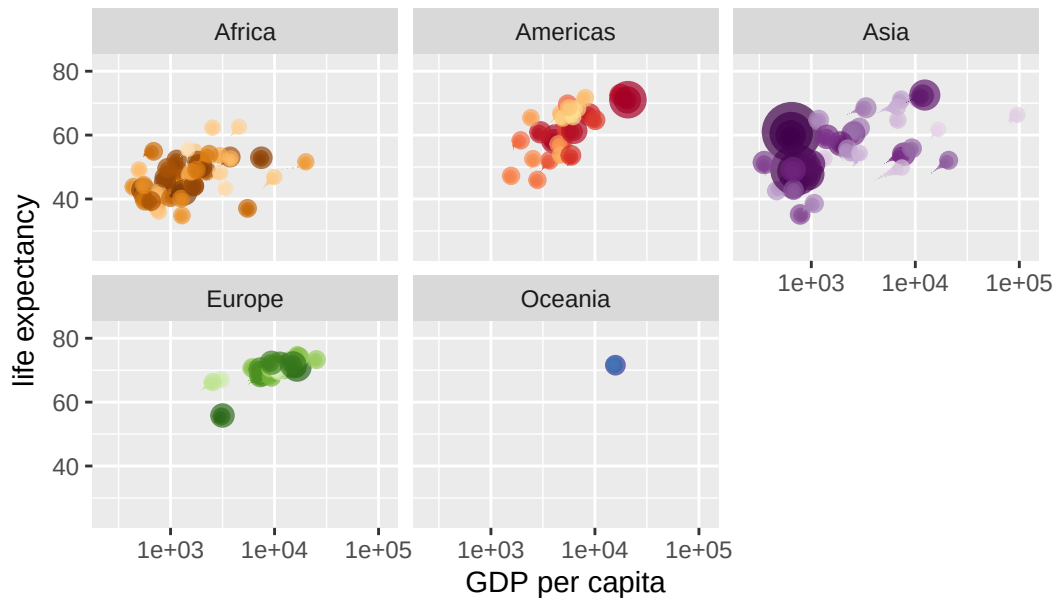
Year: 1969



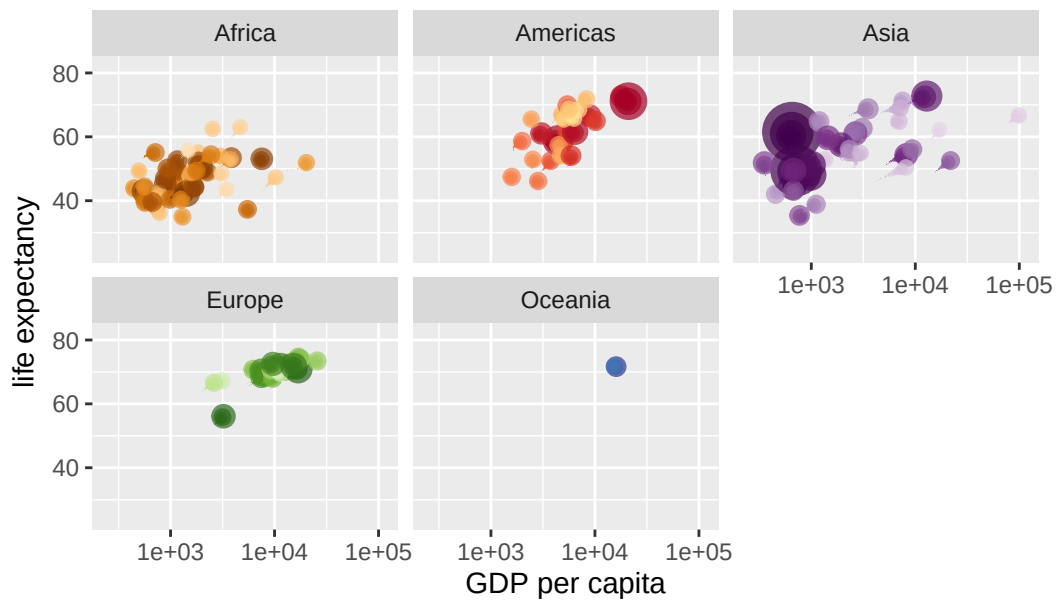
Year: 1969



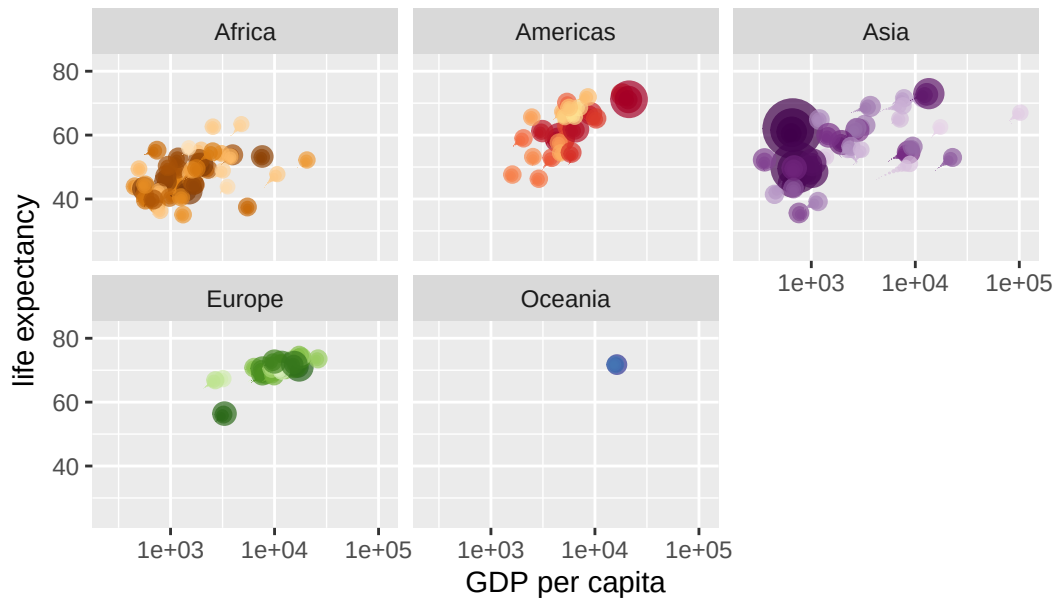
Year: 1970



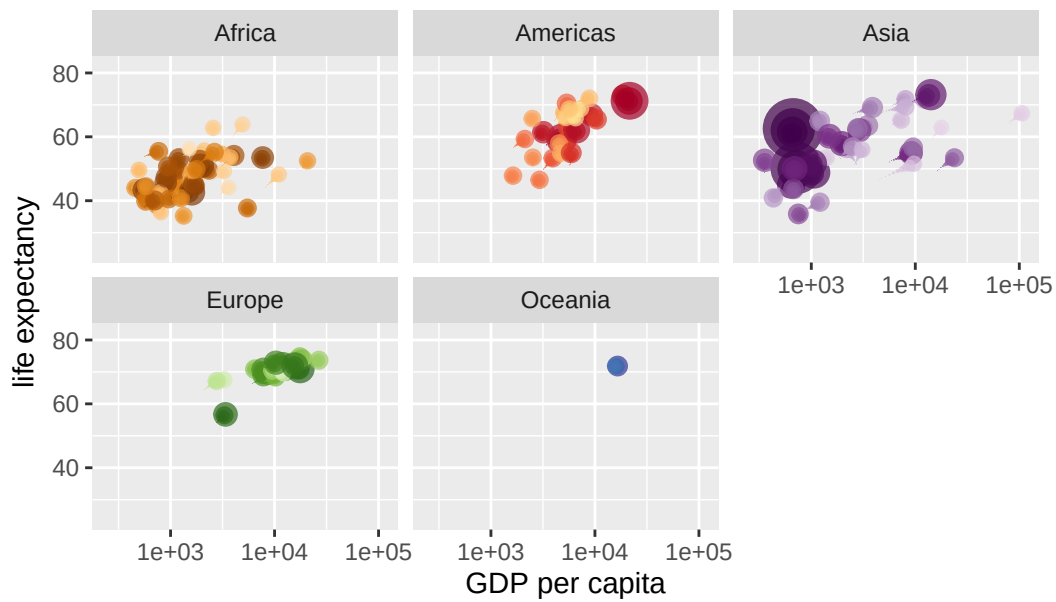
Year: 1970



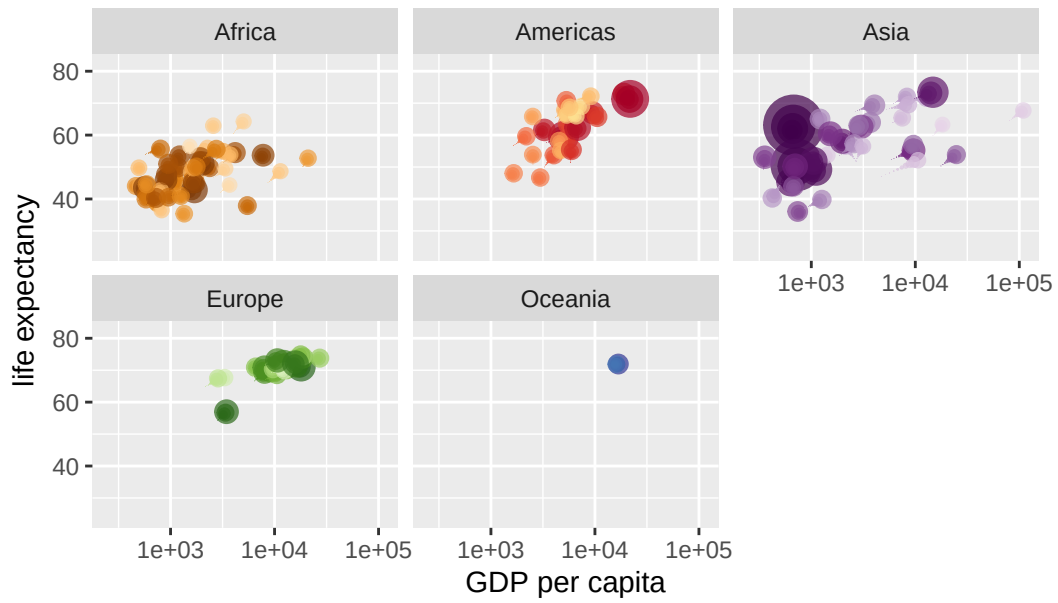
Year: 1971



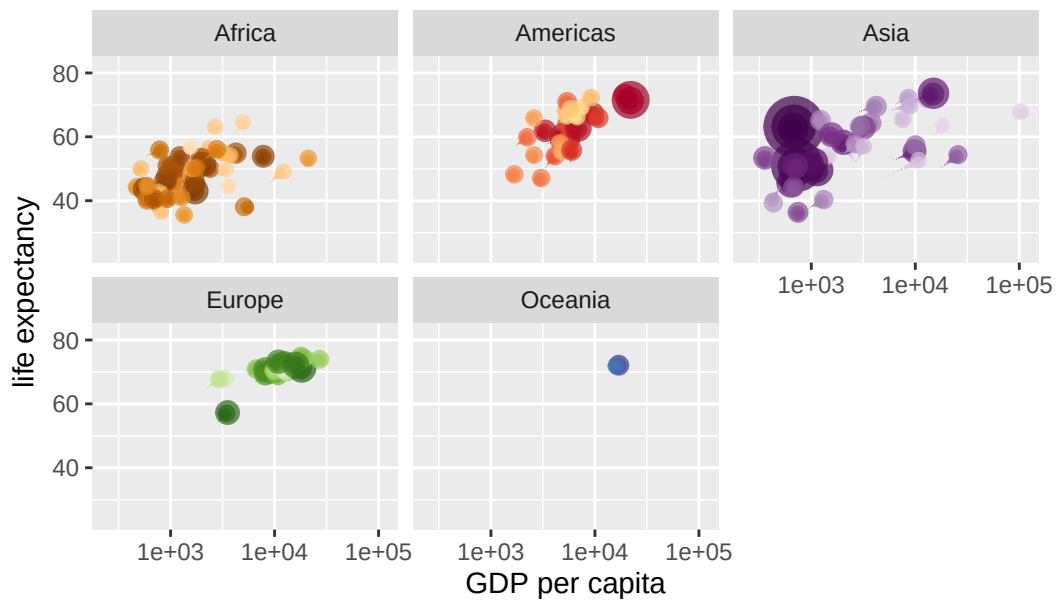
Year: 1971



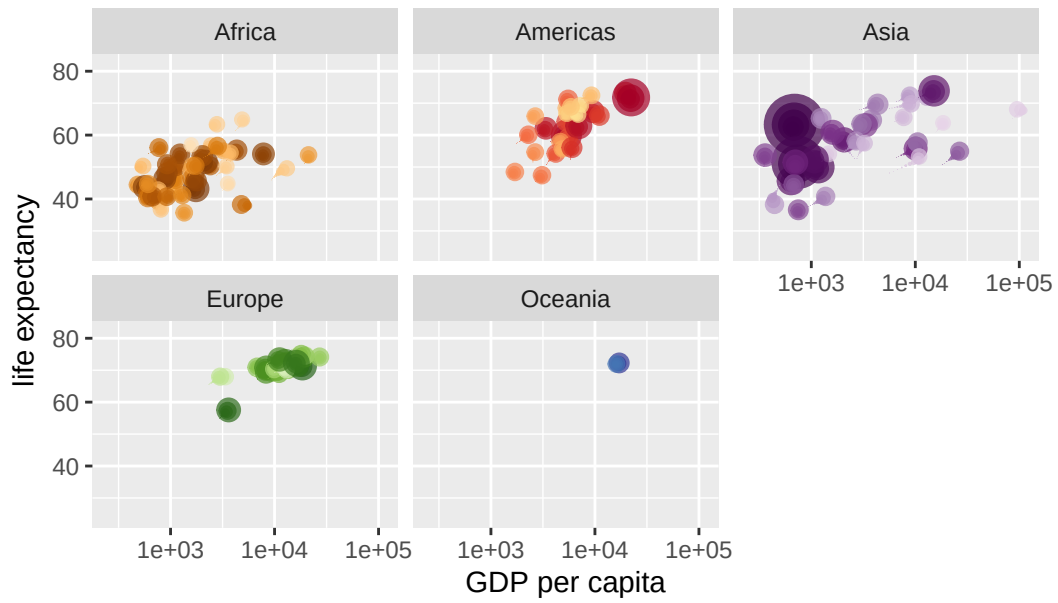
Year: 1972



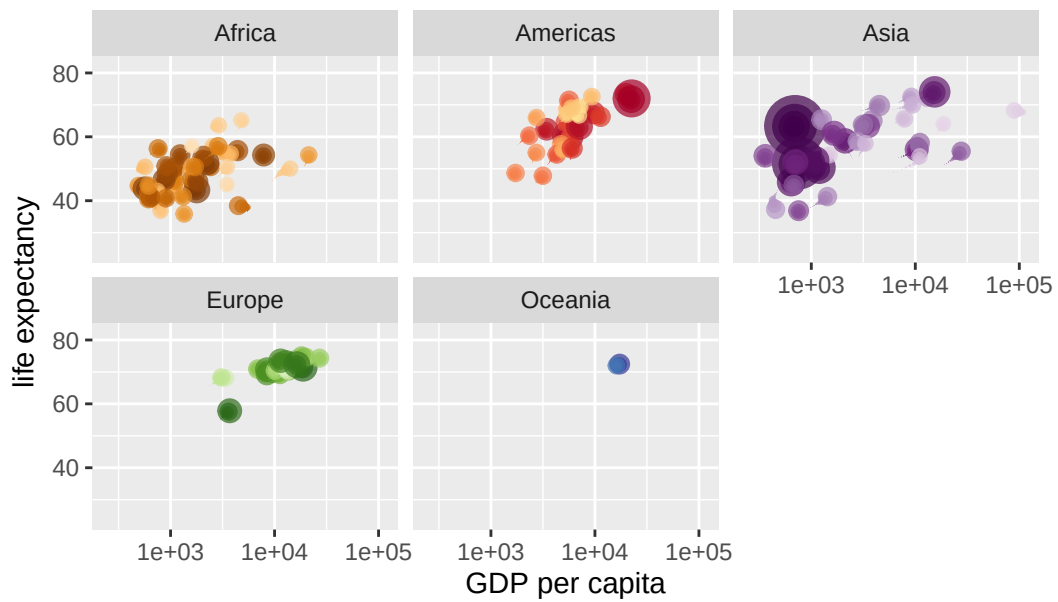
Year: 1973



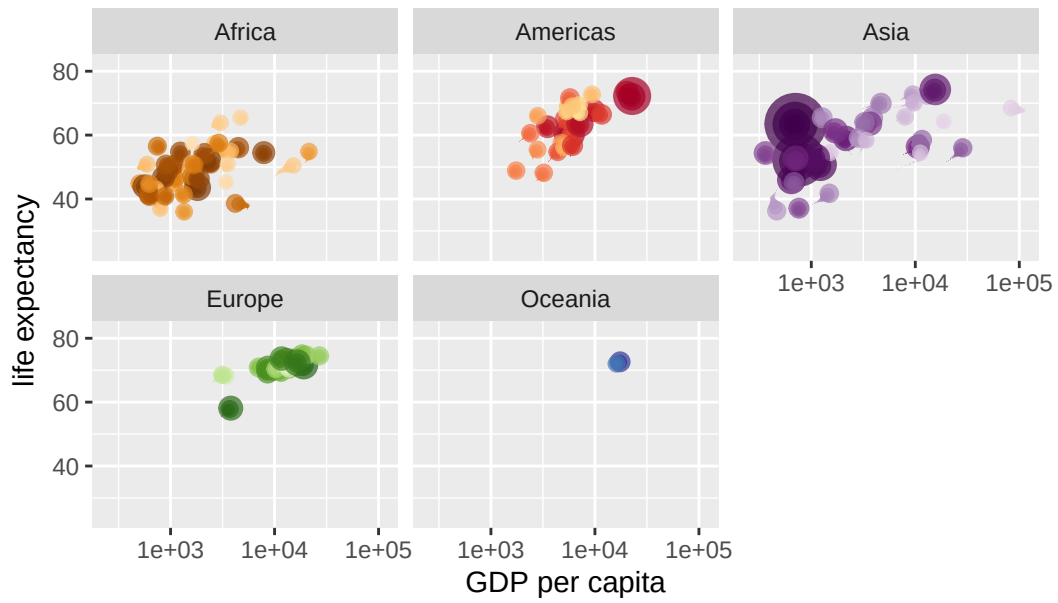
Year: 1973



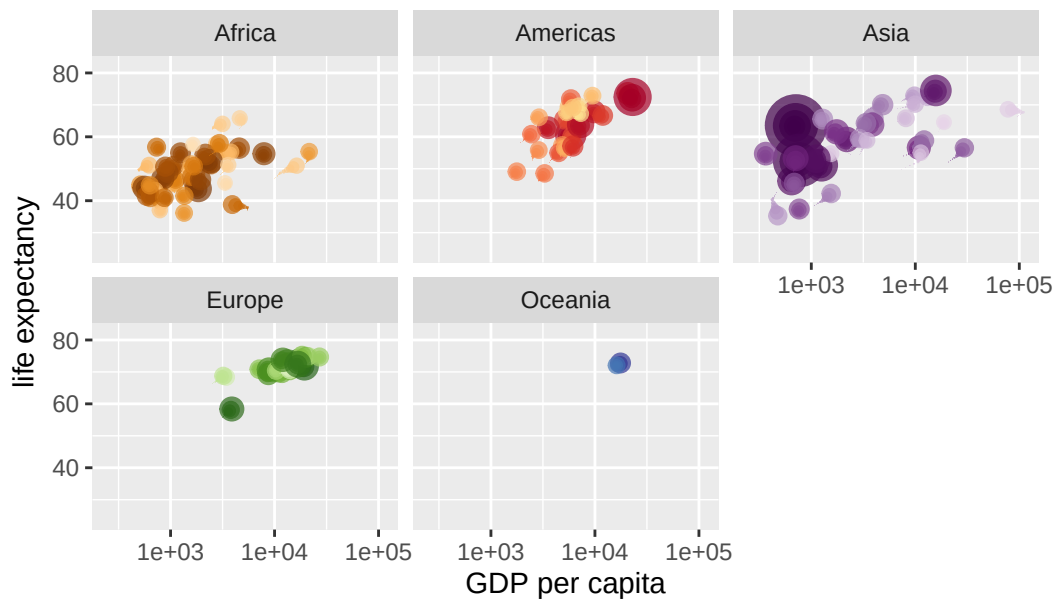
Year: 1974



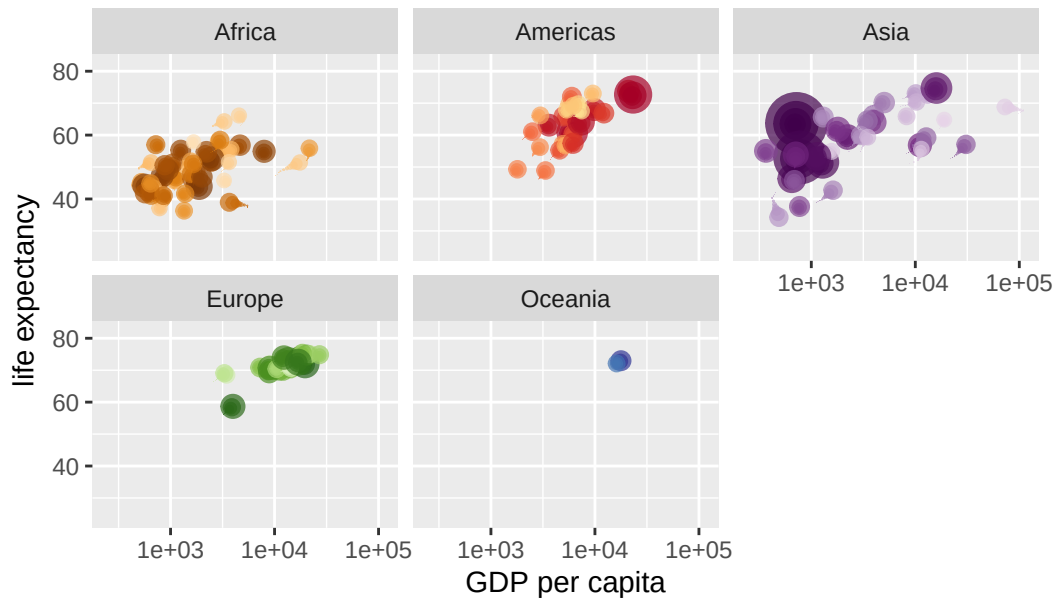
Year: 1974



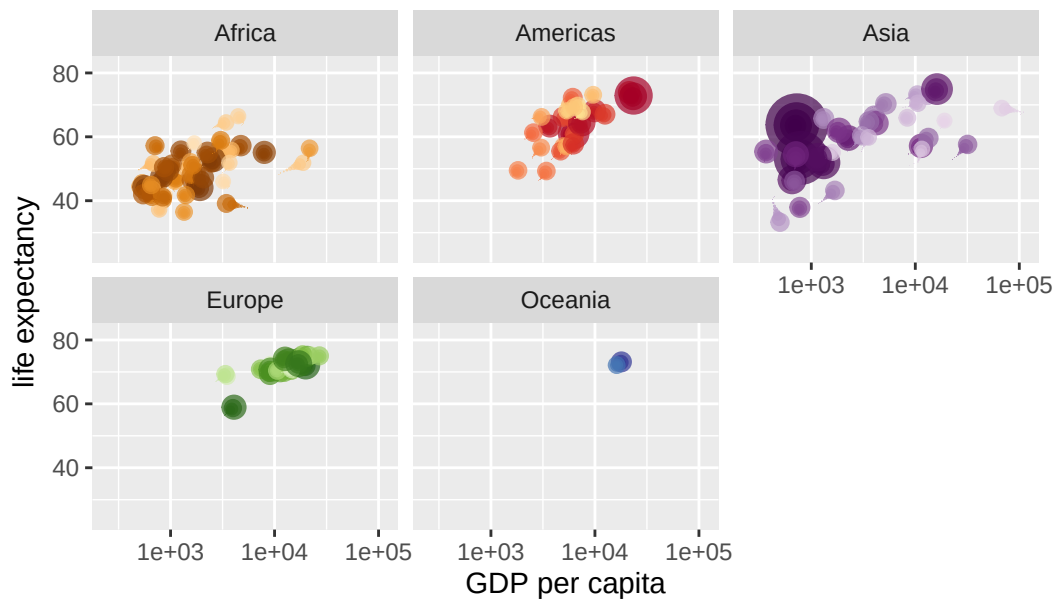
Year: 1975



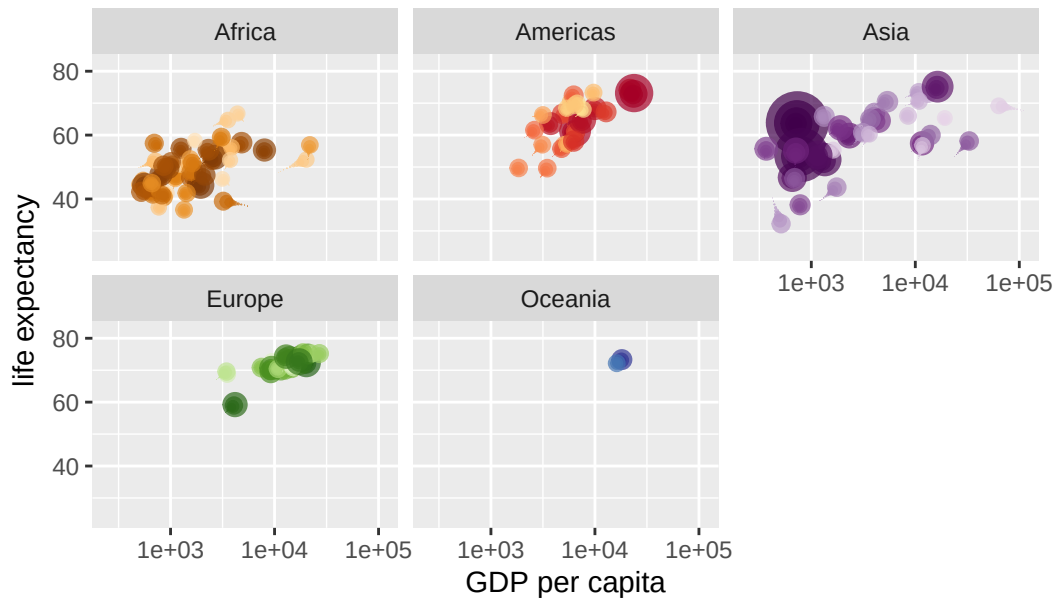
Year: 1975



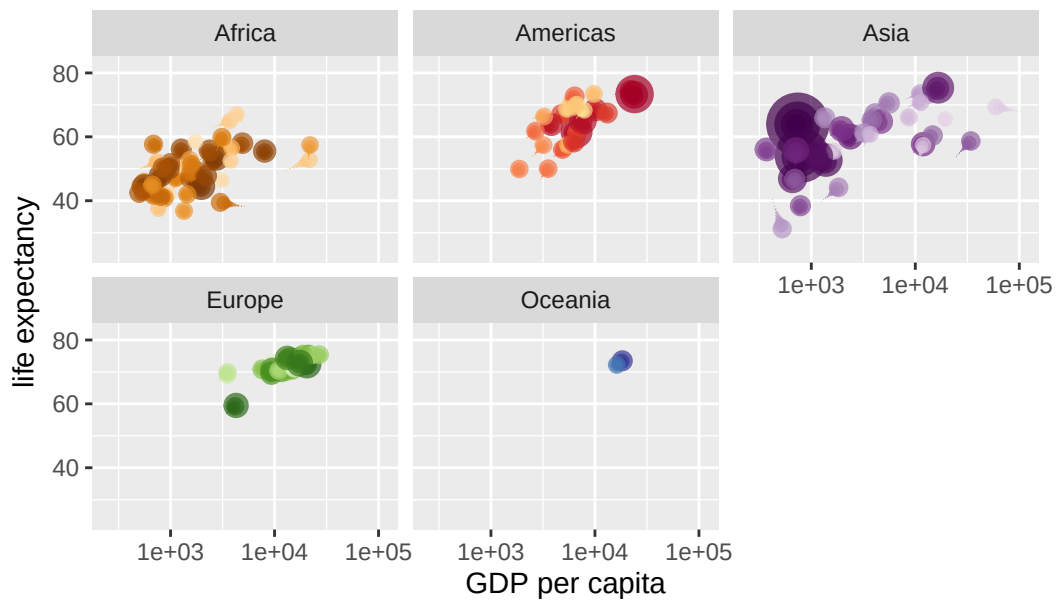
Year: 1976



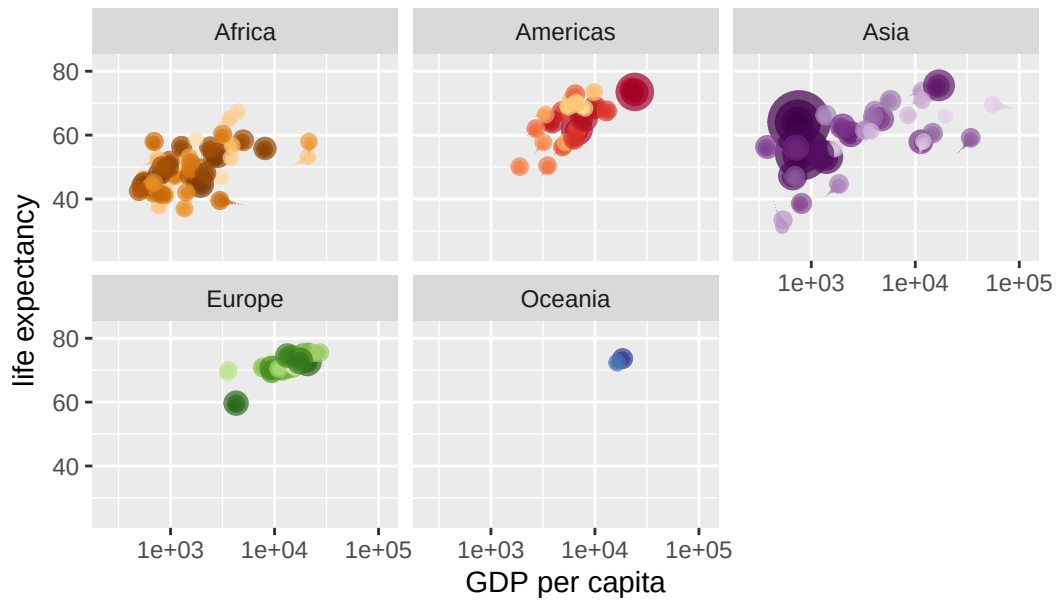
Year: 1976



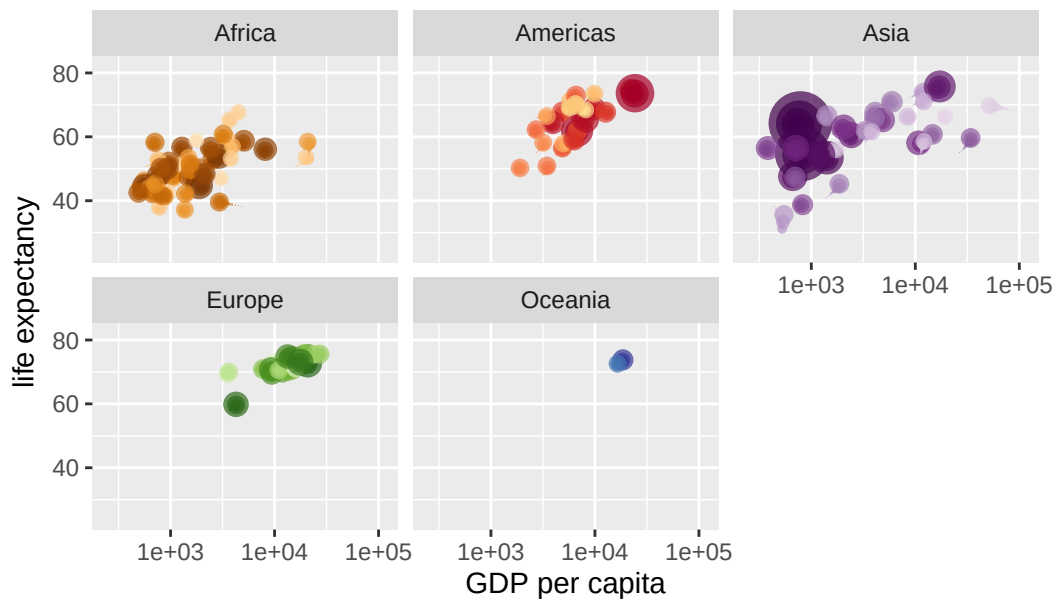
Year: 1977



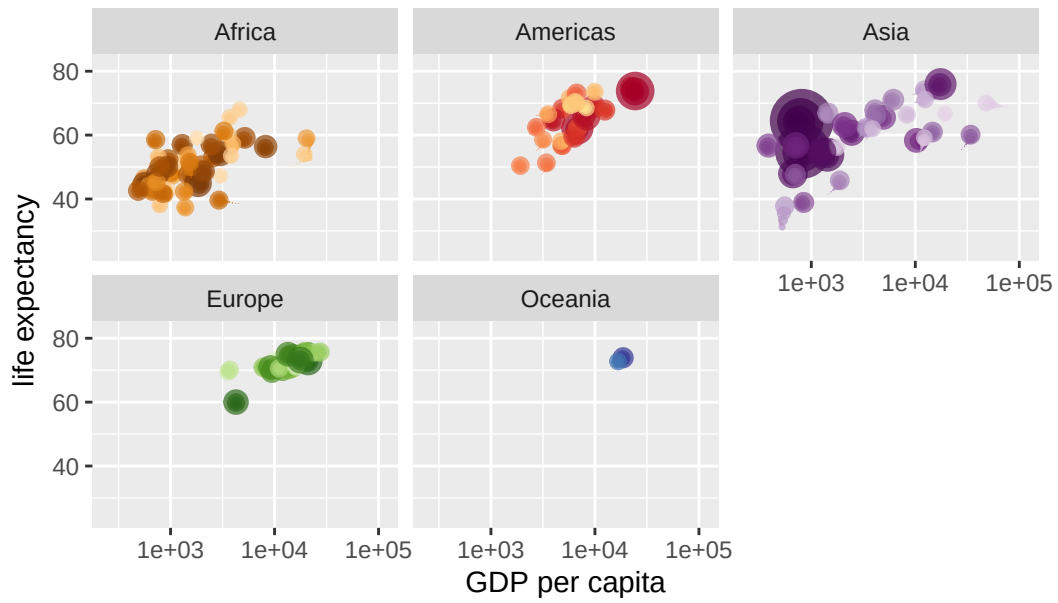
Year: 1978



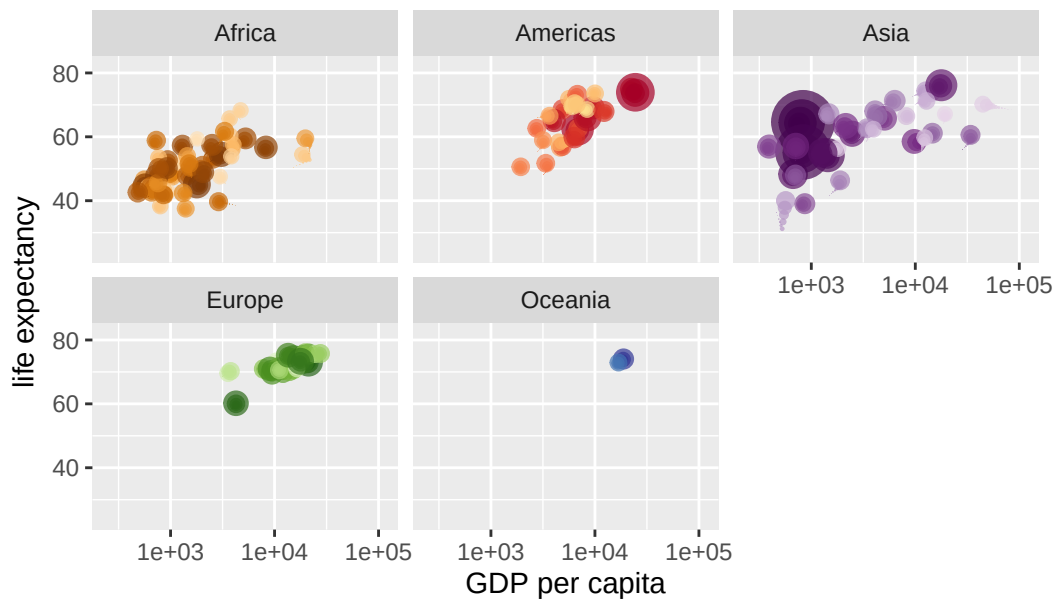
Year: 1978



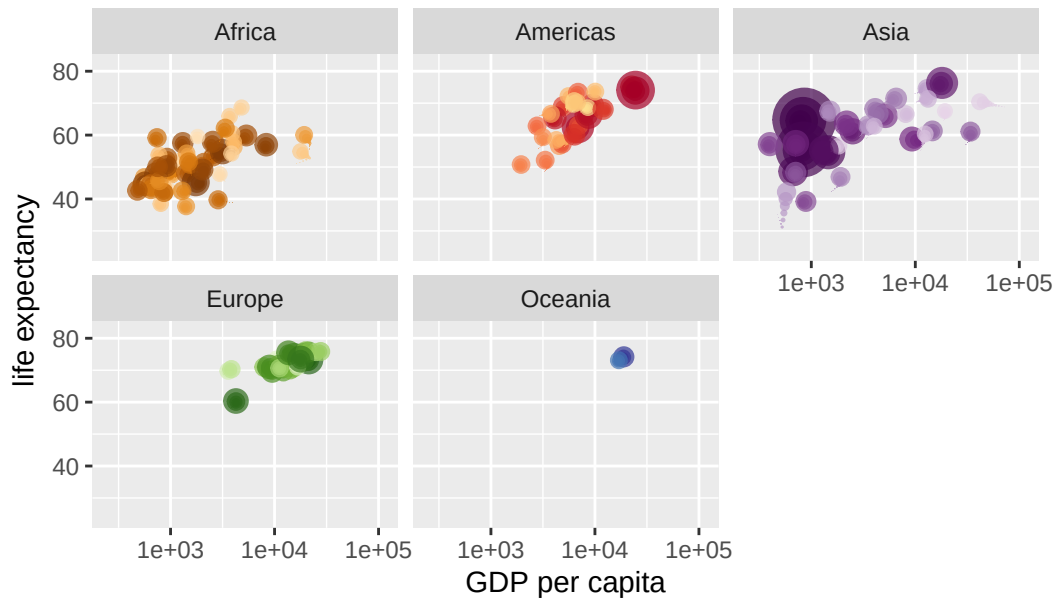
Year: 1979



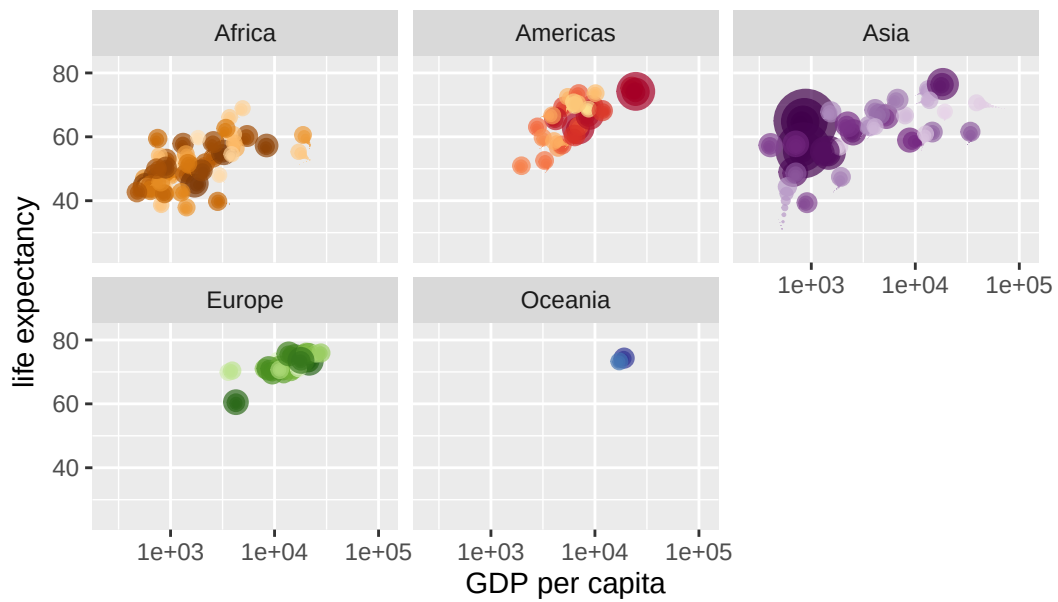
Year: 1979



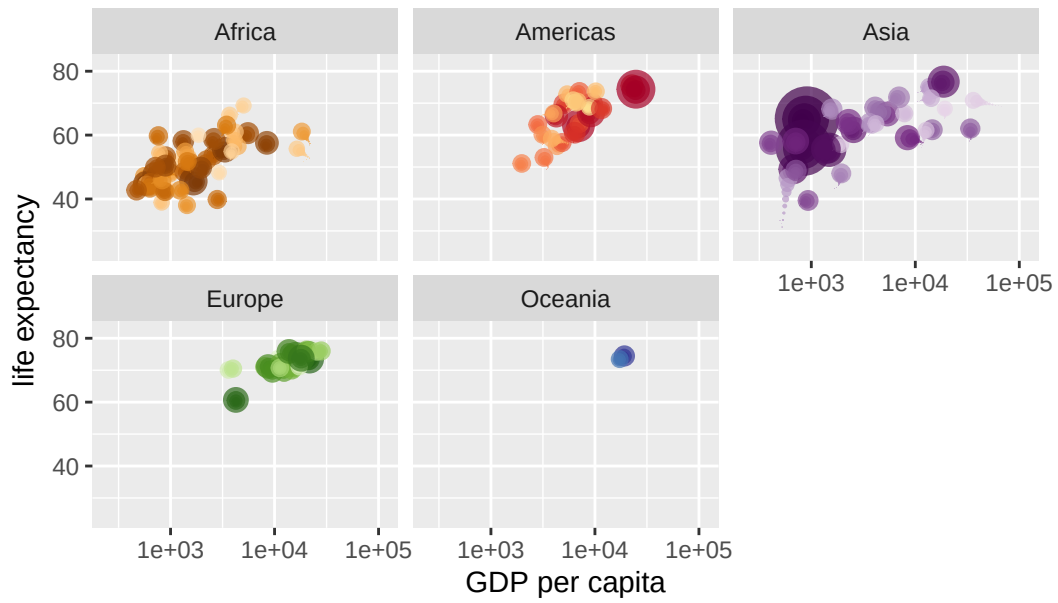
Year: 1980



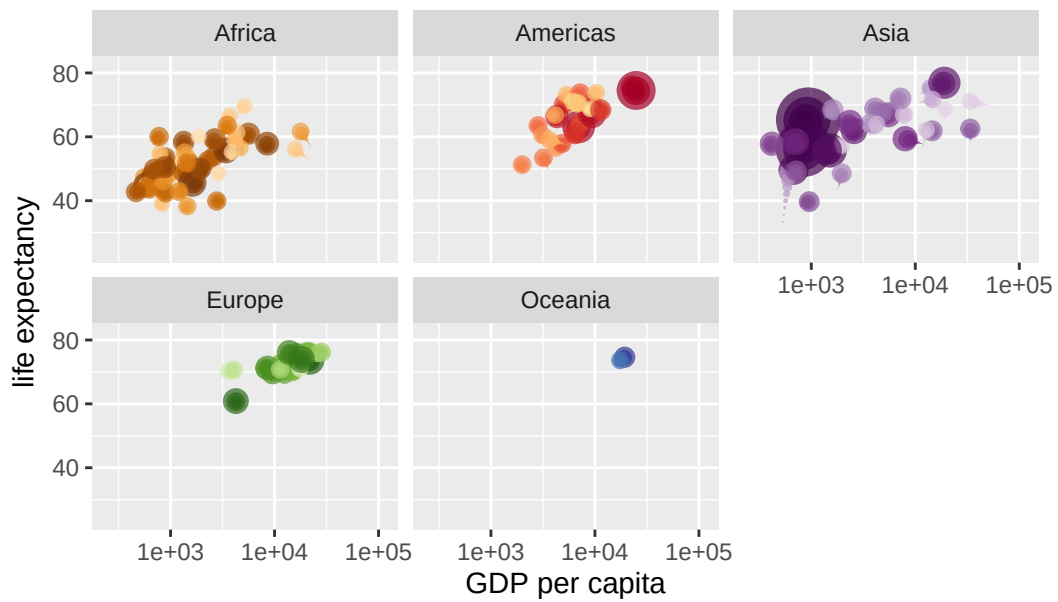
Year: 1980



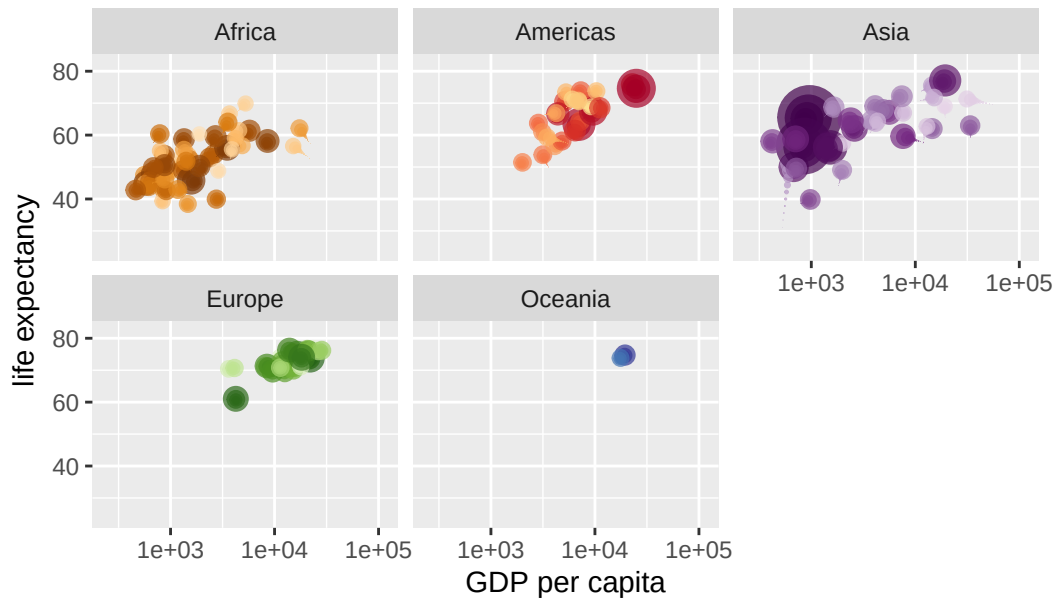
Year: 1981



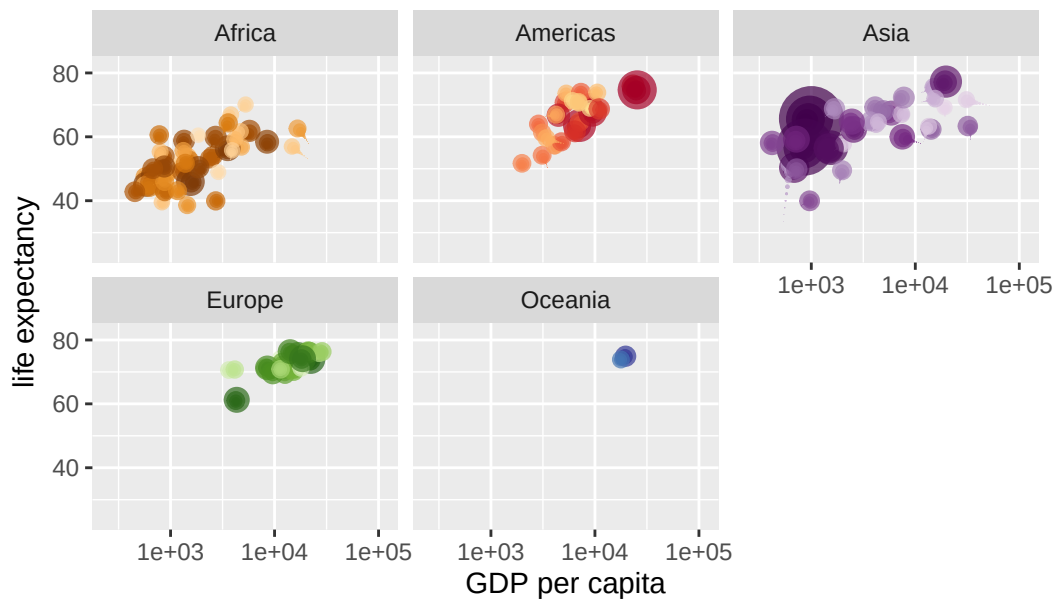
Year: 1981



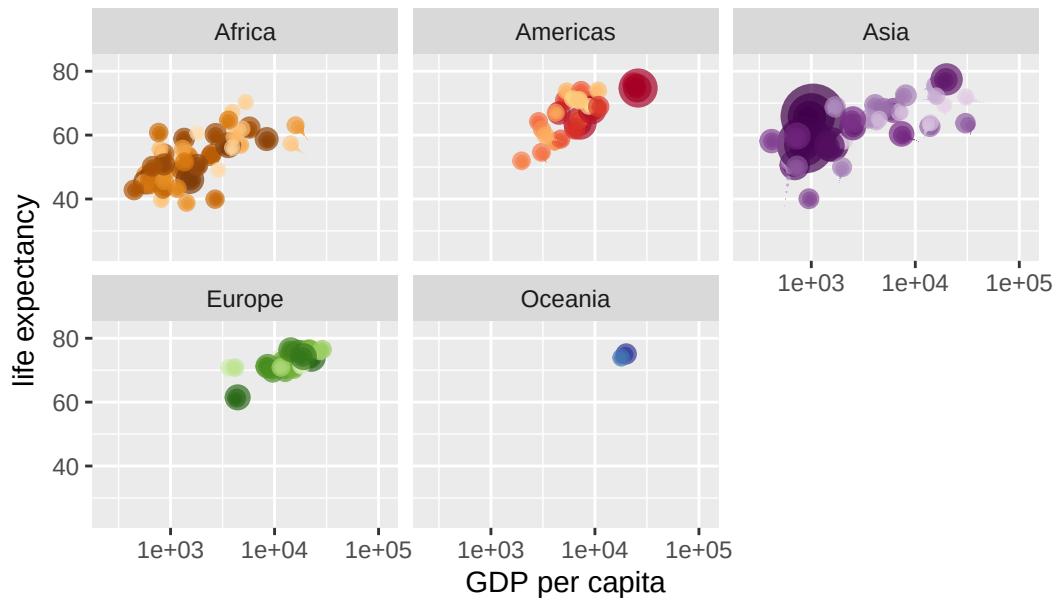
Year: 1982



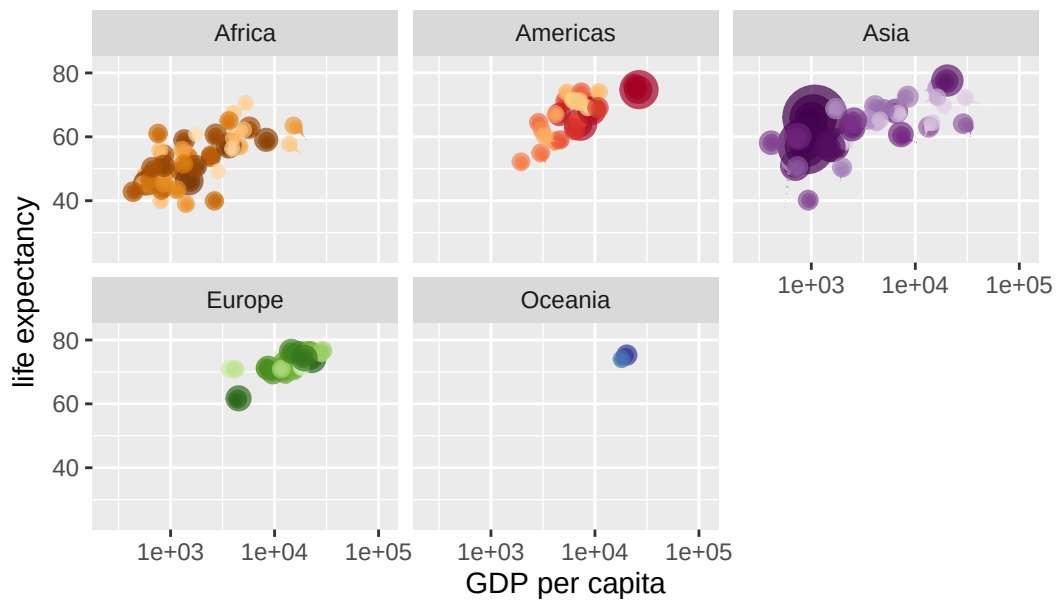
Year: 1983



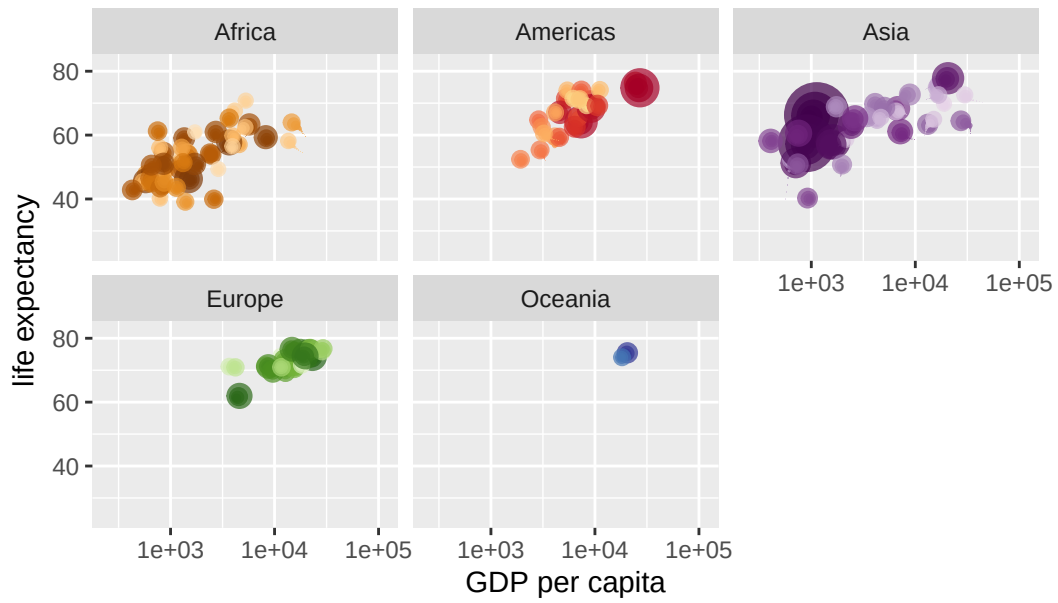
Year: 1983



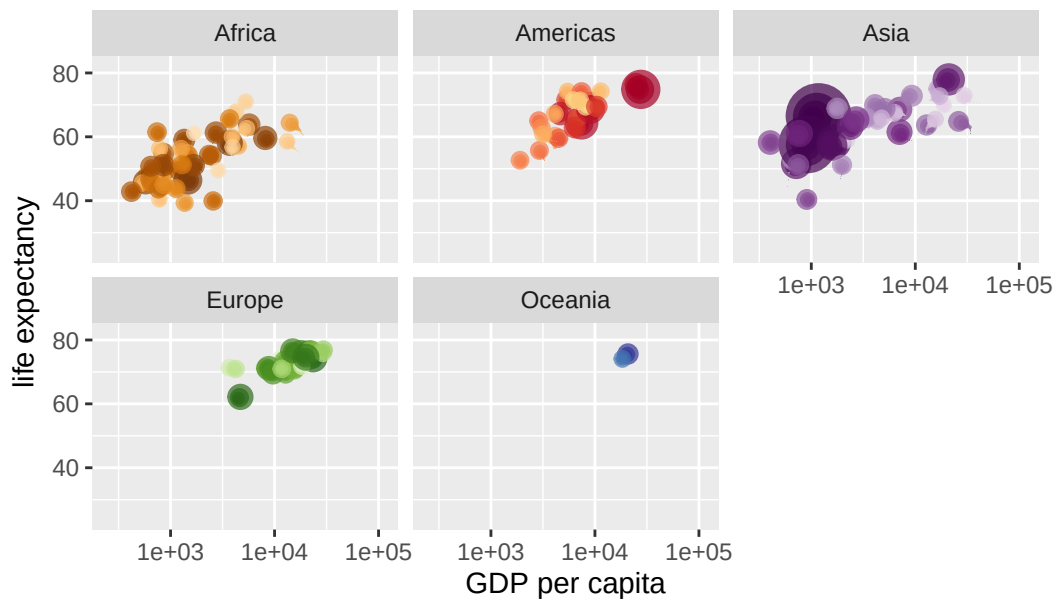
Year: 1984



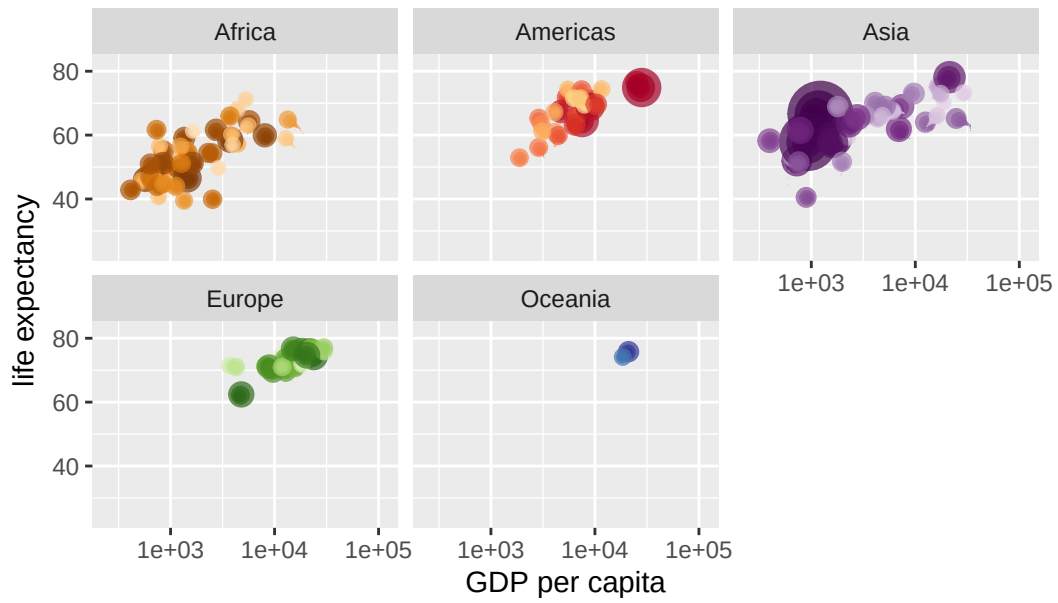
Year: 1984



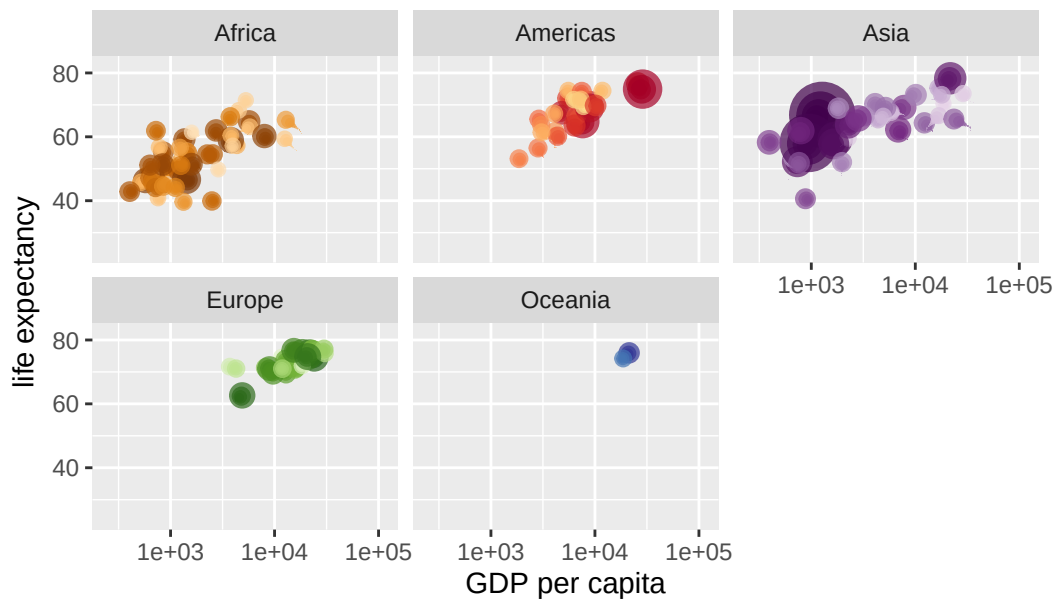
Year: 1985



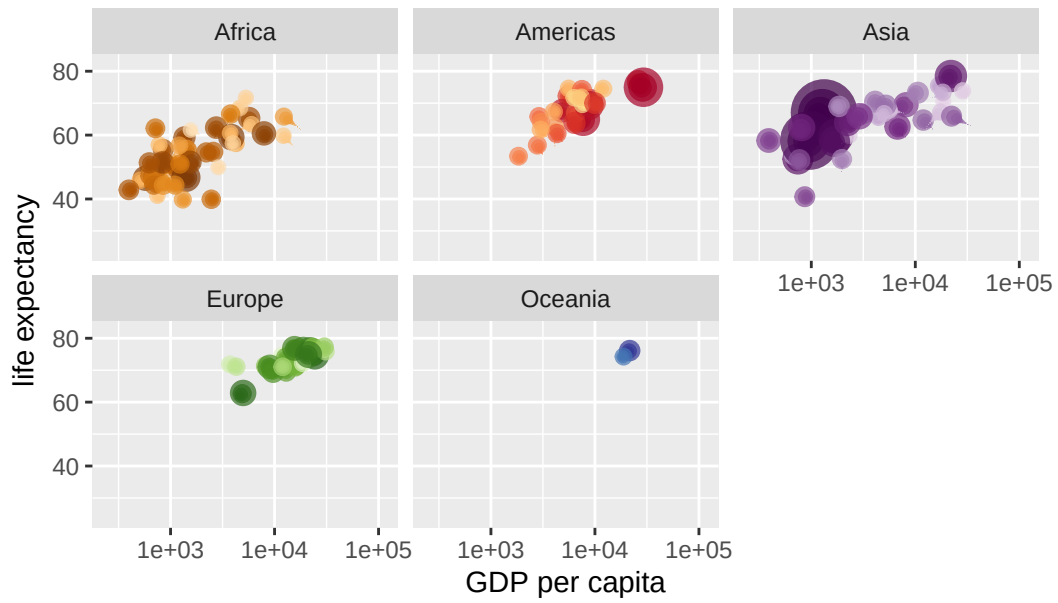
Year: 1985



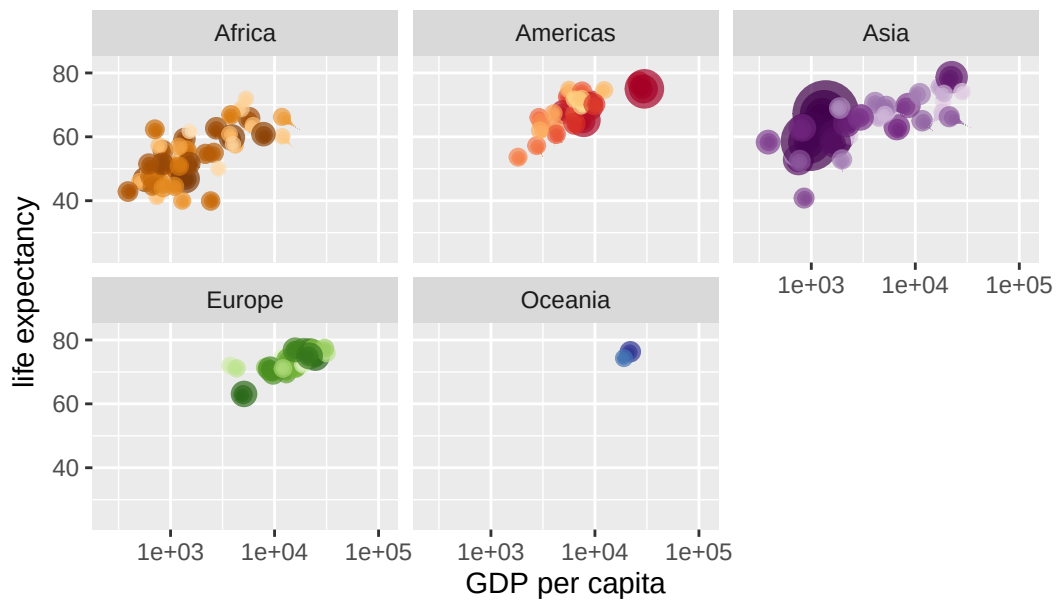
Year: 1986



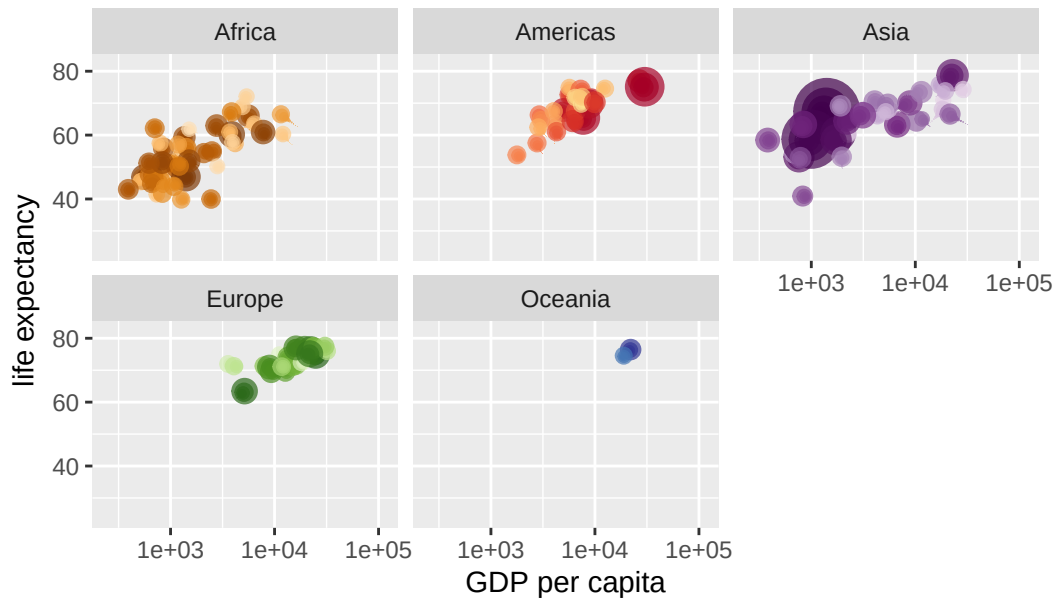
Year: 1986



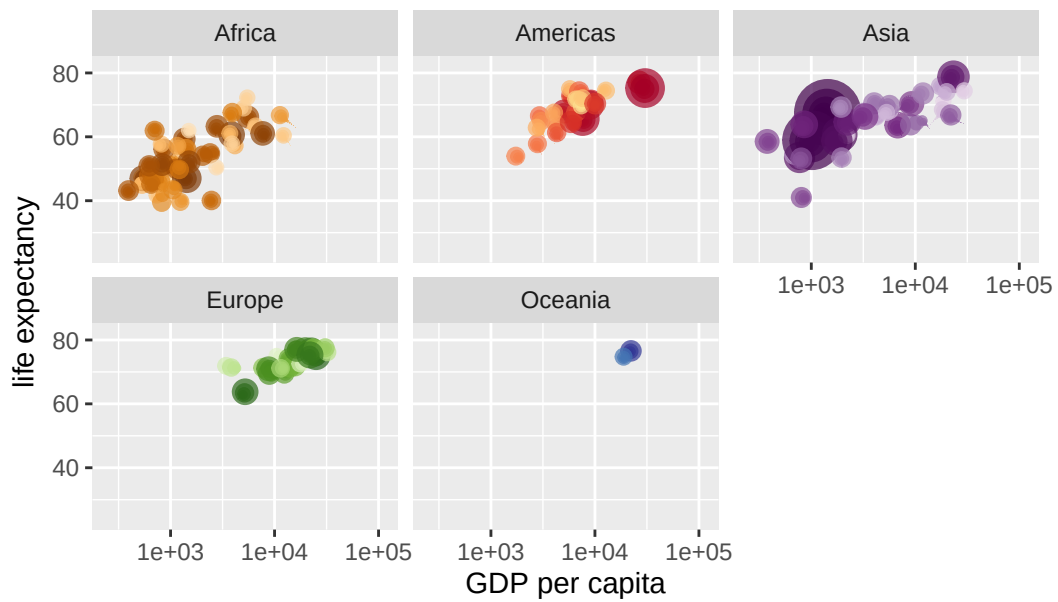
Year: 1987



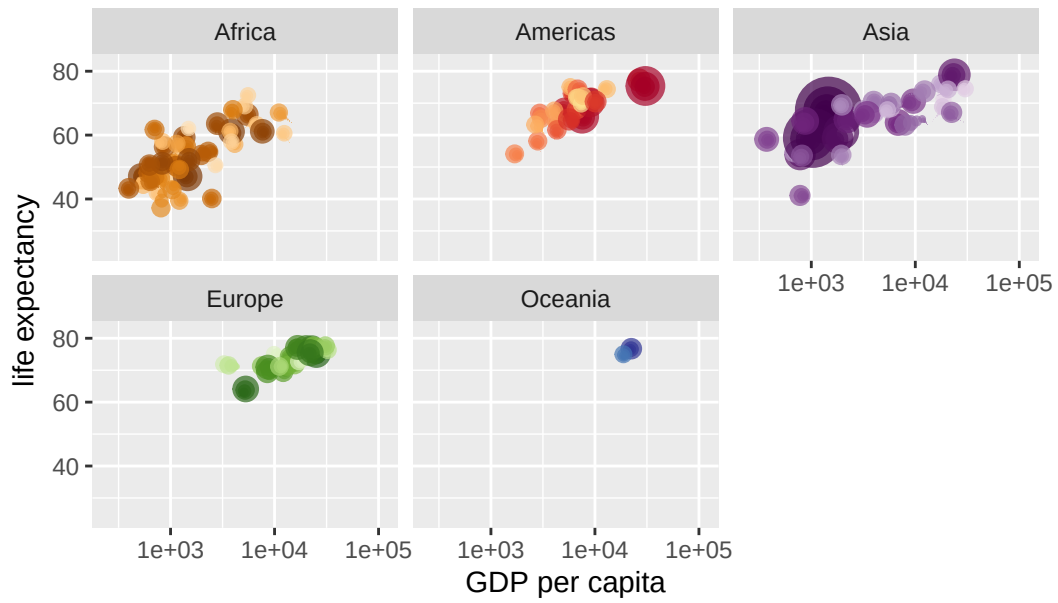
Year: 1988



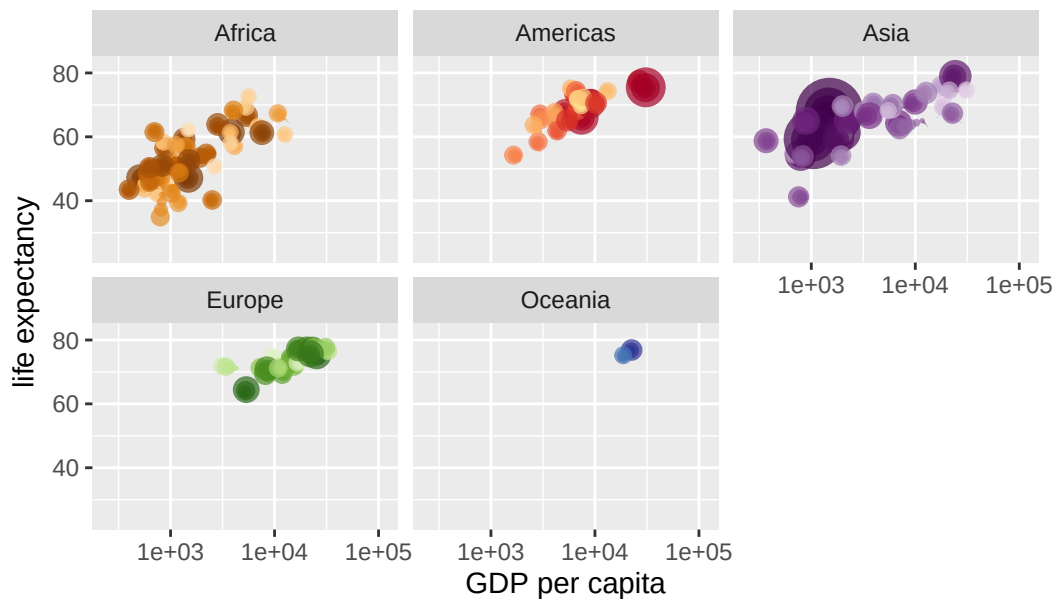
Year: 1988



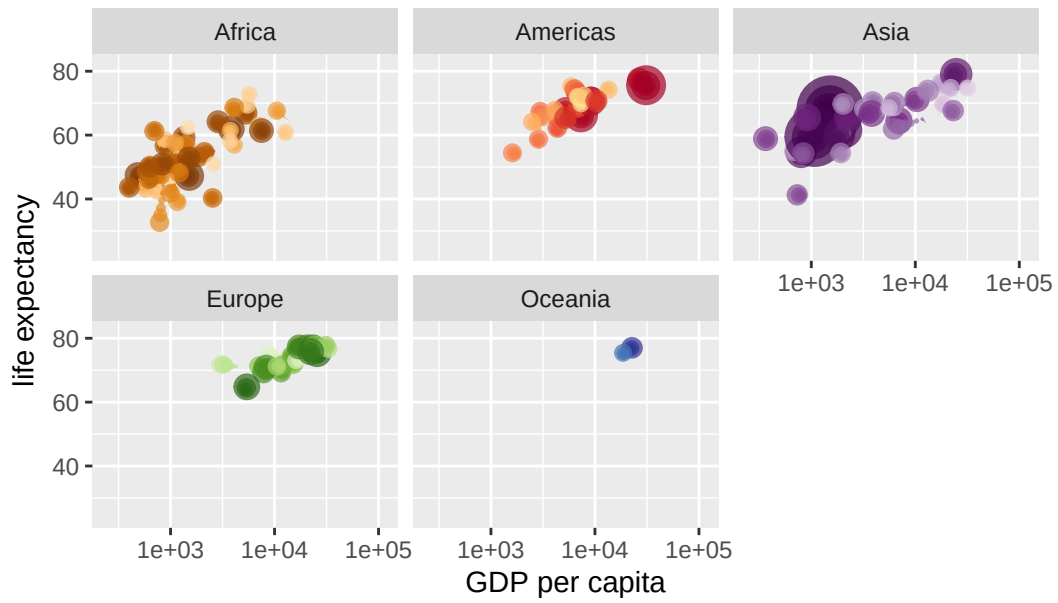
Year: 1989



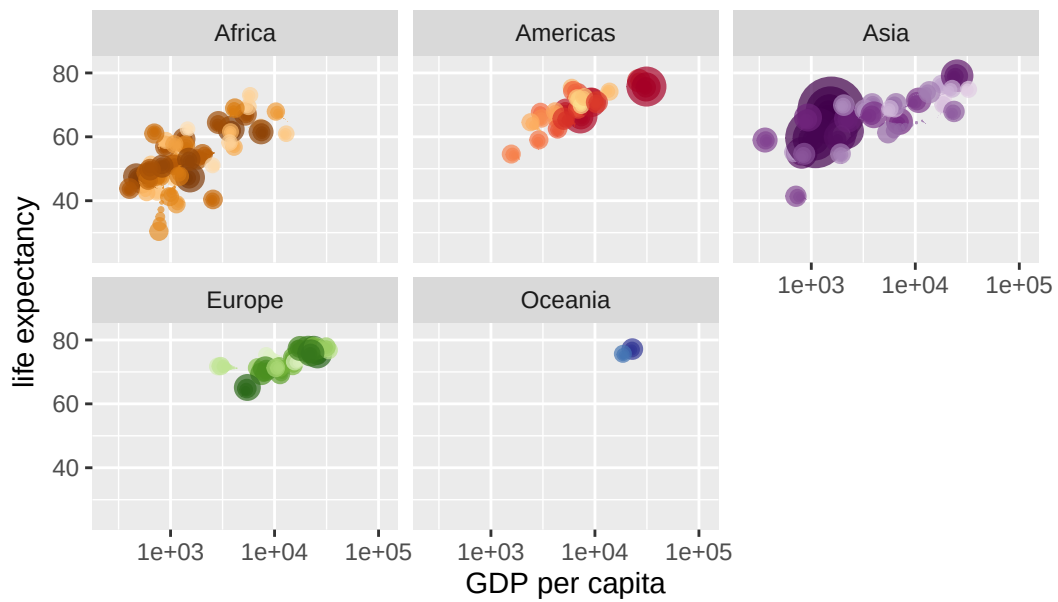
Year: 1989



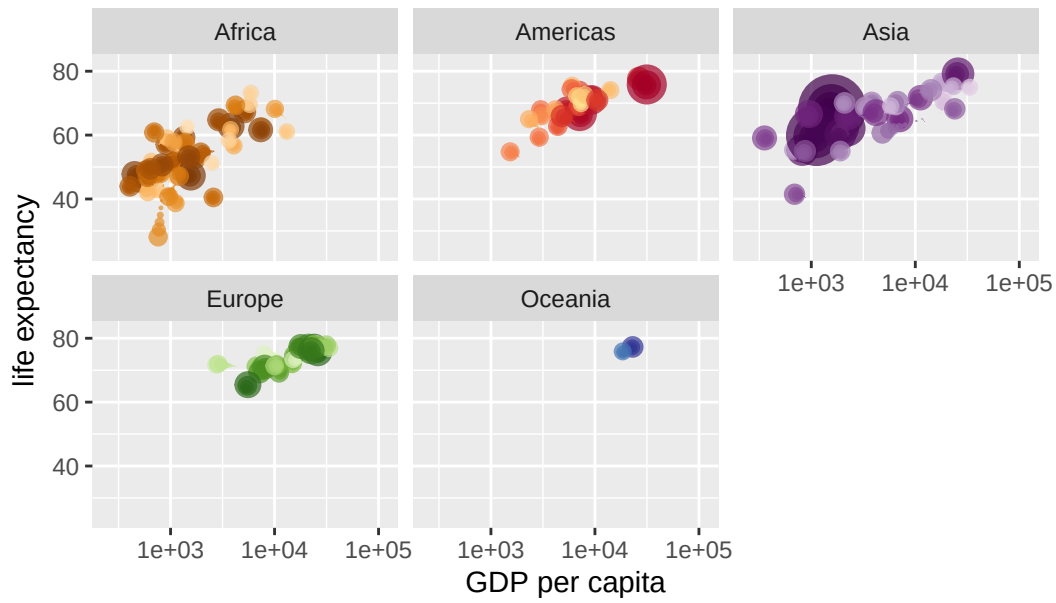
Year: 1990



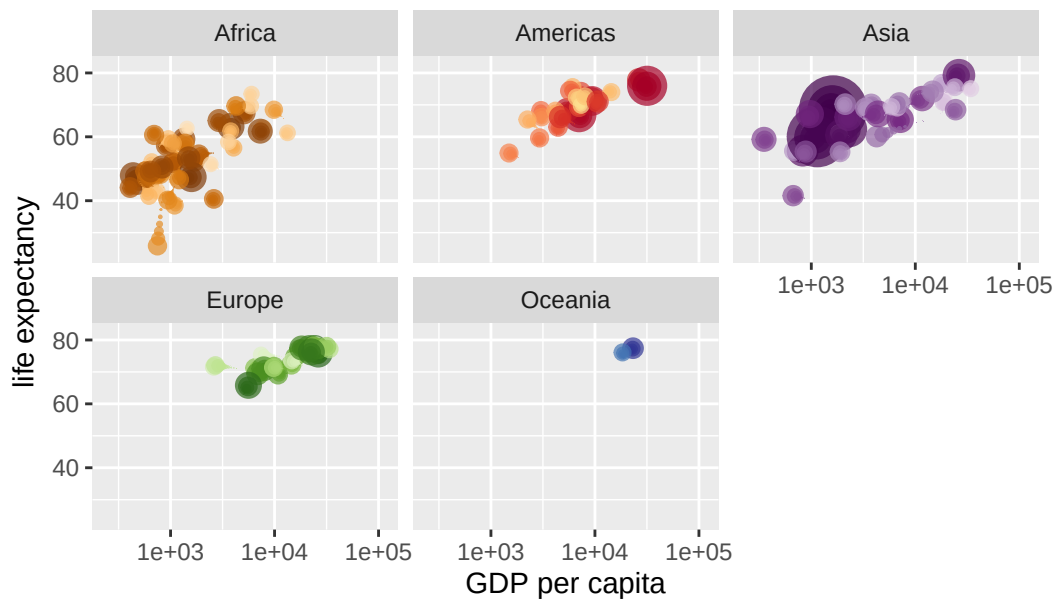
Year: 1990



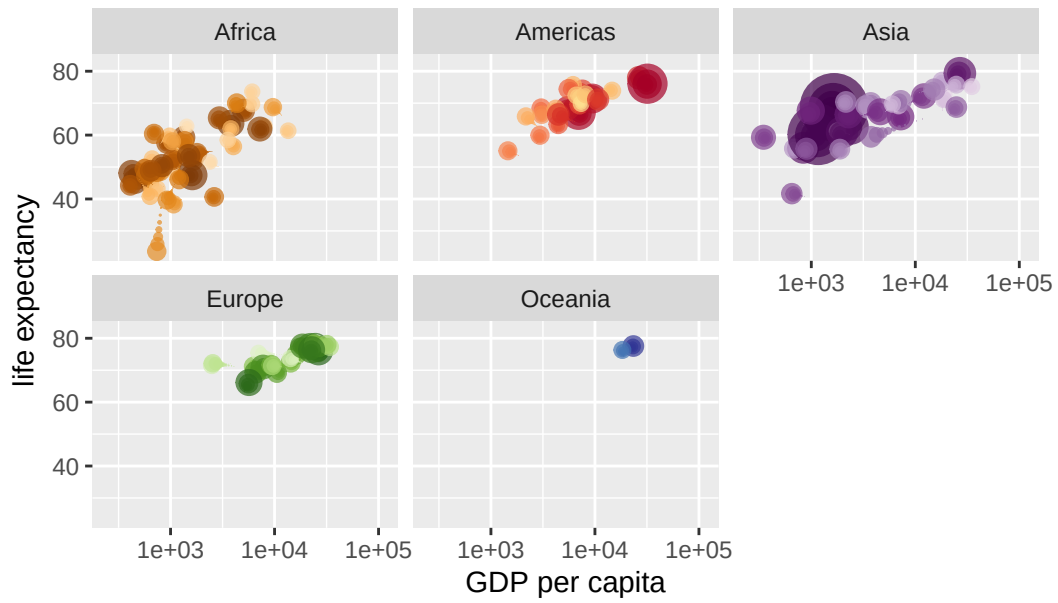
Year: 1991



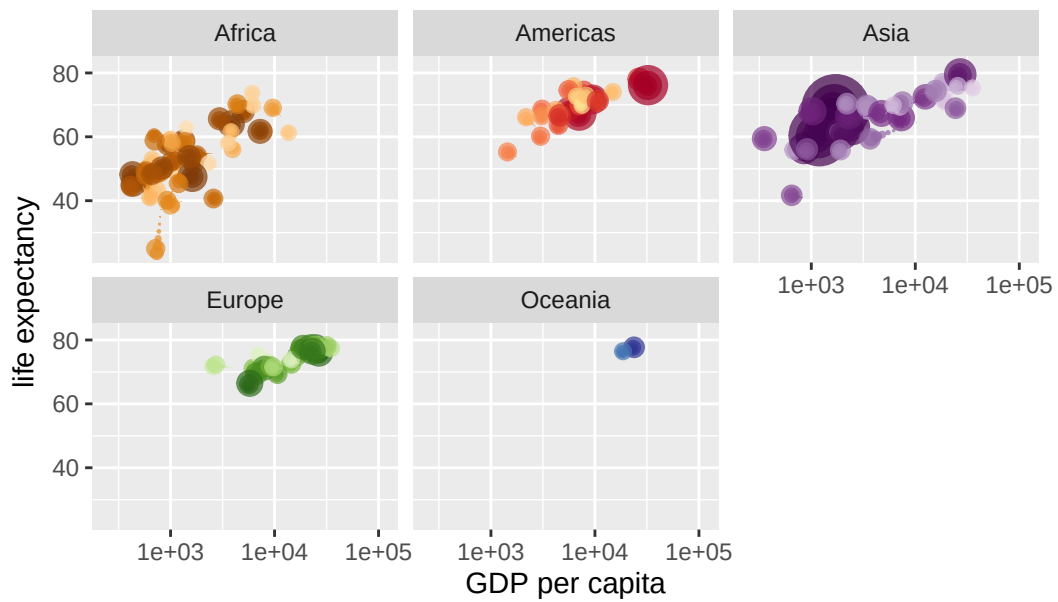
Year: 1991



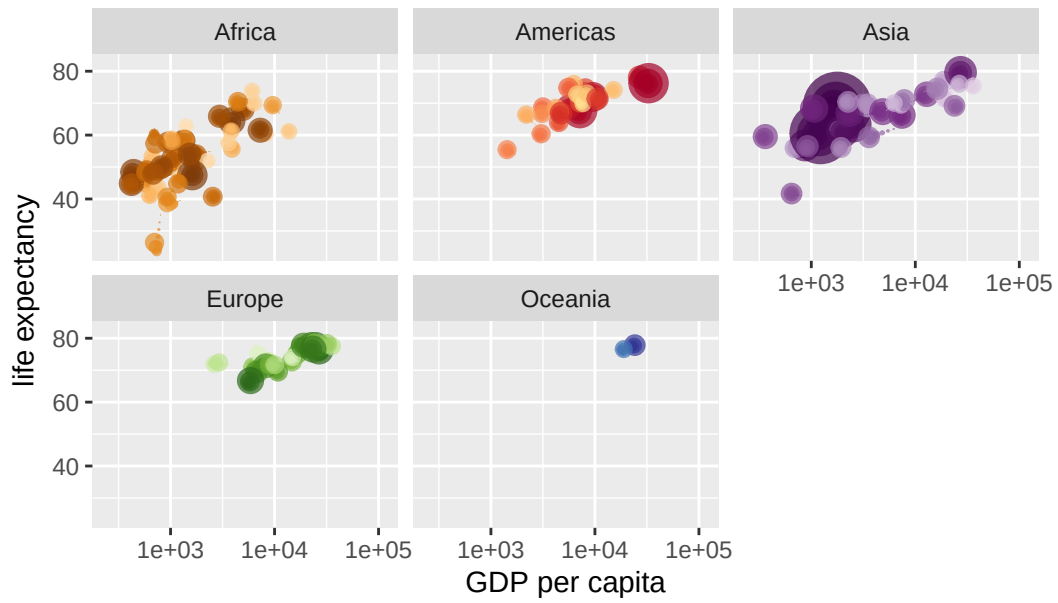
Year: 1992



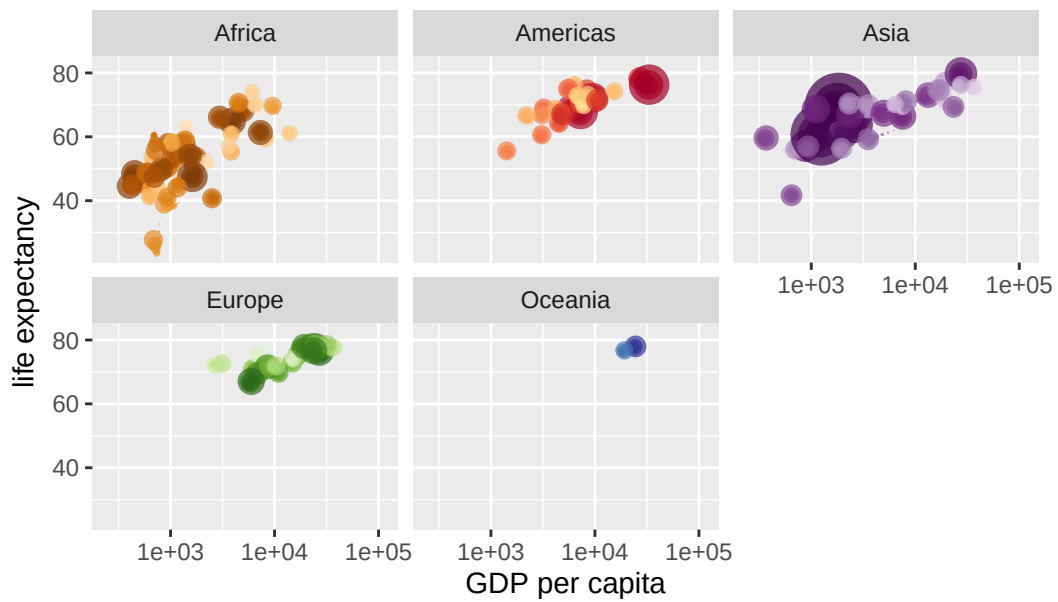
Year: 1993



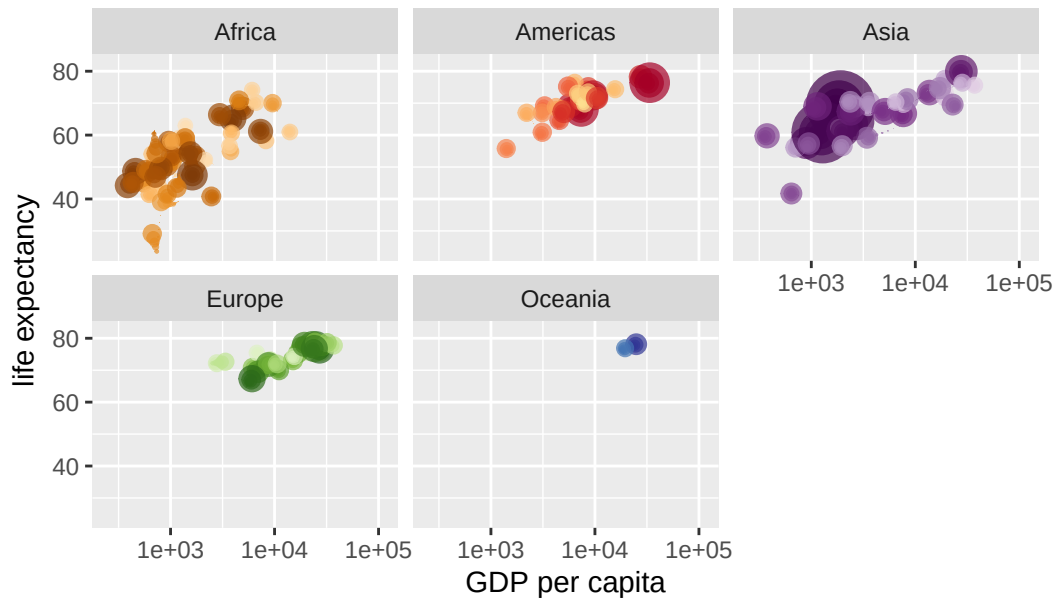
Year: 1993



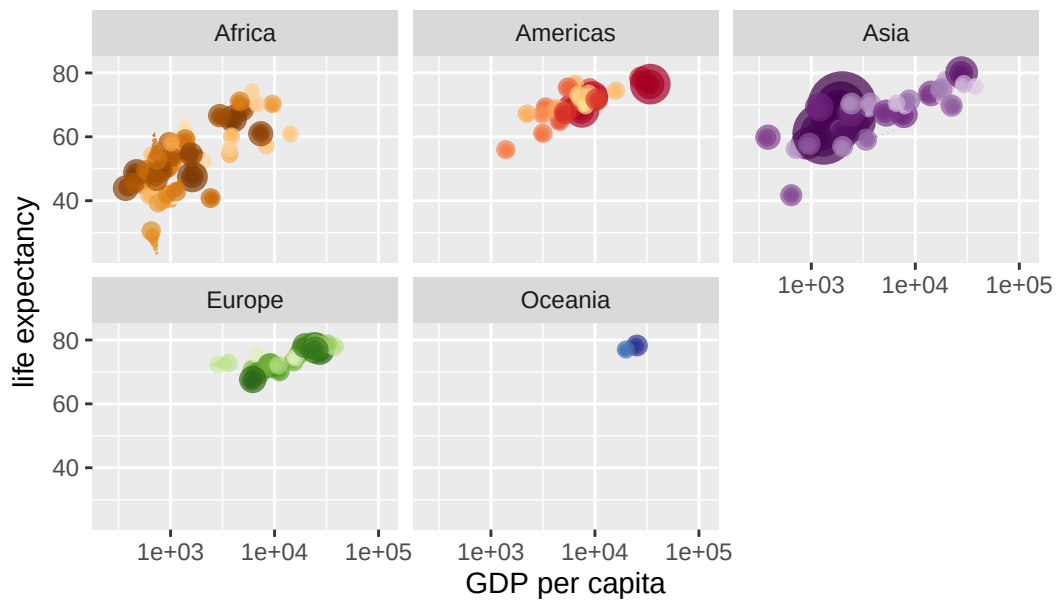
Year: 1994



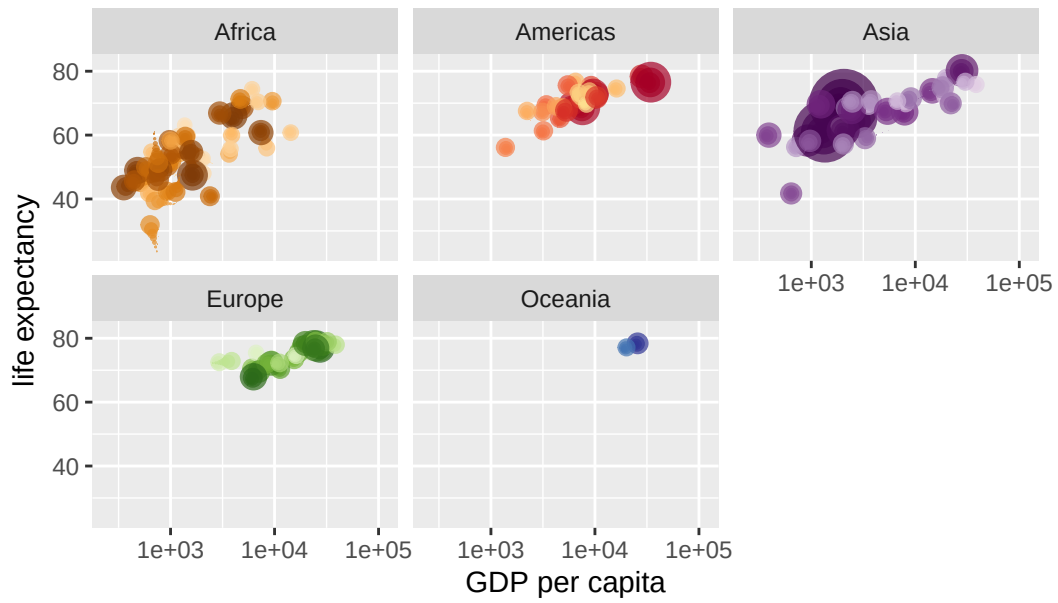
Year: 1994



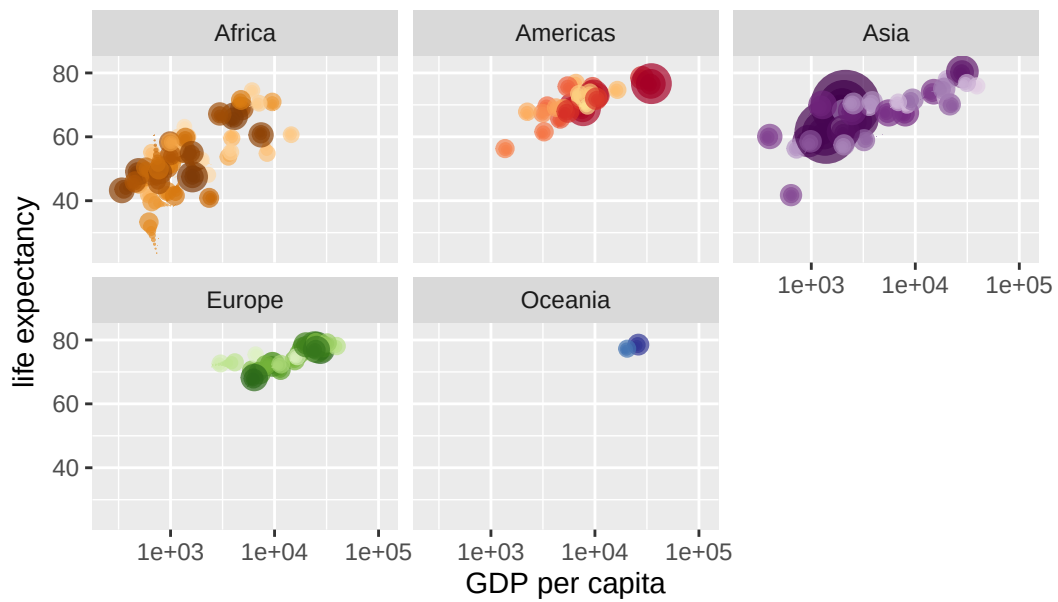
Year: 1995



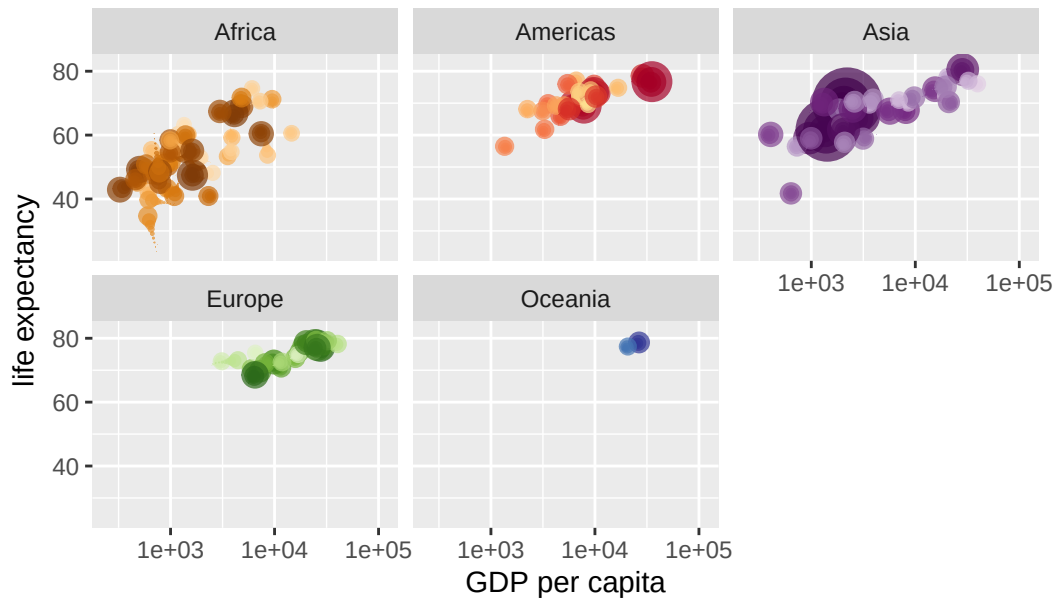
Year: 1995



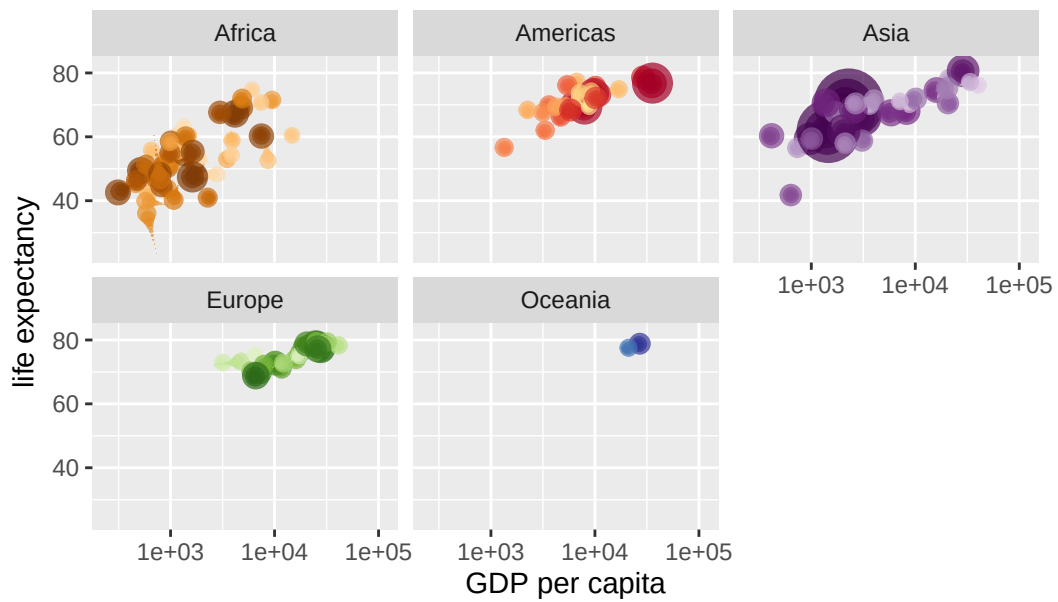
Year: 1996



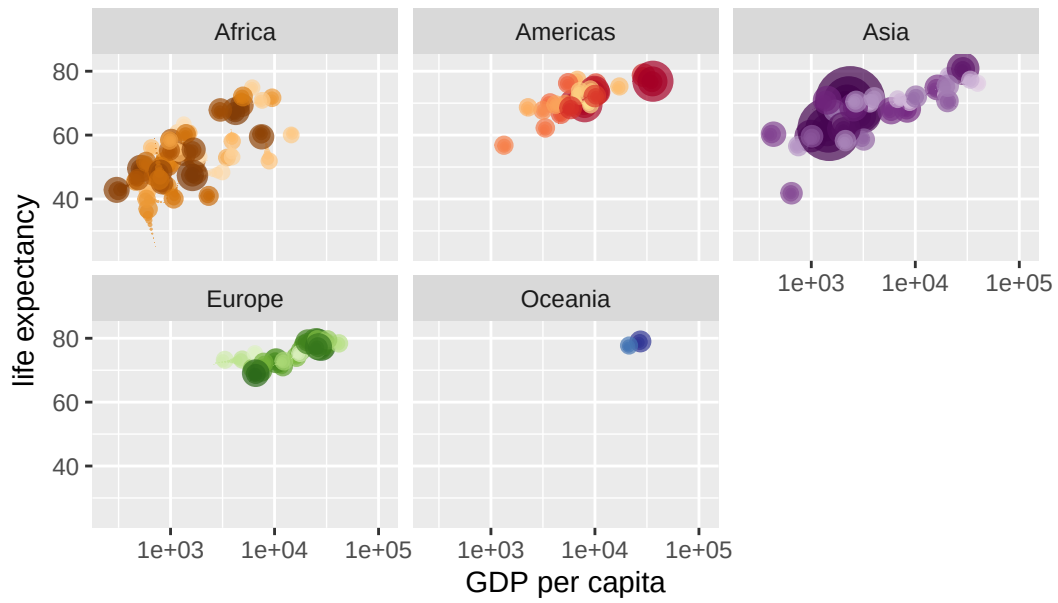
Year: 1996



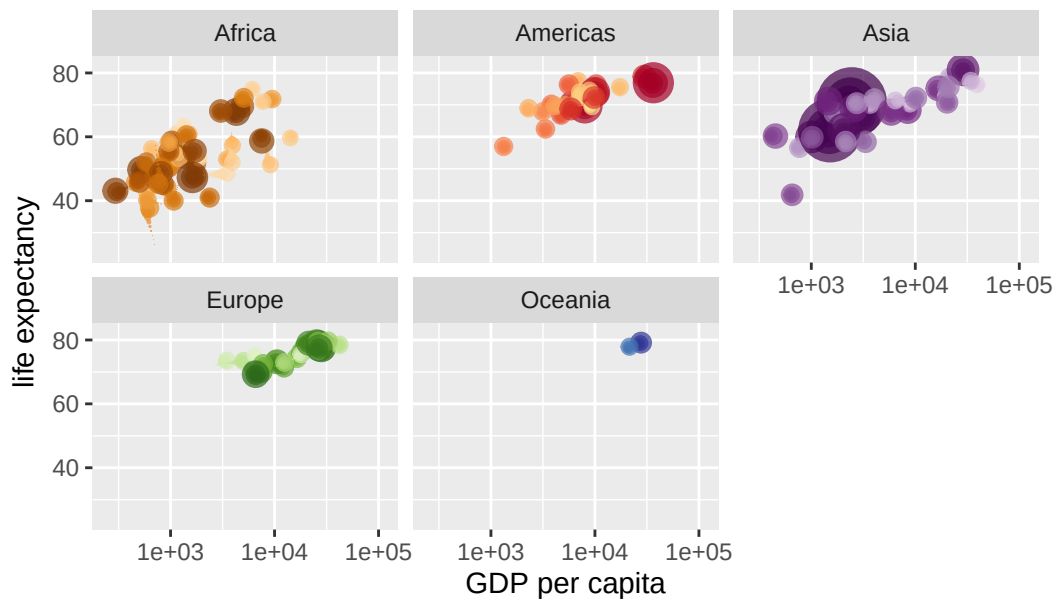
Year: 1997



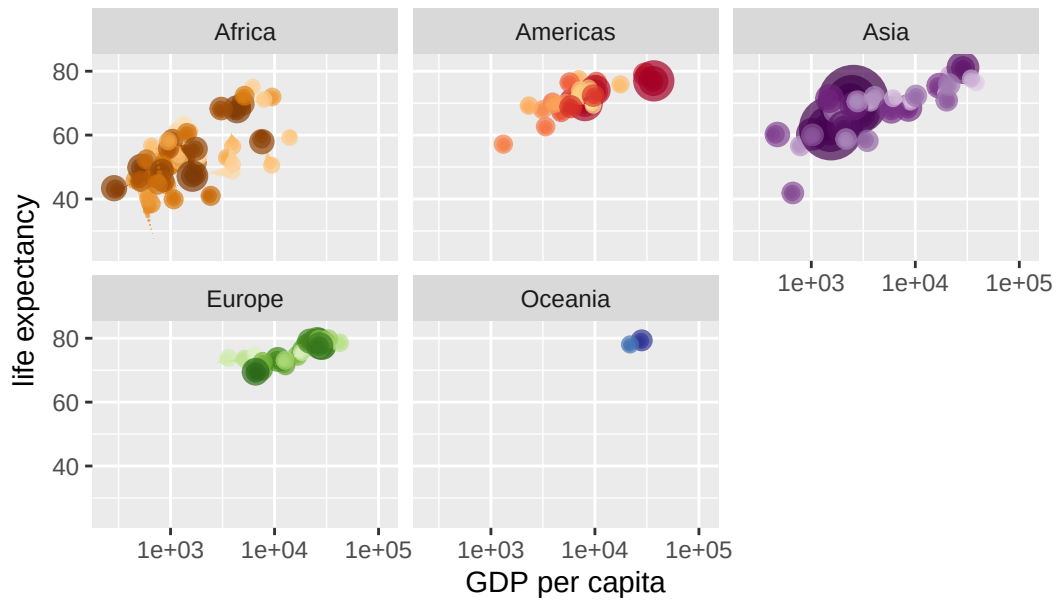
Year: 1998



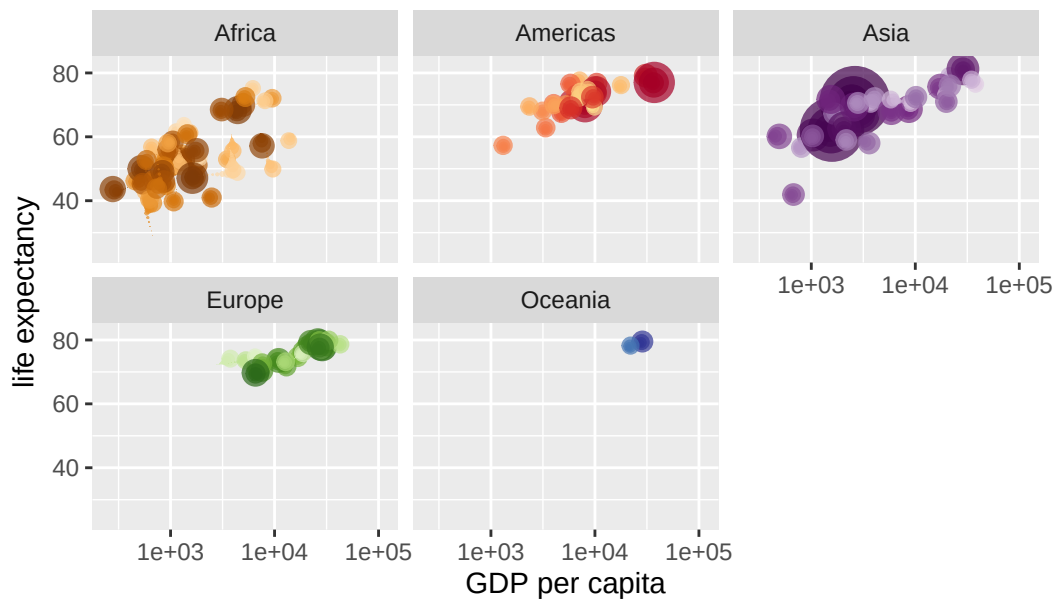
Year: 1998



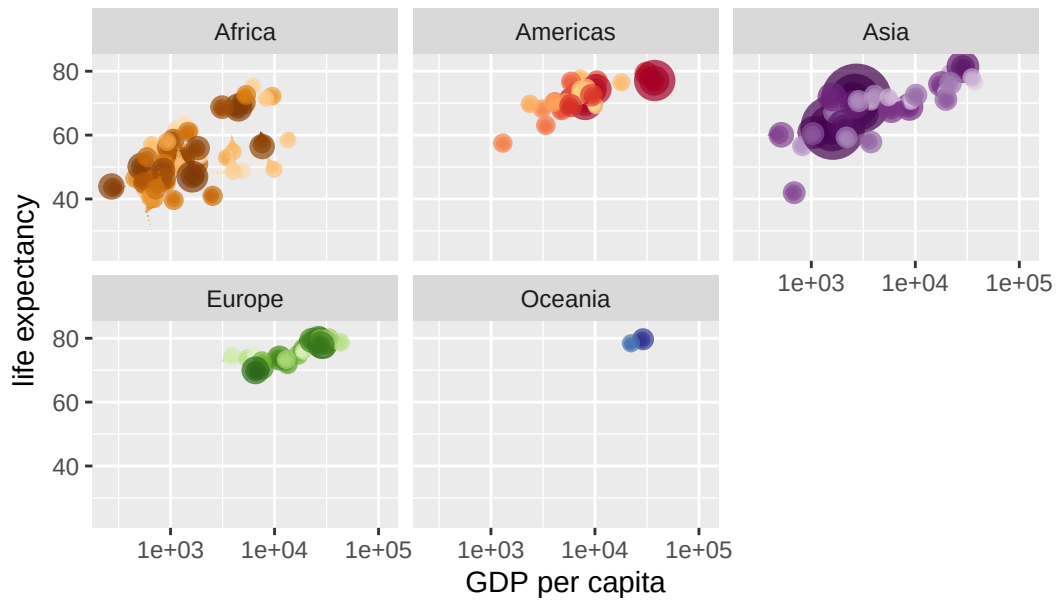
Year: 1999



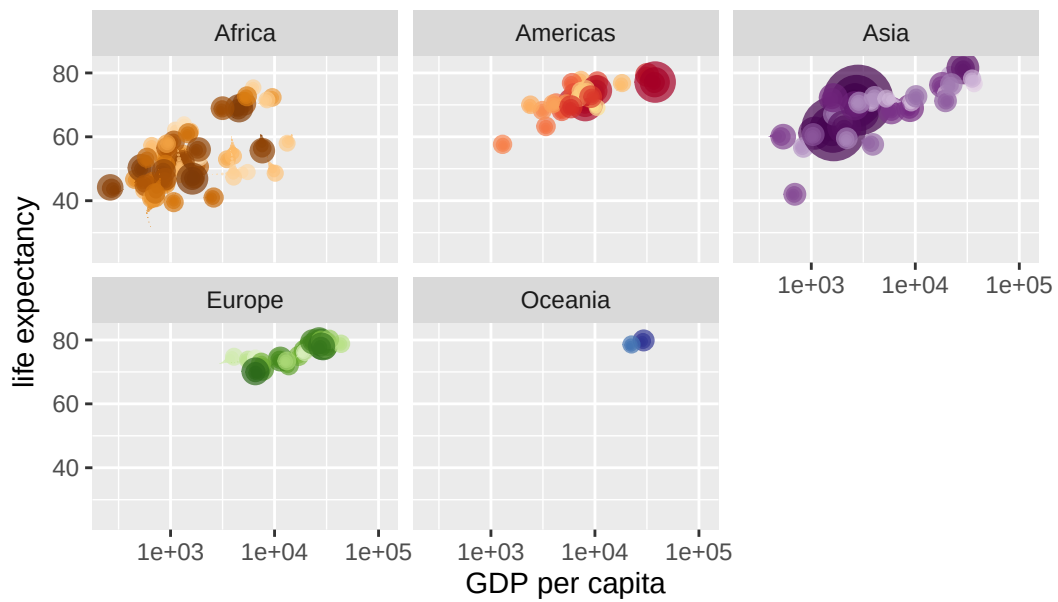
Year: 1999



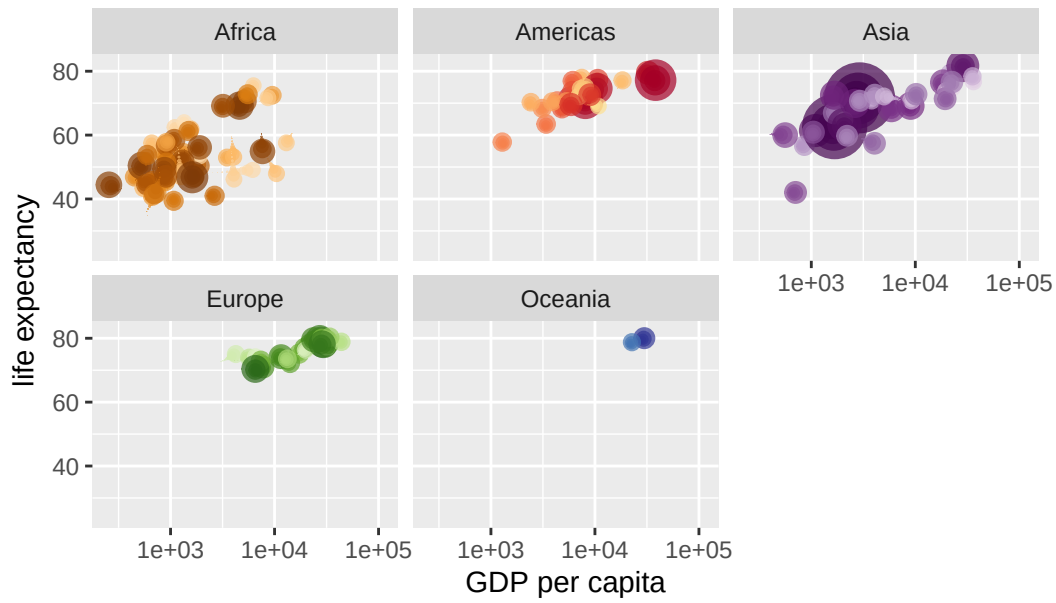
Year: 2000



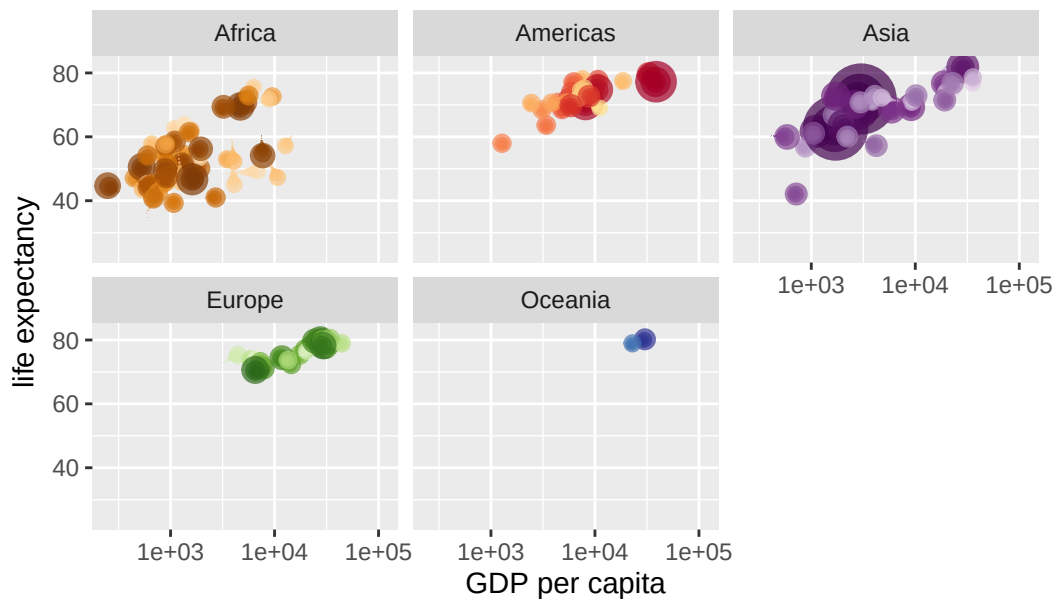
Year: 2000



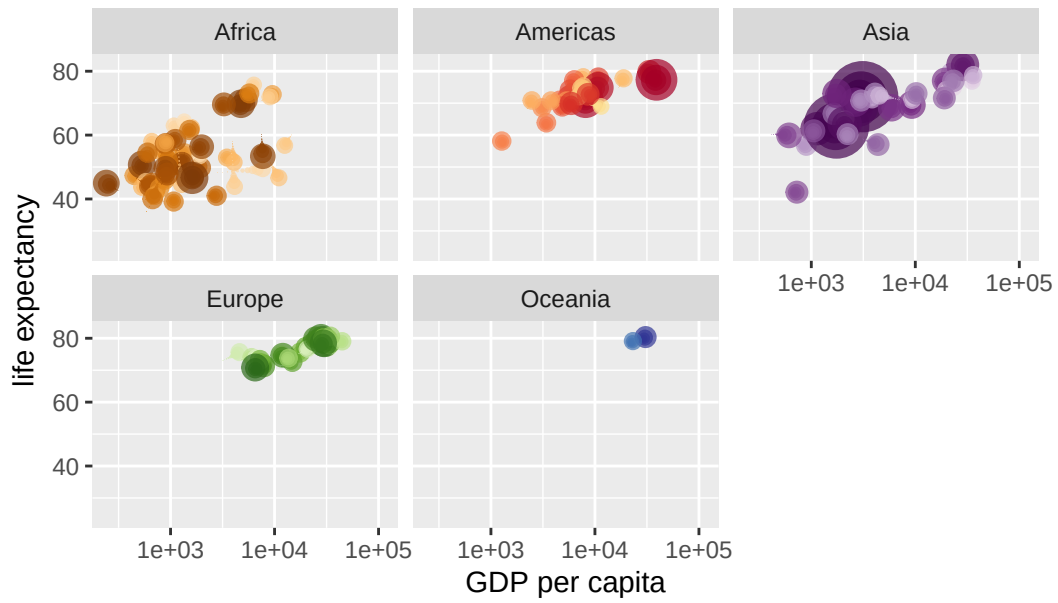
Year: 2001



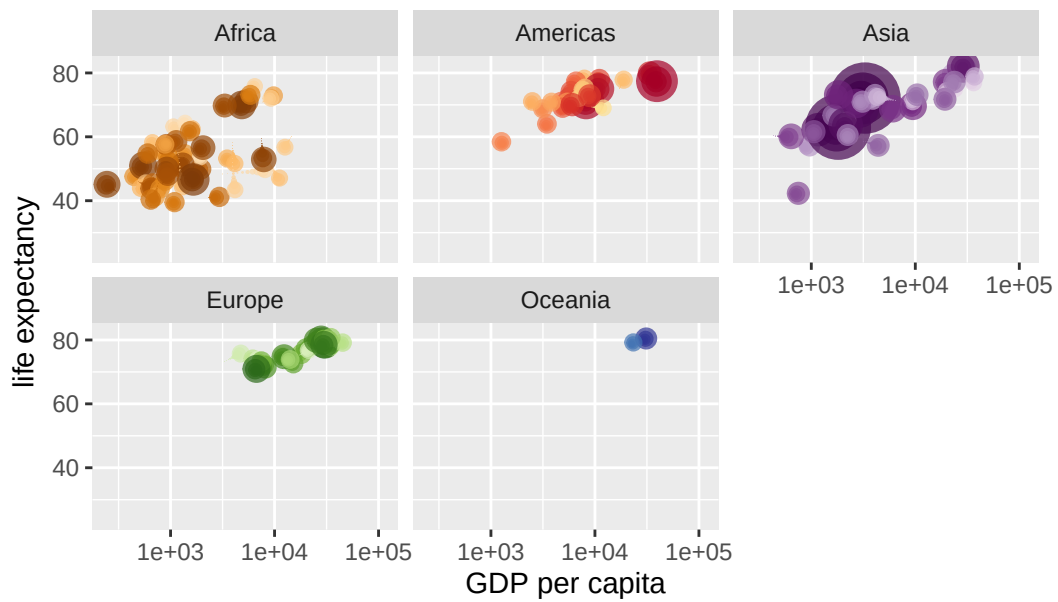
Year: 2001



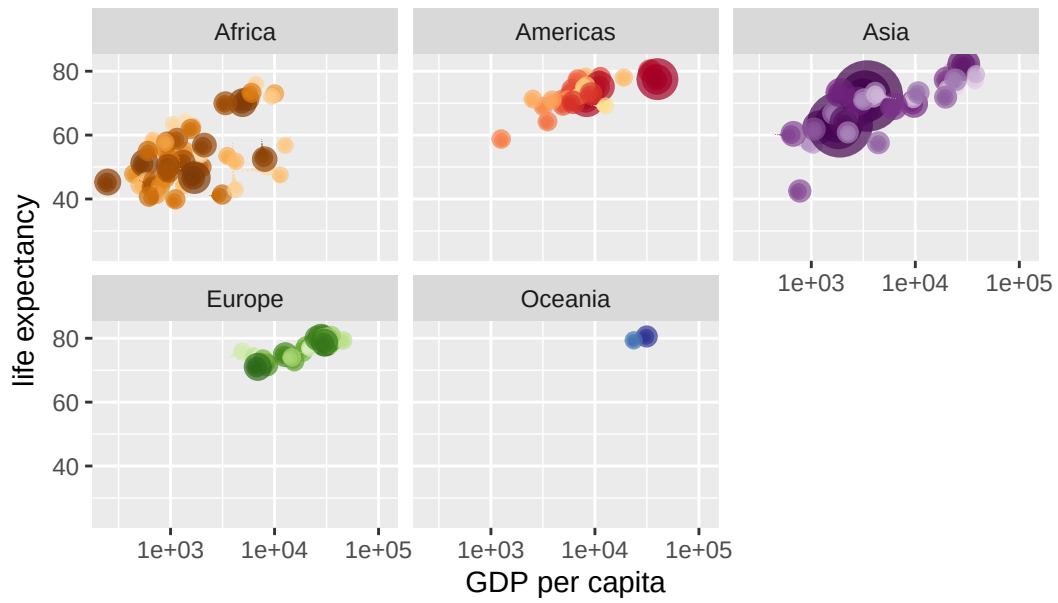
Year: 2002



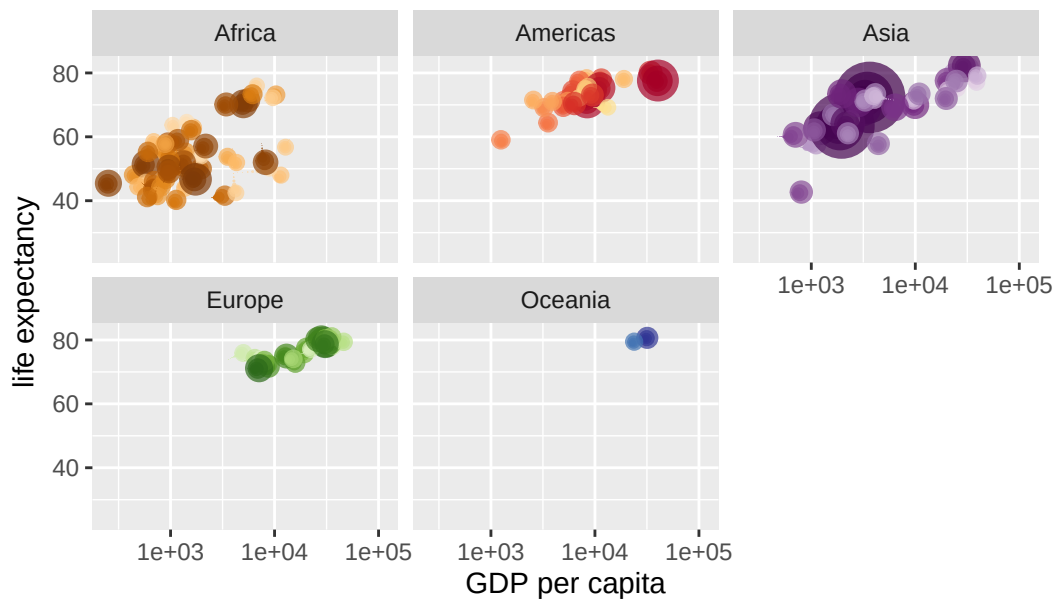
Year: 2003



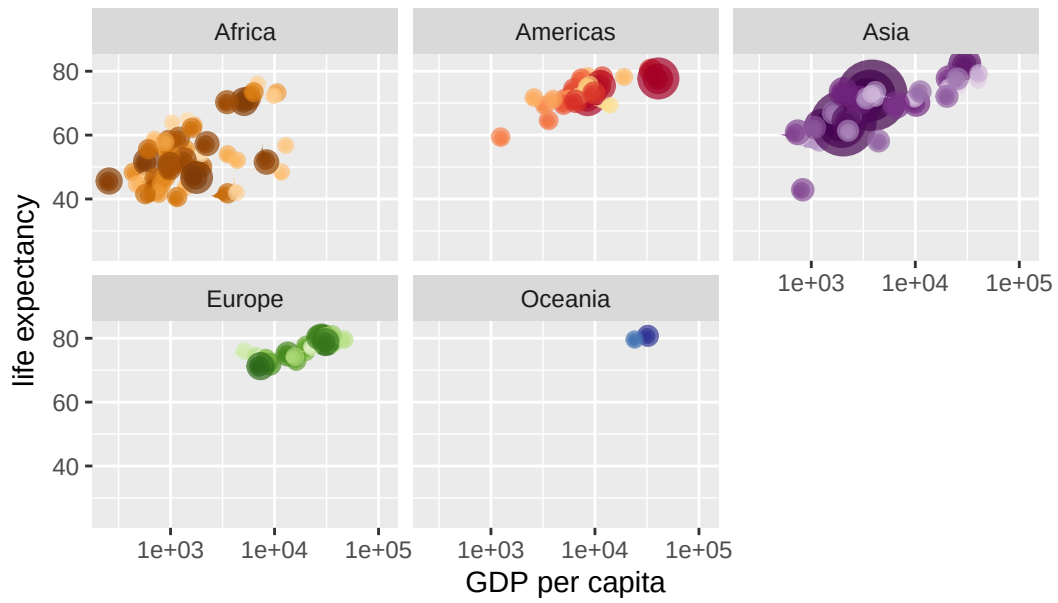
Year: 2003



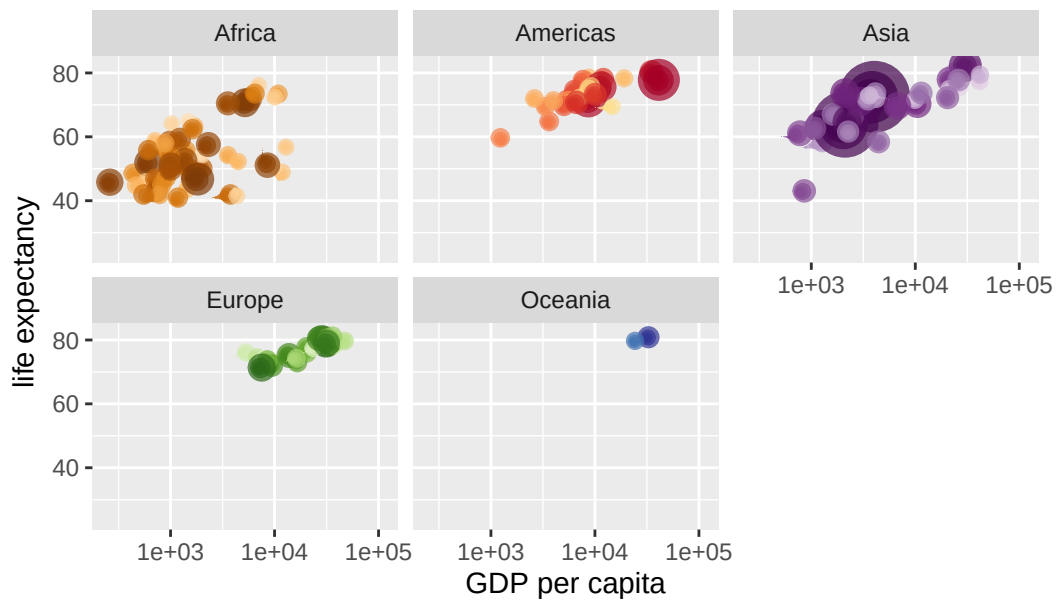
Year: 2004



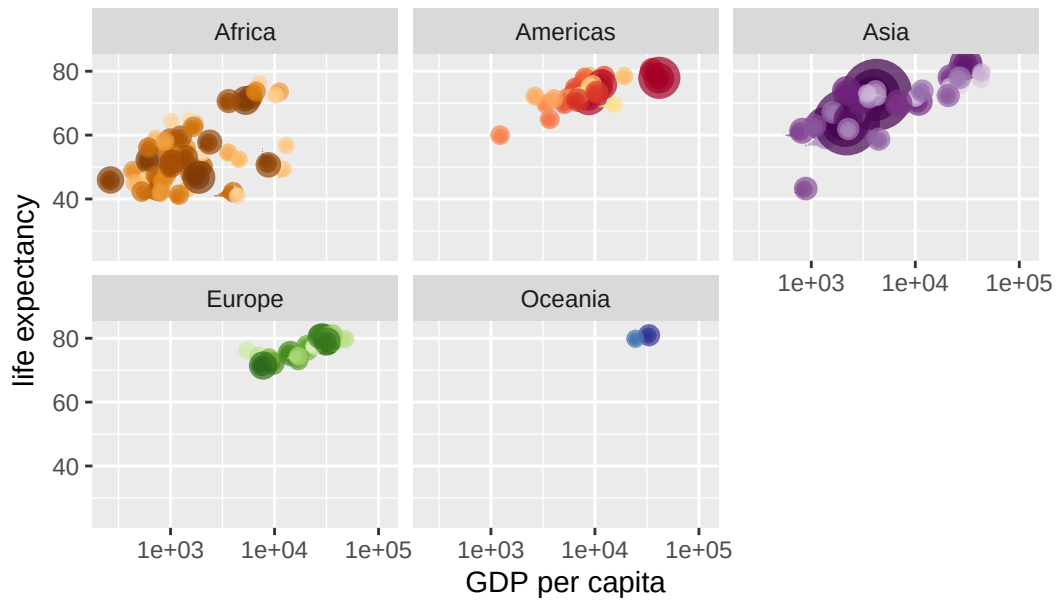
Year: 2004



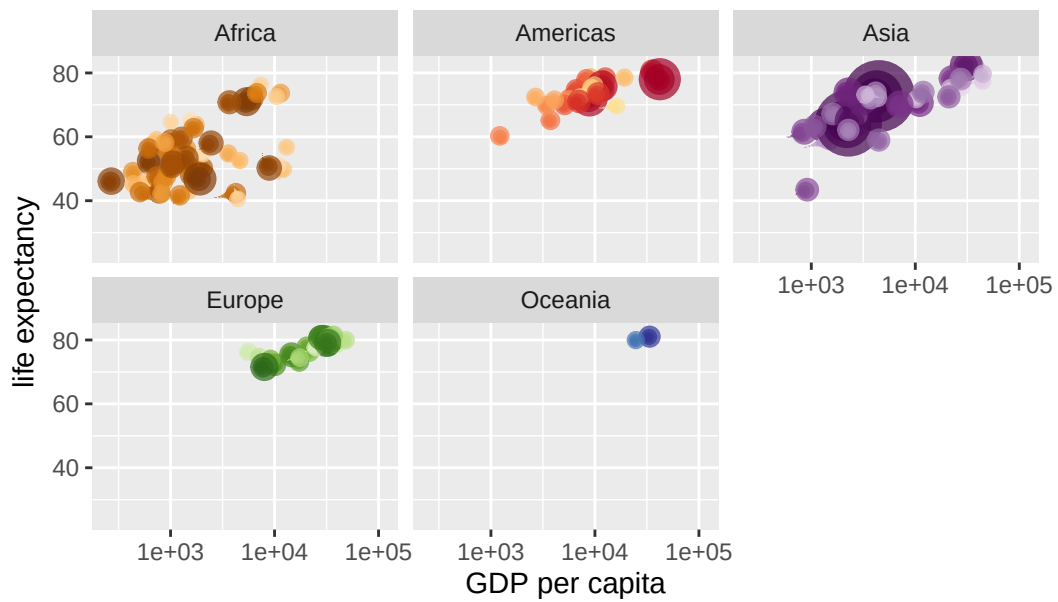
Year: 2005



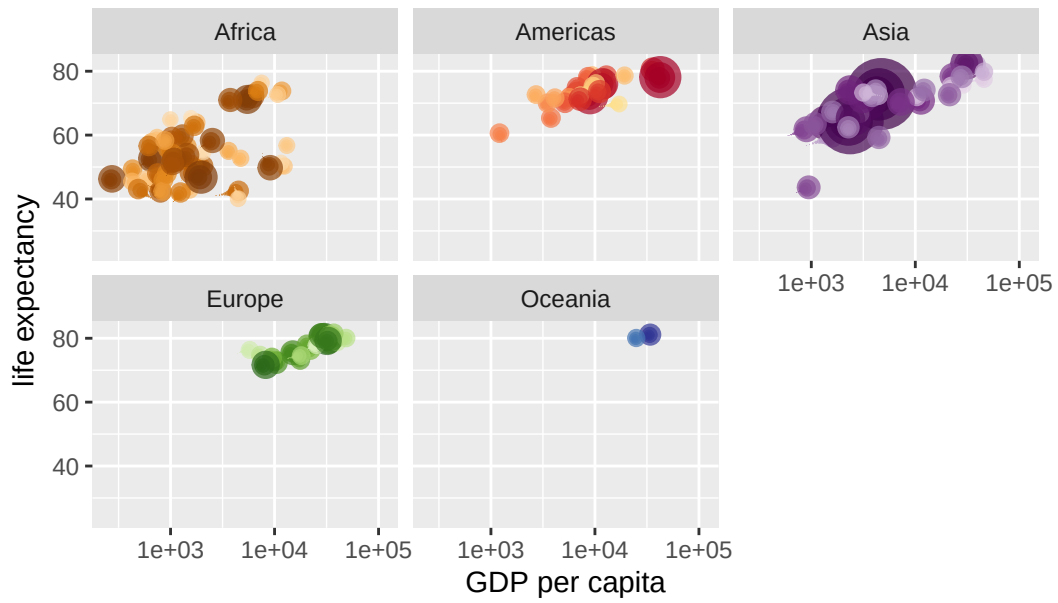
Year: 2005



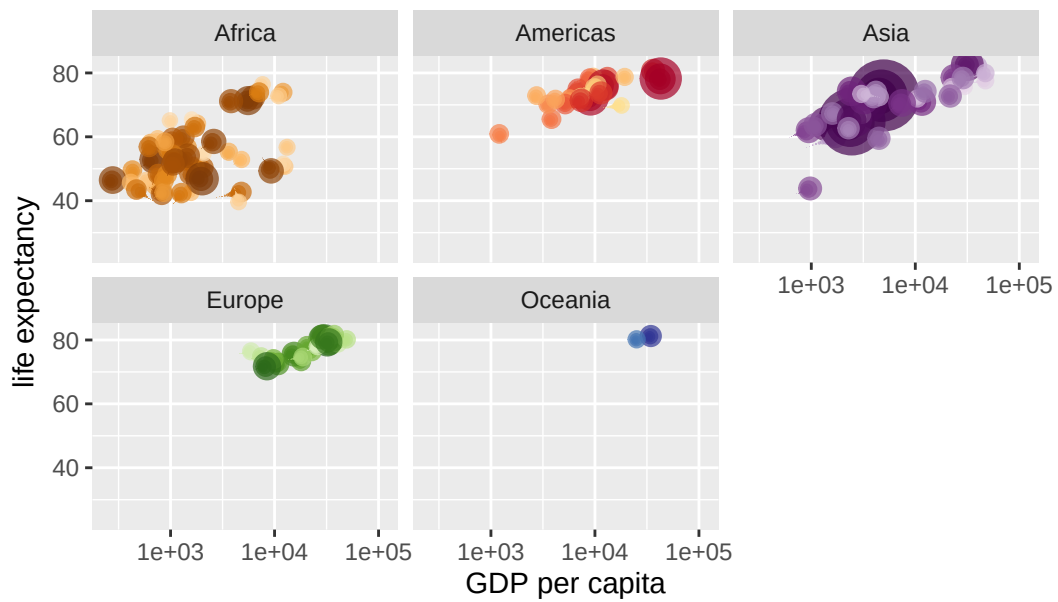
Year: 2006



Year: 2006



Year: 2007



Section 10

```
library(patchwork)

# Setup some example plots
p1 <- ggplot(mtcars) + geom_point(aes(mpg, disp))
p2 <- ggplot(mtcars) + geom_boxplot(aes(gear, disp, group = gear))
p3 <- ggplot(mtcars) + geom_smooth(aes(dis, qsec))
p4 <- ggplot(mtcars) + geom_bar(aes(carb))

# Use patchwork to combine them here:
(p1 | p2 | p3) /
  p4
```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'

